

CLINICAL STUDY PROTOCOL

CTA-2101

PHASE 1B/2A STUDY OF CTO1681 FOR THE PREVENTION AND TREATMENT OF CYTOKINE RELEASE SYNDROME IN PATIENTS WITH DIFFUSE LARGE B-CELL LYMPHOMA RECEIVING CHIMERIC ANTIGEN RECEPTOR T-CELL THERAPY

Sponsor: CytoAgents, Inc.

Investigational Product: CTO1681-IR-OSD-10 µg Immediate Release
Tablets

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INVESTIGATOR STATEMENT

I understand that all documentation provided to me by CytoAgents, Inc. or its designated representative concerning this study that has not been publicly published previously will be kept in the strictest confidence. This documentation includes the study protocol, investigator brochure, case report forms, and other proprietary medical and scientific data.

This study will not commence without the prior written approval of a properly constituted Institutional Review Board. No changes will be made to the study protocol without the prior written approval of CytoAgents, Inc. and the Institutional Review Board, except where necessary to eliminate an immediate hazard to the patient.

I have read, understood, and agree to abide by all the conditions and instructions contained in this protocol.

Investigator Name

Investigator Signature

Date

Investigational site or name of institution and location (printed)

CLINICAL STUDY SYNOPSIS

Name of Sponsor/Company: CytoAgents, Inc.
Name of Investigational Product: CTO1681-IR-OSD-10 µg Immediate Release Tablets
Name of Active Ingredient: CTO1681-DS
Title of Study: A Phase 1b/2a Study of CTO1681 for the Prevention and Treatment of Cytokine Release Syndrome in Patients with Diffuse Large B-cell Lymphoma Receiving CAR T-cell Therapy
Study Centers: Phase 1b: Up to 6 Phase 2a: Up to 20
Phase of Development: 1b/2a
Introduction: The safety and tolerability of CTO1681 (formerly known as GP1681) are well understood due to the clinical experience accrued with it as the active component of beraprost sodium. Beraprost is a well-studied agonist of the prostacyclin (IP) receptor and, like other prostacyclins, is a vasodilator and inhibitor of platelet aggregation. Beraprost has been approved and marketed in various Asian countries for decades as a treatment for pulmonary arterial hypertension and chronic arterial occlusive disease. CTO1681 downregulates the production of over 20 proinflammatory cytokines, many of which are directly involved in cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), (ie, interleukin [IL]-1α, -2, -6, -10, granulocyte-macrophage colony-stimulating factor [GM-CSF], monocyte chemoattractant protein-1 [MCP-1], macrophage inflammatory protein [MIP-1α], tumor necrosis factor alpha [TNFα] and interferon gamma [IFN-γ]) (Teachey, Lacey et al. 2016, Gust, Hay et al. 2017, CTO1681 [Investigators Brochure]). The mechanism of action for CTO1681 cytokine regulation is through its function as a prostaglandin E2 (PGE2) and prostaglandin I2 (PGI2) agonist. CTO1681 binds the PGE2 receptor 4 (EP4) and IP prostanoid receptors. EP4 regulation is responsible for the majority of prostanoid anti-inflammatory processes and hence the site of the majority of CTO1681 regulation of cytokine production. Cytokine suppression through the EP4 receptor is mediated through control of nuclear factor kappa light chain enhancer of activated B cells (NFκB) stimulation of cytokine gene transcription. Activation of the EP4 signaling pathway at the EP4 receptor stabilizes the interaction of the EP4 receptor-associated protein. The EP4 receptor-associated protein inhibits phosphorylation and degradation of NFκB1 p105 through direct binding of p105, which in turn prevents the activation of NFκB and its translocation to the nucleus, thus inhibiting the transcription of NFκB-dependent proinflammatory cytokines. Chimeric antigen receptor (CAR) T-cell therapy has emerged as a promising therapeutic approach for hematological and solid tumor malignancies, and overall response rates with durable remission can be achieved in patients with refractory or relapsed B-cell malignancies. Chimeric antigen receptor T-cell products targeting cluster of differentiation (CD) 19 have been approved for B-cell non-Hodgkin lymphoma (B-NHL). However, CAR T-cell therapy has been known to result in life-threatening immune-mediated toxicities, namely CRS and neurotoxicity, referred to as ICANS.

Cytokine release syndrome is a systemic inflammatory response that can be triggered as a response to immune-mediated therapies due to the initial release of proinflammatory cytokines like IFN- γ , TNF- α , IL-1, IL-2, and IL-6. These cytokines lead to an activation of bystander immune cells and endothelial cells, which in turn activate more immune cells culminating in a cytokine storm. Activated monocytes and macrophages may play an important role in CRS pathophysiology and are major contributors to the “amplification” of the inflammatory response in CAR T-cell therapy.

Immune effector cell-associated neurotoxicity syndrome is a clinical neuropsychiatric syndrome that can occur following immunotherapy, particularly associated with T-cell-engaging therapies, and can often occur concurrently with or immediately following CRS. Cytokine release syndrome and ICANS are potentially life-threatening toxicities that are often caused by CAR T-cell therapy, and frequently occur within the first week following CAR T-cell therapy but may have later onset. For the CD19-directed CAR T-cell therapies currently commercially available, the reported median time to onset of CRS has a range between 2 and 5 days, with a maximum reported range of onset of 1 to 51 days. For ICANS, the reported median time to onset has a range between 4 and 8 days; the maximum reported range of onset is 1 to 359 days. For this Phase 1b/2a study, only patients with planned axicabtagene ciloleucel (ie, YESCARTA[®]) therapy will be enrolled to manage potential variability between CAR T-cell therapy in response and adverse reaction rates. Cytokine release syndrome and ICANS are not unique to axicabtagene ciloleucel, and CTO1681 studies in the future may involve additional CAR T-cell treatments.

The investigational product CTO1681 is a broad-based anti-inflammatory drug. Early administration of CTO1681, prior to the initial release of proinflammatory cytokines (eg, IFN- γ , TNF- α , IL-1, IL-2, and IL-6), may mitigate the risk of CRS and ICANS associated with CAR T-cell therapy. Through excess cytokine suppression, CTO1681 could potentially prevent the onset or escalation of CAR T-cell-induced immune-mediated toxicities, thereby reducing associated morbidities and hospitalization. The purpose of this study is to evaluate the safety and tolerability of CTO1681 and determine preliminary CTO1681 effectiveness, measured by reduction of CAR T-cell-induced toxicities including CRS and ICANS.

Nearly 90% of non-Hodgkin lymphoma types develop in B cells. Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of non-Hodgkin lymphoma in the United States, accounting for approximately 30% of all cases and is the predominate lymphoma subtype treated with CAR T-cell therapy. Of the commercially available CAR T-cell therapies, axicabtagene ciloleucel has a notably high incidence of CRS, with over 90% of patients with large B-cell lymphoma developing any grade of CRS, and 9% of patients developing Grade 3 or higher CRS ([Yescarta 2022](#)).

This Phase 1b/2a study will enroll adult patients with DLBCL who are scheduled to receive CD19-directed CAR T-cell therapy. The Phase 1b portion of the study will include an initial dose escalation, and open-label safety and toxicity assessment on a sample of patients. The Phase 2a portion of the study will be a randomized, proof-of-concept assessment to investigate the potential benefit of CTO1681, initiated prior to CAR T-cell therapy, versus placebo, in mitigating or reducing CAR T-cell-induced toxicity.

Methodology:

This is a Phase 1b/2a study designed to explore the safety and effectiveness of CTO1681 in preventing or reducing CAR T-cell-induced toxicities.

Phase 1b: The Phase 1b portion of the study is a multicenter, open-label, dose-escalating, safety and pharmacokinetic (PK) study of multiple ascending doses of CTO1681 in patients with DLBCL who receive commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy.

This portion of the study will be conducted using a rolling 6 design to inform dose escalation among cohorts of patients with DLBCL. The Phase 1b portion of the study will include the following sequential steps for every patient in each cohort: Screening, treatment enrollment, pretreatment with lymphodepleting (LD) chemotherapy, initiation of study drug (CTO1681), CAR T-cell infusion, continuing treatment with CTO1681 (for a total of 15 days), and safety follow-up. Study Eligibility will be determined after leukapheresis and prior to LD chemotherapy.

Timing of treatment with CTO1681 will occur according to the following schedule in relation to CAR T-cell infusion (CAR T-cell infusion is on “Day 0”):

- Days –12 to –3: LD chemotherapy (if warranted per YESCARTA package insert [PI]).
- Day –1: Initiate study drug (CTO1681) approximately 24 hours before CAR T-cell infusion. Study drug will be continued orally (PO), 3 times daily (TID) for a total of 15 days.

The total daily dose of CTO1681 will be 30, 60, or 90 µg (administered in TID increments of 10, 20 or 30 µg/dose, respectively), depending on the cohort into which each patient is enrolled. Study drug must be taken within 30 minutes after a meal or snack.

- Day 0: CAR T-cell infusion (per YESCARTA PI).

Patients will be monitored to document tumor response for up to 6 months using the Lugano Criteria for Malignant Lymphoma. In addition, patients will be monitored throughout the study for efficacy and safety.

A rolling 6 design will be implemented to advise on dose escalation. The dose-limiting toxicity (DLT) observation period is defined as 43 days following the first dose of study drug (Day –1) which is 42 days following initiation of CAR T-cell therapy (Day 0), ie, Day +41.

Up to 3 dose cohorts will be considered with a maximum of 18 patients enrolled during the rolling 6 dose escalation procedure. Each dose level will be enrolled with a minimum of 3 and up to 6 patients. A minimum of 3 patients must complete the DLT observation period (through Day +41) prior to data review for dose escalation. A maximum of 0/3, 0/4, 0/5 or 1/6 DLT may be recorded to allow escalation to the next higher dose level. Recommendations for next dose level will be made by the Safety Review Committee (SRC) based on safety, laboratory, and any PK and pharmacodynamic (PD) data that are available at the time of review, according to the rolling 6 design. If adequate safety data are not available from 1 or more patients in a given cohort, additional patients will be enrolled in that cohort, and only patients for whom there is adequate safety data may be included in the safety evaluation. If 1 DLT has been observed, any new patients will be entered at the same dose level until 6 patients are enrolled and have completed the DLT observation period, or at least 2 DLTs are observed, and therefore the maximum tolerated dose (MTD) has been exceeded.

The MTD is defined as the maximum CTO1681 dose achieved where patients are treated safely. The dose may be escalated to the next level of dose if no more than 0 DLT observed in 3-5 patients or 1 DLT observed in 6 patients. If a dose level is too toxic (2 or more DLTs observed in 3-6 patients), then the MTD stopping rules for that cohort/dose level may be applied. Any or all of the cohorts deemed safe by the SRC following the dose escalation procedures may be backfilled up to a maximum of 15 additional patients at the discretion of the Medical Monitor and Sponsor. The backfill sample will allow further safety, antitumor response, PK, and PD information to better select a recommended

Phase 2 dose (RP2D) at, or below the MTD for Phase 2a. The maximum sample size for Phase 1b is 54 patients.

Daily Dose Level	Number of Patients in Rolling 6 Dose Escalation	Dose Escalation Rules	MTD Stopping Rules	Maximum Backfill Patients ^a
30 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15
60 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15
90 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15

Abbreviations: DLT = dose limiting toxicity; MTD = maximum tolerated dose; SRC = safety review committee

^a Backfill patients will be enrolled at the discretion of the Medical Monitor and Sponsor if the dose level is determined safe according to the MTD stopping rules and by recommendation of SRC. A maximum of 18 patients may be treated at each dose level.

Phase 2a: The Phase 2a portion of the study will be conducted as a randomized, double-blind, placebo-controlled study and will enroll approximately 100 patients. Patients will be treated during Phase 2a at the RP2D of CTO1681, which was established during Phase 1b, or with placebo. Patients will be randomized in a 1:1 ratio between the CTO1681 and placebo arms. Patients will receive commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy. Each patient will follow the same study steps as noted above for the Phase 1b portion. Patients enrolled in this Phase 2a portion of the study will be monitored for tumor response for up to 12 months (using the Lugano Criteria for Malignant Lymphoma) following CAR T-cell infusion. Sparse PK samples will be collected in all patients in Phase 2a to perform population PK and exploratory exposure-response analyses.

For both the Phase 1b and Phase 2a portions of the study, safety, efficacy, PK, and PD assessments will be conducted at protocol-specified intervals per the Schedule of Events.

Study patients will be hospitalized for observation and management of CAR T-cell-related toxicities based on institutional guidelines. All CRS and ICANS grading will be according to American Society of Transplantation and Cellular Therapy (ASTCT) criteria. Detailed management guidelines for CRS and ICANS are provided in the full protocol and are consistent with standard of care (SOC) management.

Management of CRS and ICANS:

CTO1681 should be continued in patients who develop CRS and/or ICANS, except as noted under the section for Management of DLTs. Detailed management guidelines for CRS and ICANS will be provided to the Investigator to standardize care across the participating study sites. These guidelines will be consistent with the SOC outlined in the PI for the selected CAR T-cell therapy, except that discretionary use of corticosteroids or tocilizumab prophylactically with Grade 1 CRS or ICANS will not be allowed. For Grade 2 or higher CRS and ICANS, treatment with corticosteroids and tocilizumab, as well as other supportive therapies may be considered.

Dose-Limiting Toxicities:

Key DLT criteria are:

- Hypotension: Defined as documented, prolonged (≥ 2 hours) systolic blood pressure < 80 mm Hg unresponsive to basic medical therapy (ie, fluid challenge).
- Bleeding: Any clinically significant bleeding event (graded per Common Terminology Criteria for Adverse Events v5.0 [CTCAE]) Grade ≥ 3 (ie, excluding accidental cuts, periodontal, menstrual, or hemorrhoidal bleeding, or easily controlled epistaxis), including any episode of hemoptysis.
- Hepatic dysfunction: Any occurrence of transaminase elevation of $> 5 \times$ upper limit of normal (ULN; CTCAE Grade ≥ 3) that does not improve to $\leq$ Grade 2 within 7 days.
- QTc prolongation: Development of QT interval corrected using Fridericia's formula (QTcF) ≥ 550 msec or a > 60 msec change from baseline, confirmed by repeat 12-lead electrocardiogram (ECG).
- Ventricular tachycardia: CTCAE Grade ≥ 3 requiring urgent medical intervention.
- Other:
 1. Any other CTCAE Grade ≥ 3 non-hematological toxicity that does not improve to CTCAE $\leq$ Grade 2 within 72 hours.
 2. Any death not due to underlying malignancy (Note: All CTCAE Grade 5 events will trigger a study pause to collect and assess the event for DLT determination. Expedited safety reporting to the Food and Drug Administration [FDA] would be included).

Exemptions from DLT criteria are based on anticipated high confounding prevalence in the specific study patient population and will include (per CTCAE):

- Grade 3 or 4 anemia, lymphopenia, or thrombocytopenia.
- Grade 3 or 4 fever or febrile neutropenia lasting ≤ 14 days.
- Grade 3 or 4 tumor lysis syndrome lasting ≤ 14 days.
- Grade 3 peripheral sensory neuropathy will not be considered a DLT in subjects with a prior history of peripheral sensory neuropathy.
- Grade 3 fatigue and anorexia that does not result in hospitalization, tube feeding, or total parenteral nutrition.

Objectives:**Phase 1b:**

- Determine the preliminary safety profile of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.

- Determine the RP2D and, as appropriate, MTD of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.

Phase 2a:

- Determine preliminary CTO1681 effectiveness in preventing CRS or reducing CRS severity and duration in comparison to placebo.
- Determine preliminary CTO1681 effectiveness in preventing ICANS or reducing ICANS severity and duration in comparison to placebo.
- Determine the expanded safety profile of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.
- Investigate the potential impact of CTO1681 on antitumor activity of CAR T-cell therapy in comparison to historical data and placebo.

Endpoints:

Phase 1b:

Primary:

- Adverse events (AEs), including DLTs, and laboratory and ECG abnormalities, as characterized by type, frequency, timing, severity (graded per CTCAE v5.0), seriousness, and relationship to CTO1681.

Exploratory:

- Incidence of CRS (any grade).
- Time to resolution of CRS (any grade).
- Incidence of ICANS (any grade).
- Time to resolution of ICANS (any grade).
- Biomarkers relating to the mechanism of action (MOA) of CTO1681 and immune response.
- Frequency and duration of hospitalizations and intensive care unit (ICU) admissions due to CRS or ICANS.
- Use of tocilizumab, steroids, and other anticytokine therapies for treatment of CRS or ICANS (amount and duration).
- PK concentration profile of CTO1681.
- Levels and persistence of CAR T-cells.
- Immune reconstitution (ie, B-cell aplasia).
- Antitumor activity including complete response (CR), overall response rate (ORR), duration of response (DOR), and progression-free survival (PFS) according to Investigator review, and overall survival (OS).

Phase 2a (Active vs Placebo):

Primary:

- Incidence of Grade ≥ 2 CRS.

Secondary:

- Additional assessments of CRS incidence and severity:
 - Incidence of CRS (any grade).
 - Incidence of Grade ≥ 3 CRS.
 - Time to CRS resolution.

- Additional assessments of ICANS incidence and severity:
 - Incidence of ICANS (any grade).
 - Incidence of Grade ≥ 2 ICANS.
 - Incidence of Grade ≥ 3 ICANS.
 - Time to ICANS resolution.
- AEs and laboratory abnormalities, and ECG abnormalities as characterized by type, frequency, timing, severity (graded per CTCAE v5.0), seriousness, and relationship to CTO1681.
- Frequency and duration of hospitalizations and ICU admissions due to CRS or ICANS.
- Use of tocilizumab, steroids, and other anticytokine therapies for treatment of CRS or ICANS.
- Antitumor activity including CR rate, ORR, DOR, and PFS according to Investigator review, and OS.

Exploratory:

- Levels and persistence of CAR T-cells.
- PK concentration profile of CTO1681.
- Biomarkers relating to MOA of CTO1681 and immune response.
- Immune reconstitution.

Number of Patients (planned):

Phase 1b: 9-18 patients in dose escalation; a maximum of 54 patients including backfill expansion

Phase 2a: Approximately 100 patients randomized 1:1 to CTO1681 (n = 50) or placebo (n = 50)

Inclusion Criteria:

1. Age 18 years or older.
2. Undergone leukapheresis and scheduled to receive protocol-specified commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy for DLBCL without corticosteroid prophylaxis for CRS and/or ICANS. All patients must have relapsed or refractory disease to at least 1 prior line of systemic therapy.
3. Met all inclusion criteria for CAR T-cell therapy per institutional guidelines.
4. Adequate organ function defined as:
 - Estimated Creatinine Clearance per Cockcroft-Gault formula ≥ 60 mL/min.
 - Serum alanine aminotransferase/ aspartate aminotransferase $\leq 2.5 \times$ ULN.
 - Total bilirubin $\leq 1.5 \times$ ULN.
 - Left ventricular ejection fraction $\geq 40\%$ on echocardiogram or multigated acquisition scan and no clinically significant pericardial effusion.
 - Platelets $\geq 50,000/\text{mm}^3$.
 - Absolute neutrophil count $> 1000/\mu\text{L}$.
 - Absolute lymphocyte count $> 100/\mu\text{L}$.
5. Documented measurable lymphoma disease adequate to judge by Lugano Criteria.
6. Eastern Cooperative Oncology Group performance status 0 to 1.
7. Female participants of childbearing potential and all male participants must agree to use Investigator-approved methods of birth control while on study drug and for 30 days thereafter.
8. Patients who are willing to provide written informed consent before the predose procedures, or patients who have a legal representative capable of providing informed consent on their behalf.

Exclusion Criteria:

1. Any cytotoxic chemotherapy within 14 days prior to leukapheresis.
2. Clinically significant malabsorption syndromes and swallowing difficulties which are inadequately controlled with medication (eg, odynophagia, dysphagia, gastroesophageal reflux disease) as per Investigator assessment.
3. Grade 2 or greater electrolyte imbalance, per CTCAE v5.0:
 - Potassium < 3.0 or > 5.5 mmol/L
 - Sodium < 130 or > 150 mmol/L
 - Calcium < 8.0 or > 11.5 mg/dL
 - Magnesium < 0.5 or > 1.23 mmol/L
4. Clinically significant ECG abnormality at Screening or Baseline (Day -1) including, but not limited to, a confirmed QTcF value > 470 msec. Patients to be excluded include those with QTcF readings that are borderline or difficult to interpret because of a condition such as bundle branch block, or in those where the end of the T wave is difficult to measure. This also includes any Grade 2 or greater conduction block disorder, atrial, or ventricular arrhythmia.
5. History of clinically significant arrhythmia and/or requiring anticoagulation/antiplatelet treatment at therapeutic dose.
6. Any clinically significant (ie, active) cardiovascular disease, including cerebral vascular accident/stroke (< 6 months before enrollment), myocardial infarction (< 6 months before enrollment) or unstable angina, and congestive heart failure $\geq$ New York Heart Association Classification Class III.
7. Uncontrolled thromboembolic events or recent severe hemorrhage within the last 6 months.
8. Known history of any bleeding disorder.
9. Requirement for ongoing therapeutic doses of anticoagulant therapy, antiplatelet or fibrinolytic agents (low molecular weight heparin prophylaxis is allowed).
10. Baseline systolic blood pressure < 100 mm Hg.
11. History of autoimmune disease/graft versus host disease requiring immunosuppressive therapy within the last 2 years. However, physiologic steroids (prednisone equivalent) may be given at a dose of 5 mg or less.
12. Patients who, in the opinion of the Investigator, would be unlikely to comply with study procedures or are otherwise unsuitable for enrollment.

Investigational Product, Dosage, and Mode of Administration:

Phase 1b: CTO1681 will be orally administered as a 10- μ g tablet. In Cohort 1, CTO1681 tablets will be administered PO TID for 15 days at a total daily dose of 30 μ g. Cohorts 2 and 3 will receive CTO1681 PO TID for 15 days at total daily doses of 60 and 90 μ g, respectively. Neither inpatient dose modification nor dose modification within a cohort is permitted, except as specified in the management of DLTs.

Phase 2a: CTO1681 will be orally administered as 10- μ g tablets, potentially multiple tablets per dose, depending on the RP2D dose determined by the Phase 1b results.

NOTE: As CTO1681 has a positive food effect, it should be administered within 30 minutes after completion of a meal or snack.

Duration of Study (per patient):

Phase 1b: approximately 9 months (including Screening)

Phase 2a: approximately 15 months (including Screening)

Reference Therapy, Dosage, and Mode of Administration:

Phase 1b: None

Phase 2a: Placebo tablets will be administered PO, TID in a manner identical to active treatment. Placebo tablets will be identical in size, shape, color, and packaging to CTO1681 10-μg tablets.

Criteria for Pausing or Stopping the Study:

If any of the events referenced below as Pausing criteria or Stopping criteria occur within 43 days after the first exposure to the study drug (ie, Day +41), the SRC (in Phase 1b) or the IDMC (in Phase 2a) will be immediately informed, and an ad hoc meeting scheduled to review all available safety data. A pause to study recruitment will commence at that time. The committee will then provide recommendations as to how to proceed.

Pausing /Stopping criteria:

Occurrence of any DLT will prompt an ad hoc meeting of SRC (Phase 1b) or IDMC (Phase 2a):

- Hypotension: Defined as documented, prolonged (≥ 2 hours) systolic blood pressure < 80 mm Hg unresponsive to basic medical therapy (ie, fluid challenge).
- Bleeding: Any clinically significant bleeding event (graded per CTCAE v5.0) Grade ≥ 3 (ie, excluding accidental cuts, periodontal, menstrual, or hemorrhoidal bleeding, or easily controlled epistaxis), including any episode of hemoptysis.
- Hepatic dysfunction: Any occurrence of transaminase elevation of $> 5 \times$ upper limit of normal (ULN; CTCAE Grade ≥ 3) that does not improve to $\leq$ Grade 2 within 7 days.
- QTc prolongation: Development of QTcF ≥ 550 msec or a >60 msec change from baseline, confirmed by repeat 12-lead electrocardiogram (ECG).
- Ventricular tachycardia: (CTCAE Grade ≥ 3) requiring urgent medical intervention.
- Other:
 1. Any other CTCAE Grade 3 $\geq$ non-hematological toxicity that does not improve to CTCAE $\leq$ Grade 2 within 72 hours.
 2. Any death not due to underlying malignancy (Note: All CTCAE Grade 5 events will trigger a study pause to collect and assess the event for DLT determination. Expedited reporting to the FDA would be included).

SRC (Phase 1b) or IDMC (Phase 2a) will recommend stopping the study if:

- CTCAE Grade 4 toxicity (unmanageable, unexpected) occurring in $> 25\%$ of dosed patients that is possibly, probably, or definitely related to CTO1681 (excluding fever, hematological toxicities, and CRS/ICANS).
- Death resulting from a CTO1681-associated DLT not related to underlying disease, CAR T-cell therapy, or an extrinsic event (unrelated to the investigational product or study procedures). Specifically:
 - Fatal bleeding event
 - Refractory hypotension resulting in death
 - Refractory ventricular tachycardia resulting in death

Note: safety stopping criteria for Phase 2a will also take into account the data from subjects from Phase 1b who were treated at the RP2D as well as with lower dosages.

The Sponsor may decide to pause, stop, or make adaptations to the study based on recommendations of the SRC/IDMC and the totality of evidence. The SRC may recommend that the study be paused or stopped if it is determined that patient safety may be compromised by continuing the study. The IDMC may recommend that the study be stopped based on safety or reduced antitumor response.

Statistical Methods:

Sample size considerations:

Phase 1b: The maximum sample size of 54 (maximum 18 per dose level) was selected to support the dose escalation and RP2D selection of CTO1681 dose level in accordance with the rolling 6 dose escalation procedure and backfill expansion to ensure sufficient safety data for Phase 2a dose selection.

Phase 2a: This study was designed with a sample size selected to provide sufficient power for the primary endpoint. With a sample size of 100 patients (50 per group), there is 80% power with a 1-sided type I error rate of 0.05 to detect a difference in proportion of 40% of control patients experiencing Grade ≥ 2 CRS versus 16% of CTO1681 patients.

Analysis populations:

- **Modified Intent-to-treat (mITT) Population:** The mITT Population will include all patients who receive both CAR T-cell therapy and at least 1 dose of study drug (CTO1681/ placebo). Patients will be grouped by dose group for Phase 1b analyses and as randomized for the Phase 2a analyses.
- **Safety Population:** The Safety Population will include all patients who receive at least 1 dose of study drug (CTO1681). Patients will be grouped as treated.
- **DLT Population:** The DLT Population will include all patients in Phase 1b who receive both CAR T-cell therapy and at least 1 dose of CTO1681. Patients who withdraw before experiencing a DLT and before completing the DLT observation period (through Day +41, 43 days after the first dose of CTO1681 or placebo) for reasons unrelated to study procedures, conduct, or treatment, will not be included in this population.

Additional analysis populations may be defined in the Statistical Analysis Plan (SAP). A standard intent-to-treat (ITT) population including all randomized patients may be analyzed as an evaluation of mITT robustness.

Data presentation/descriptive statistics:

All statistical analyses will be described in a detailed SAP that will be executed prior to the unblinding of the Phase 2a data. Analysis of the Phase 1b portion of the study occurring before SAP finalization will be described in a separate Phase 1b/interim analysis plan. In the event of discrepancies between the protocol and the SAP, the SAP will be used to determine correct analyses and methodologies.

Demographic, safety, PK, PD, and efficacy parameters will be summarized by phase of study, presented overall and by dose group for Phase 1b, and overall and by treatment group for Phase 2a, and will be supported by graphical presentations, as appropriate. Descriptive summaries will include the number of patients (n), mean, standard deviation, minimum, and maximum for continuous parameters, and frequency and percentage for categorical parameters. Longitudinal data will be presented by appropriate time intervals depending on the endpoint being summarized. Time-to-event endpoints will be analyzed using Kaplan-Meier methods and summarized with quartiles (25th, 50th [median], and 75th percentiles). Confidence intervals (CI) will be presented, as appropriate.

Phase 1b:

In keeping with the exploratory nature of a Phase 1b study, no inferential hypothesis testing will be performed. All endpoint summaries will be presented with descriptive statistics. Tabulations will be summarized by dose level and overall. Summaries will be performed on the mITT and DLT Populations.

An end of Phase 1b interim analysis will be performed on completion of Phase 1b once all subjects in the last cohort have either discontinued from the study or completed the DLT evaluation period (ie, Day +41, 43 days after the first dose of CTO1681).

Phase 2a:

Primary and secondary endpoint analyses will be performed on the mITT Population. Analysis of safety parameters will be performed on the Safety Population.

The primary Phase 2a endpoint, incidence of Grade ≥ 2 CRS, will be summarized as the frequency and proportion of patients with the event and compared between treatment groups using a Fisher's exact test assessed at a 1-sided type I error rate of 0.05. Exact 90% CI will be presented.

Other binary efficacy endpoints will be summarized using frequency, proportion, and exact 90% CIs. Comparisons between treatment groups will be made for exploratory purposes using Fisher's exact test. Ordinal variables, including maximum severity of CRS, will be analyzed using a Wilcoxon-Mann-Whitney test, as appropriate. Time to CRS resolution and time to ICANS resolution will be analyzed using Kaplan-Meier methods and compared between treatment groups using a logrank test. Continuous endpoints will be summarized using descriptive statistics.

Analysis of safety parameters will include summary of AEs, clinical laboratory measurements, vital signs, ECG results, and concomitant medications.

An end-of-Phase 2a interim analysis will be performed on completion of the Phase 2a portion of the study, once all subjects have either discontinued from the study or completed the DLT evaluation period (ie, Day +41, 43 days after the first dose of CTO1681).

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LIST OF ABBREVIATIONS AND DEFINITION OF TERMS

Abbreviation	Definition
AE	Adverse event
ASTCT	American Society of Transplantation and Cellular Therapy
AUC	Area under the curve
B-NHL	B-cell non-Hodgkin lymphoma
BBB	Blood-brain barrier
BP	Blood pressure
CAR	Chimeric antigen receptor
CBC	Complete blood count
CI	Confidence interval
C _{max}	Maximum serum concentration
CR	Complete response
CRP	C-reactive protein
CRS	Cytokine release syndrome
CSF	Cerebrospinal fluid
CT	Computerized tomography
CTCAE	Common Terminology Criteria for Adverse Events
DLBCL	Diffuse large B-cell lymphoma
DLT	Dose limiting toxicity
DOR	Duration of response
eCRF	Electronic data capture form
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
EFS	Event-free survival
EP4	PGE2 receptor 4
FDA	The Food and Drug Administration
FL	Follicular lymphoma
GCP	Good clinical practice
GM-CSF	Granulocyte-macrophage colony-stimulating factor
HCT	Hematopoietic cell transplant
IB	Investigator's brochure

IDMC	Independent data monitoring committee
ICANS	Immune effector cell-associated neurotoxicity syndrome
ICE	Immune effector cell associated encephalopathy
ICU	Intensive care unit
IEC	Independent ethics committee
IFN- γ	Interferon gamma
IgG	Immunoglobulin G
IL	Interleukin
IND	Investigational New Drug Application
INR	International normalized ratio
IP	Prostacyclin
ITT	Intent-to-Treat
IRB	Institutional review board
IWRS	Interactive Web Response System
LBCL	Large B-cell lymphoma
LD	Lymphodepleting
MCP-1	Monocyte chemoattractant protein-1
MIP-1 α	Macrophage inflammatory protein-1 alpha
mITT	Modified Intent-to-treat
MOA	Mechanism of action
MTD	Maximum tolerated dose
NE	Neurological event
NF κ B	Nuclear factor kappa light chain enhancer of activated B cells
NHL	Non-Hodgkin lymphoma
ORR	Overall response rate
OS	Overall survival
PET	Positron emission tomography
PFS	Progression-free survival
PD	Pharmacodynamic
PGE2	Prostaglandin E2
PGI2	Prostaglandin I2
PI	Package insert

PK	Pharmacokinetic(s)
PMBL	Primary mediastinal large B-cell lymphoma
PO	Orally
PR	Partial response
PT	Prothrombin time
PTT	Partial thromboplastin time
QTcF	QT interval corrected using Fridericia's formula
R/R	Relapsed/refractory
RP2D	Recommended Phase 2 dose
SAE	Serious adverse event
SAP	Statistical analysis plan
SOC	Standard of care
SoE	Schedule of events
SRC	Safety review committee
$t_{1/2}$	Half-life
TBNK	Lymphocyte subsets enumeration
TID	Three times daily
t_{max}	Time to reach maximum serum concentration (C_{max})
TNF- α	Tumor necrosis factor alpha
ULN	Upper limit of normal

1. INTRODUCTION

1.1 Background

1.1.1 Non-Hodgkin Lymphoma Overview

Non-Hodgkin lymphoma (NHL) is a heterogeneous group of lymphoproliferative malignancies with differing patterns of behavior and responses to treatment ([Shankland, Armitage et al. 2012](#)). Similar to Hodgkin lymphoma, NHL usually originates in lymphoid tissues and can spread to other organs. Non-Hodgkin lymphoma, however, is much less predictable than Hodgkin lymphoma and has a far greater predilection to disseminate to extranodal sites. In the United States in 2022, there were 80,470 new cases of NHL diagnosed and 20,250 deaths attributed to the disease ([ACS 2022](#)). The prognosis of a patient diagnosed with NHL depends on the histologic type, stage, and treatment.

Non-Hodgkin lymphoma can be divided into 2 prognostic groups: indolent lymphomas and aggressive lymphomas. The more common indolent NHLs include follicular lymphoma (FL; a B-cell lymphoma) ([Ciobanu, Stanca et al. 2013](#)), while aggressive NHLs include diffuse large B-cell lymphoma (DLBCL) and mediastinal large B-cell lymphoma (primary mediastinal large B-cell lymphoma [PMBL]) ([Wienand and Chapuy 2021](#)). Nearly 90% of non-Hodgkin lymphoma types develop in B cells. Diffuse large B-cell lymphoma is the most common subtype of non-Hodgkin lymphoma in the United States, accounting for approximately 30% of all cases ([Yescarta 2022](#)).

In general, with modern treatment of patients with NHL, the overall survival (OS) rate at 5 years is over 60%. Of patients with aggressive NHL, more than 50% can be cured. Most relapses occur in the first 2 years after therapy. The risk of late relapse is higher in patients who manifest both indolent and aggressive histologies ([Cabanillas, Velasquez et al. 1992](#)).

While indolent NHL is responsive to immunotherapy, radiation therapy, and chemotherapy, a continuous rate of relapse is usually seen in advanced stages. Patients, however, can often be re-treated with considerable success if the disease histology remains low grade. Patients who present with, or convert to aggressive forms of NHL may have sustained complete responses (CR) with combination chemotherapy regimens or aggressive consolidation with marrow or stem cell support ([Yuen, Kamel et al. 1995](#), [Bastion, Sebban et al. 1997](#)).

In patients with relapsed or refractory (R/R) FL or DLBCL, autologous hematopoietic cell transplant (HCT) is a treatment option in those who achieve a CR after second-line therapy ([Lerner, Thomas et al. 2007](#), [Gisselbrecht, Glass et al. 2010](#), [Cuccuini, Briere et al. 2012](#)). However, there continues to be an unmet medical need in patients with R/R FL or DLBCL who are unable to achieve a CR with salvage chemotherapy, as well as in patients with R/R PMBL ([Sehn, Antin et al. 1998](#)).

In recent years, chimeric antigen receptor (CAR) T-cell therapy has emerged as a promising therapeutic approach for patients with hematological and solid tumor malignancies. Durable remission can be achieved in patients with R/R B-cell malignancies ([Cappell, Sherry et al. 2020](#), [Westin and Sehn 2022](#)). Chimeric antigen receptor T-cell products targeting CD19 have been approved for B-cell non-Hodgkin lymphomas (B-NHLs), and DLBCL is the predominate subtype treated with CAR T-cell therapy. Of the commercially available CAR T-cell therapies,

axicabtagene ciloleucel is commonly used and has a notably high incidence of cytokine release syndrome (CRS), with 90% of patients developing any grade of CRS, and 9% of those patients experiencing CRS developing Grade 3 or higher ([Yanez, Sanchez-Escamilla et al. 2019](#), [Yescarta 2022](#)).

The first CAR T-cell therapy approved in the United States by Food and Drug Administration (FDA) for patients with R/R B-NHL was axicabtagene ciloleucel (axi-cel; YESCARTA®) in 2017 ([Westin, Kersten et al. 2021](#), [Yescarta 2022](#)). In 2018, tisagenlecleucel (tisa-cel; KYMRIAH®) was approved in patients with B-NHL, with the exception of patients with PMBL ([Kymriah 2022](#)). Most recently, in 2021, lisocabtagene maraleucel (liso-cel; BREYANZI®) was approved ([Breyanzi 2021](#)).

1.1.2 CAR T-cell Therapy for Non-Hodgkin Lymphoma

Pivotal clinical trials for axi-cel (ZUMA-1), tisa-cel ([Schuster, Bishop et al. 2019](#)), and liso-cel (TRANSCEND-NHL-001 [TRANSCEND]) were conducted, establishing the efficacy and safety of these CAR T-cell therapies in patients with R/R B-NHL. All trials recruited heavily pretreated patients, the majority of whom were chemorefractory (76% in ZUMA-1, 55% in JULIET, and 67% in TRANSCEND) with a diagnosis of DLBCL (76% in ZUMA-1, 80% in JULIET, and 51% in TRANSCEND) ([Schuster, Bishop et al. 2019](#), [Abramson, Palomba et al. 2020](#), [Locke, Miklos et al. 2022](#)). The CR rate ranged from 32% to 58% ([Locke, Ghobadi et al. 2019](#), [Breyanzi 2021](#), [Kymriah 2022](#)). Overall response rates (ORRs) ranged from 52% to 74% with 1-year OS rates of 48% to 59%, demonstrating that CAR T-cell therapies have altered the natural history of chemorefractory DLBCL, in comparison with nonrandomized historical controls ([Locke, Ghobadi et al. 2019](#), [Schuster, Bishop et al. 2019](#), [Westin, Tam et al. 2019](#), [Abramson, Palomba et al. 2020](#)).

Retrospective studies have reported on the use of axi-cel and tisa-cel as the standard of care (SOC) in patients with R/R B-NHL. Three such studies of axi-cel showed similar outcomes to ZUMA-1, which reported a CR of 58% ([Locke, Ghobadi et al. 2019](#)). Jacobson et al, Nastoupil et al, and Riedell et al reported CRs of 50%, 64% , and 39% (assessed at 90 days), respectively ([Jacobson, Hunter et al. 2020](#), [Nastoupil, Jain et al. 2020](#), [Riedell, Walling et al. 2020](#)). Pasquini et al. reported a CR of 52% in patients with 6 months of follow-up ([Pasquini, Locke et al. 2019](#)). Similar to the pivotal clinical trials, the majority of real-world CAR T-cell therapy patients had DLBCL; a smaller proportion of patients had transformed FL, PMBL, or other lymphomas.

Two retrospective studies ([Pasquini, Hu et al. 2020](#), [Riedell, Walling et al. 2020](#)) reported efficacy outcomes for tisa-cel. Complete response rates—39% (at 90 days for Riedell et al) and 40% (for Pasquini et al) ([Pasquini, Hu et al. 2020](#), [Riedell, Walling et al. 2020](#)), respectively, were comparable with the CR of 40% reported in the JULIET study ([Schuster, Bishop et al. 2019](#)). At the 2021 meeting of the American Society of Hematology, results of two Phase 3 trials (ZUMA-7 and TRANSFORM) were shared revealing CAR T-cell therapy has the potential to replace chemoimmunotherapy as second-line treatment of patients with R/R large B-cell lymphoma (LBCL). Both trials were designed to test whether moving CAR T-cell therapy forward into the second line could improve outcomes. For patients with LBCL who failed first-line therapy, chemoimmunotherapy followed by high-dose chemotherapy and autologous HCT has been the SOC, but only about 25% of patients who are candidates for HCT achieve

durable remissions. At a median follow-up of 24.9 months in the ZUMA-7 trial, patients randomly assigned to receive CAR T-cell therapy with axi-cel had a median event-free survival (EFS) of 8.3 months, compared with 2 months for patients randomly assigned to SOC chemoimmunotherapy (Locke, Miklos et al. 2022). In the TRANSFORM trial, which compared liso-cel with SOC second-line chemotherapy, median EFS was 10.1 versus 2.3 months for liso-cel and SOC (Kamdar, Solomon et al. 2022).

The efficacy of CAR T-cell therapy has been demonstrated across several trials, but the use of this therapeutic approach may be limited because of its association with substantial and even life-threatening toxicities, namely CRS and immune effector cell-associated neurotoxicity syndrome (ICANS), which may limit its use.

1.1.3 CRS and ICANS Related to CAR T-cell Therapy

Cytokine release syndrome is a systemic inflammatory response that can be triggered by immune-mediated therapies due to the initial release of proinflammatory cytokines like interferon gamma (IFN- γ), tumor necrosis factor alpha (TNF- α), interleukin (IL)-1, IL-2, and IL-6 (Shimabukuro-Vornhagen, Godel et al. 2018). These cytokines lead to an activation of bystander immune cells and endothelial cells, which in turn activate more immune cells culminating in a cytokine storm. Activated monocytes and macrophages may play an important role in CRS pathophysiology and are major contributors to the “amplification” of the inflammatory response in CAR T-cell therapy.

The pathophysiology of CRS is comprised of several phases. In the first phase, CAR T-cells migrate to the tumor site after infusion and there is CAR-mediated recognition of antigen-expressing target cells. This is followed by a characteristic proliferation of CAR T-cells at the tumor site, in situ cytokine production by both activated CAR T-cells and cellular components of the tumor microenvironment, activation of “bystander” endogenous immune cells, direct and indirect tumor cell killing, and onset of CRS. There is then an increase in proinflammatory cytokine levels and expansion of CAR T-cell populations in the peripheral blood, which is associated with a systemic inflammatory response that can lead to endothelial injury and vascular leakage in multiple tissues and organs, resulting in associated effects such as hypoxia, hypotension and organ damage. During this phase, endothelial cell activation may lead to a breakdown of the blood-brain barrier (BBB) and diffusion of cytokines into the cerebrospinal fluid and central nervous system, which may contribute to the development of ICANS. The pathophysiology of ICANS is less understood, but it is associated with elevated levels of proinflammatory cytokines and may develop concurrently with CRS onset, although ICANS usually develops after CRS, and often after CRS has resolved (Castaneda-Puglianini and Chavez 2021). Finally, activation-induced cell death of T-cells after eradication of the tumor results in decreased serum cytokine levels and a reduced systemic inflammatory response, the end of CRS and/or ICANS symptoms and, potentially, the persistence of long-term memory CAR T-cells (Shimabukuro-Vornhagen, Godel et al. 2018, Freyer and Porter 2020).

Cytokine release syndrome is hallmarked by the incidence of specific adverse events (AEs) resulting from the on-target action of CAR T-cell therapy. Adverse events of fever, hypotension, and hypoxia are signs of CRS. Cytokine release syndrome can also lead to multiple organ dysfunctions, such as persistent cytopenias, cardiac complications, acute kidney injury and

coagulation system abnormalities. In the pivotal axi-cel, liso-cel, and tisa-cel trials CRS had a median time to onset of 2 to 5 days after CAR T-cell therapy infusion (Neelapu, Locke et al. 2017, Abramson, Palomba et al. 2019, Bachanova, Westin et al. 2019). Median time to resolution ranged from 5 to 8 days (Neelapu, Locke et al. 2017, Schuster, Bishop et al. 2019). Overall rates of CRS reported in the trials varied and were 92% (axi-cel), 58% (tisa-cel), and 42% (liso-cel). In addition, rates of Grade 3/4 CRS were ~10%, 22%, and 2% (axi-cel, tisa-cel, and liso-cel, respectively). It should be noted the grading scales used in the axi-cel (ZUMA-1) and liso-cel (TRANSCEND) trials (Lee criteria) were not directly comparable with the grading system used in the tisa-cel trial (JULIET; grading scale and management guide developed by investigators at the University of Pennsylvania (Porter, Hwang et al. 2015). Retroactive regrading of liso-cel patients to the Lee et al criteria resulted in fewer patients designated as Grade 3 (revision of 14% to 9%) and more patients designated Grade 1 or 2 (revision of 36% to 40%); 1 patient was declared as not having CRS at all (Schuster, Maziarz et al. 2020).

Immune effector cell-associated neurotoxicity syndrome is a clinical neuropsychiatric syndrome that can occur after immunotherapy, particularly associated with T-cell engaging therapies and can often occur concurrently with or immediately following CRS. Cytokine release syndrome and ICANS are potentially life-threatening toxicities that are often caused by CAR T-cell therapy and frequently occur within the first week after CAR T-cell therapy but may have later onset. For the CD19-directed CAR T-cell therapies currently commercially available, the reported median time to onset of CRS has a range between 2 and 5 days, with a maximum reported range of onset of 1 to 51 days. For ICANS, the reported median time to onset has a range between 4 and 8 days; the maximum reported range of onset is 1 to 359 days (Breyanzi 2021, Kymriah 2022, Yescarta 2022).

Immune effector cell-associated neurotoxicity syndrome is characterized by neurological events (NE). The NEs may be relatively mild confusion and aphasia, more serious encephalopathy or status epilepticus, and occasionally life-threatening cerebral edema. In the pivotal trials, the onset of ICANS was typically after the peak of CRS (median of 5, 6, and 9 days after administration of axi-cel, tisa-cel, liso-cel therapy, respectively). As with CRS, virtually all NEs resolved (median of 11 to 17 days across studies) (Locke, Ghobadi et al. 2019, Schuster, Bishop et al. 2019, Abramson, Palomba et al. 2020). Neurological events were documented in 67%, 21%, and 30% of patients who received axi-cel, tisa-cel, and liso-cel therapy, respectively. In these pivotal trials, severity of NEs was determined using Common Terminology Criteria for Adverse Events (CTCAE) v4.03; most NEs were Grade 1/2. Grade 3/4 NEs occurred in 32%, 11%, and 10% of patients who received axi-cel, tisa-cel, and liso-cel therapy, respectively (Schuster, Bishop et al. 2018, Locke, Ghobadi et al. 2019, Abramson, Palomba et al. 2020). Real-world studies of axi-cel revealed 82% to 91% of patients developed CRS, which was Grade ≥ 3 in 7% to 9% of patients (Pasquini, Locke et al. 2019, Nastoupil, Jain et al. 2020, Riedell, Walling et al. 2020), generally similar to the rates observed in the ZUMA-1 trial. In addition, rates of Grade ≥ 3 NEs (19%-33% in the axi-cel studies) were similar to those in ZUMA-1. In patients who received commercially available tisa-cel, CRS had an incidence of 41% and 45% in the studies and NE an incidence of 14% and 18% (Pasquini, Hu et al. 2020, Riedell, Walling et al. 2020) rates, which were lower than in JULIET (CRS, 58%; NE, 21%). Though the clinical features of ICANS are easily recognized, its pathophysiology is not well understood. Recent animal models have implicated endothelial cell activation and disruption of the BBB resulting in direct neuronal cell injury, in

addition to a role for various pro-inflammatory cytokines. However, this model of CAR T-cell-mediated neurotoxicity is limited by an absence of human cytokines and hematopoietic cells, together with the occurrence of concomitant xenograft versus host disease ([Bondanza, Valtolina et al. 2006](#), [Mastaglio, Genovese et al. 2017](#), [Norelli, Camisa et al. 2018](#), [Siegler and Wang 2018](#)).

Patients with ICANS have increased levels of protein, CD4+ T-cells, CD8+ T-cells, and CAR T-cells in the cerebrospinal fluid (CSF), indicating a loss of integrity of the BBB. Clinical trials have shown an association between the number of CAR T-cells and level of cytokines in the CSF and the severity of ICANS ([Lee, Kochenderfer et al. 2015](#), [Locke, Neelapu et al. 2017](#), [Santomasso, Park et al. 2018](#)). Biochemical features of severe CRS and ICANS that are observed in patients, such as hypofibrinogenemia and increased fibrin degradation products, are shared with features of disseminated intravascular coagulation endothelial cell disruption, typically observed in sepsis and critical illness associated with increased vascular permeability ([Dyson and Singer 2009](#)). Patients with severe ICANS do show evidence of vascular leak. In a single-center study of 133 patients treated with CD19 CAR T-cells ([Gust, Hay et al. 2017](#)) changes in the angiopoietin (ANG)–TIE2 axis, which regulates endothelial cell activation ([Page and Liles 2013](#)), have been described. ANG1 is constitutively produced by platelets and perivascular cells and stabilizes the endothelium when bound to its endothelial receptor, TIE2. After endothelial cell activation by inflammatory cytokines, ANG2 is released from endothelial Weibel–Palade bodies and displaces ANG1, further increasing endothelial cell activation and microvascular permeability. Consistent with this mechanism, statistically significant increased ratios of serum ANG2 to ANG1 were observed in patients with severe ICANS compared with those with less severe neurotoxicity, together with higher concentrations of von Willebrand factor (vWF) and CXC-chemokine ligand 8 (CXCL8), all of which are produced by platelets and perivascular cells ([Schwameis, Schorgenhofer et al. 2015](#)). In addition, patients who have increased serum ANG2 to ANG1 ratios before lymphodepletion and CAR T-cell administration had a higher risk of developing ICANS. Further data point towards the sequestration of high molecular weight vWF multimers in patients with severe ICANS, resulting in coagulopathy. However, the factors that control baseline and perturbed levels of ANG2 and ANG1 remain elusive.

As CAR T-cell therapy has emerged as a promising treatment for hematologic malignancies and the unique associated toxicities of CRS and ICANS have been documented, it became increasingly difficult to assess and grade the toxicities across clinical trials and medical institutions. In an effort to harmonize definitions and grading systems, the American Society for Transplantation and Cellular Therapy (ASTCT) met in 2018 and developed uniform consensus definitions and grading systems for CRS and ICANS ([Lee, Santomasso et al. 2019](#)) (refer to Table 2 and Table 3).

1.2 Treatment of CRS and ICANS Secondary to CAR T-cell Therapy

1.2.1 Standard of Care

At present, SOC treatment for CRS is tocilizumab (a recombinant humanized monoclonal antibody directed against the IL-6 receptor [IL-6R]) or corticosteroids. Lower-grade CRS is primarily managed with tocilizumab while refractory, prolonged, or higher-grade CRS is managed with corticosteroids. Unlike corticosteroids, tocilizumab does not appear to suppress

T-cell function and/or induce T-cell apoptosis ([Paliogianni, Ahuja et al. 1993](#), [Lanza, Scudeletti et al. 1996](#)).

Across the axi-cel and tisa-cel registration trials (ZUMA-1 and JULIET, respectively), tocilizumab treatment resulted in an ORR of 70%. Within hours, decreased oxygen and vasopressor requirements were observed ([Freyer and Porter 2020](#)); median time to CRS resolution was 4 days ([Le, Li et al. 2018](#)). Tocilizumab has been approved by the FDA for the management of severe or life-threatening CRS ([Actemra 2019](#)).

There is some controversy surrounding the use of IL-6R antibody in the treatment of CRS. It remains unclear whether IL-6 blockade impairs the in vivo proliferation of CAR T-cells because IL-6 has been clearly proven to promote the proliferation of T cells ([Takeda, Kaisho et al. 1998](#)); serum IL-6 levels may rapidly rise after tocilizumab administration, presumably by preventing IL-6R mediated uptake of IL-6 into peripheral tissue ([Nishimoto, Terao et al. 2008](#)).

Standard of care treatment of ICANS is typically supportive care and corticosteroids. Tocilizumab is not an approved treatment for ICANS ([Santomasso, Nastoupil et al. 2021](#)), as the targeting of IL-6R using tocilizumab has not been shown effective at preventing or treating neurotoxicity and may worsen it ([Hunter and Jacobson 2019](#)). In addition, tocilizumab appears to have limited BBB penetration ([Freyer and Porter 2020](#)). The management of ICANS may take precedence over the management of low-grade CRS when both occur simultaneously. Although cases of low-grade neurotoxicity may resolve without intervention, corticosteroids are generally used for cases of Grade ≥ 2 ICANS ([Freyer and Porter 2020](#)).

1.2.2 Second-line Treatment

When CRS is refractory to tocilizumab, siltuximab (an IL-6 antagonist; a human-mouse chimeric monoclonal antibody that binds human IL-6) or anakinra (an IL-1 receptor antagonist) may be efficacious alternatives ([Danish and Santomasso 2021](#)). Siltuximab is under investigation as a treatment for refractory CRS, as serum IL-6 levels have been shown to increase after administration of tocilizumab, presumably by preventing IL-6R mediated uptake of IL-6 into peripheral tissue ([Neelapu, Tummala et al. 2018](#)). Anakinra has been shown to mitigate CRS in some CAR T-cell therapy recipients with high-grade CRS ([Strati, Ahmed et al. 2020](#)).

Alternate options for persistent or worsening ICANS include siltuximab and anakinra. Interleukin-1 has been shown to peak in the serum of patients with R/R LBCL within the first 7 days after axi-cel infusion ([Yanez, Sanchez-Escamilla et al. 2019](#)). Anakinra has been used as a second-line treatment and may represent a steroid-sparing strategy for the treatment of CAR T-cell therapy–associated toxicities, mainly ICANS. This strategy reduces the risk for steroid-related AEs including infection, hyperglycemia, arrhythmia, and adverse neuropsychiatric effects that may mimic ICANS, among others ([Yasir, Goyal et al. 2022](#)). Although overall outcomes were dismal when anakinra was used as a last resource, clinical trials investigating its early and/or prophylactic use are ongoing.

To date, siltuximab and anakinra have only been administered for tocilizumab-refractory CRS/ICANS in small studies. Hence, there is currently a lack of robust data on the use of either drugs as first line treatment/ SOC. Larger, prospective trials investigating upfront use of anakinra

or siltuximab alone or in combination with steroids for prophylaxis and/or treatment of CRS/ICANS are currently ongoing. For this reason, none may be currently considered as SOC.

1.2.3 Other Therapies

Other potential therapies for CRS and ICANS include the early introduction of treatment with corticosteroids, as well as treatment with ruxolitinib, dasatinib, or lenzilumab. Similar to the second-line therapies currently being studied, studies conducted investigating these other therapies have been small to date.

Ruxolitinib, an oral Janus kinase inhibitor ([Durongpisitkul, Jakrapanichakul et al. 2005](#)), is a promising drug in the treatment of CRS ([Pan, Deng et al. 2021](#)). Serum cytokines have been observed to decrease significantly after ruxolitinib intervention. Ruxolitinib has not been noted to completely abrogate CAR T-cell expansion in patients. Data suggest that ruxolitinib may be efficacious and appears to be well tolerated in steroid-refractory and even life-threatening CRS ([Pan, Deng et al. 2021](#)).

Dasatinib, a tyrosine kinase inhibitor, is known to halt cytolytic activity, cytokine production, and proliferation of CAR T-cells in vitro and in vivo. On discontinuation of dasatinib, the inhibitory effect is rapidly and completely reversed, and CAR T-cells resume their antitumor function. It has been observed that dasatinib is capable of effectively blocking cytokine secretion from a polyclonal, nonsynchronized CAR T-cell population in vivo, and of preventing morbidity and mortality from CRS ([Mestermann, Giavridis et al. 2019](#)).

The granulocyte-macrophage colony-stimulating factor (GM-CSF)-neutralizing antibody lenzilumab does not inhibit CAR T-cell function and may enhance their antitumor activity in patient-derived xenografts. In an acute lymphoblastic leukemia patient-derived xenograft model of CRS and neuroinflammation, GM-CSF neutralization resulted in a reduction of myeloid and T-cell infiltration in the central nervous system and a significant reduction in neuroinflammation and prevention of CRS ([Sternier, Sakemura et al. 2019](#)).

Currently no approved therapies exist beyond tocilizumab; treatment alternatives are needed. The IL-6 blockade created by tocilizumab may impair the in vivo proliferation of CAR T-cells and the rapid rise of IL-6 after administration may aggravate CRS. In addition, tocilizumab is only cautiously recommended for the management of CRS. In rare circumstances CRS may be refractory to tocilizumab and additional immunosuppression is required ([Si and Teachey 2020](#)). Tocilizumab has been found to be an ineffective treatment for ICANS since it is unable to cross the BBB ([Nishimoto, Terao et al. 2008](#)). The targeting of IL-6R using tocilizumab has not been shown to be effective for either prevention or treatment of neurotoxicity ([Hunter and Jacobson 2019](#)), and may worsen the effects of ICANS. Estimates have shown that 50-75% of patients treated with tocilizumab require steroids ([Si and Teachey 2020](#)). For these reasons there is a need for alternative therapies that will reduce the incidence of hospital admission and length of hospital stay and reduce mortality from CAR T-cell therapy related toxicities.

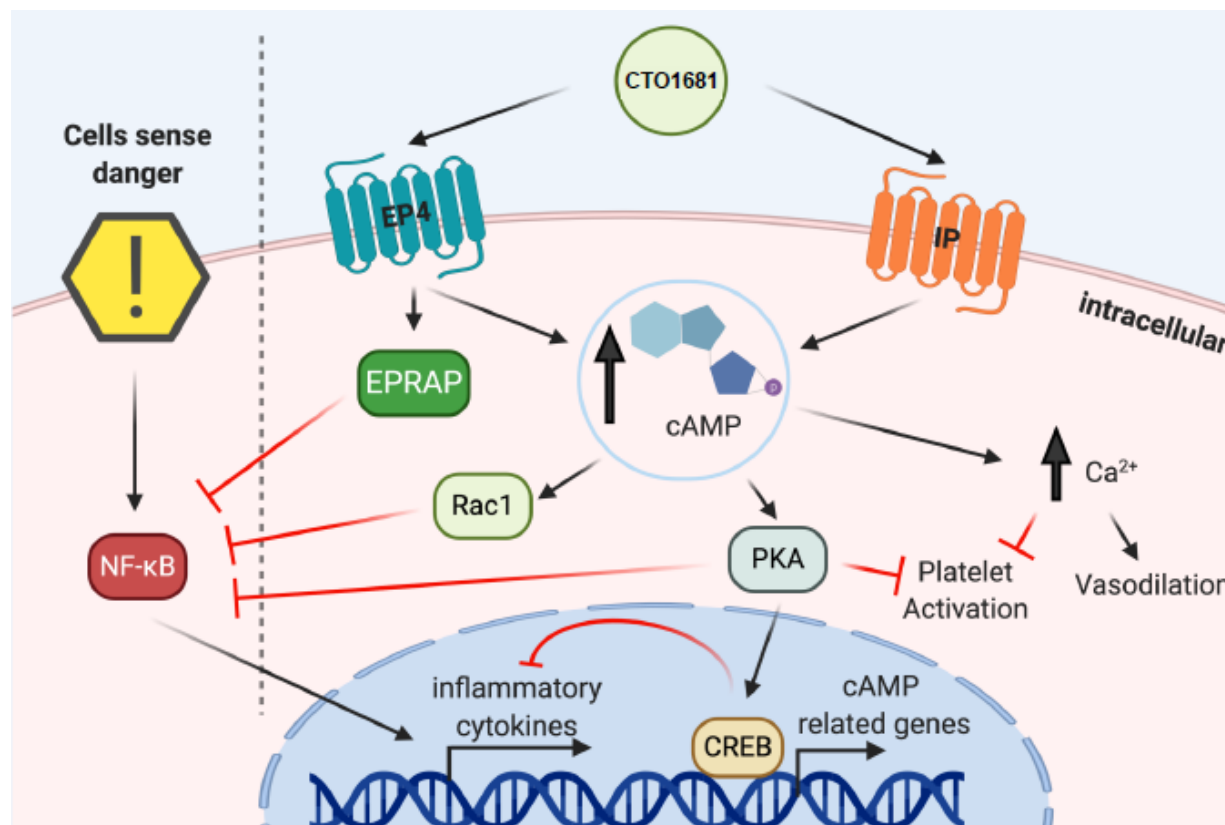
1.3 CTO1681

1.3.1 Overview

The investigational product CTO1681-IR-OSD-10 µg Immediate Release Tablets is a broad-based anti-inflammatory drug. Early administration of CTO1681, prior to the initial release of proinflammatory cytokines (eg, IFN-γ and, TNF-α, IL-1, IL-2, and IL-6), may mitigate risk of CRS associated with CAR T-cell therapy. Through excess cytokine suppression, CTO1681 could potentially prevent the onset or escalation of CAR T-cell induced immune-mediated toxicities, thereby reducing associated morbidities and hospitalization.

CTO1681 downregulates the production of over 20 proinflammatory cytokines, many of which are directly involved in CRS and ICANS, (ie, IL-1α, -2, -6, -10, GM-CSF, monocyte chemoattractant protein-1 [MCP-1], macrophage inflammatory protein [MIP-1α], TNF-α and interferon gamma [IFN γ] ([Teachey, Lacey et al. 2016](#), [Gust, Hay et al. 2017](#), [CTO1681 \[Investigators Brochure\]](#)). The mechanism of action for CTO1681 cytokine regulation is through its function as a prostaglandin E2 (PGE2) and prostaglandin I2 (PGI2) agonist. CTO1681 binds the PGE2 receptor 4 (EP4) and prostacyclin (IP) prostanoid receptors. EP4 regulation is responsible for the majority of prostanoid anti-inflammatory processes and hence the site of the majority of CTO1681 regulation of cytokine production. Cytokine suppression through the EP4 receptor is mediated through control of nuclear factor kappa light chain enhancer of activated B cells (NFκB) stimulation of cytokine gene transcription. Activation of the EP4 signaling pathway at the EP4 receptor stabilizes the interaction of the EP4 receptor-associated protein. EP4 receptor-associated protein inhibits phosphorylation and degradation of NFκB1 p105 through direct binding of p105, which then in turn prevents the activation of NFκB and its translocation to the nucleus, thus inhibiting the transcription of NFκB-dependent proinflammatory cytokines (Figure 1).

Figure 1: CTO1681 Activation of the Anti-inflammatory EP4 (and IP) Signaling Pathways



Abbreviations: cAMP = cyclic adenosine monophosphate; CREB = cAMP response element-binding protein; EP4 = prostaglandin E2 receptor 4; EPRAP = EP4 receptor associated protein; IP = prostacyclin; NFκB = Nuclear factor kappa-light-chain-enhancer of activated B cells; PKA = protein kinase A; Rac1 = ras-related C3 botulinum toxin substrate 1

Note: GP1681 = CTO1681.

Binding of prostaglandin agonists such as CTO1681 at the EP4 receptor allows interaction with EPRAP, which in turn stabilizes the NFκB subunit by preventing its phosphorylation and blocks translocation of NFκB to the nucleus, thereby down-regulating transcription of proinflammatory cytokines. Secondary means of cytokine reduction exist through EP4 and IP induction of cAMP. Furthermore, cAMP induction via the IP receptor also reduces platelet activation and increases vasodilation.

CTO1681 is the active component of the racemic mixture beraprost sodium. Beraprost is a well-studied agonist of the IP receptor and, like other prostacyclins, is a vasodilator and inhibitor of platelet aggregation ([Okada, Marchevsky et al. 1998](#), [Durongpisitkul, Laoprasitiporn et al. 2005](#)).

Beraprost is currently marketed in a number of Asian countries as a treatment for chronic arterial occlusive disease, and pulmonary arterial hypertension and has a long history of safe use in humans ([Dewachter and Naeije 2014](#), [Dorner 2015](#), [Peng, Li et al. 2016](#)). It has been well-tolerated in over 70 clinical trials conducted worldwide, in both Asian and non-Asian populations, when used acutely or chronically. It is notable that clinical trials of beraprost therapy have demonstrated reductions in circulating cytokine levels.

Peng et al found that beraprost has a protective effect on diabetic nephropathy by inhibiting inflammation and the p38 MAPK signaling pathway in high-fat diet/streptozotocin-induced diabetic rats (Peng, Li et al. 2016). It was observed that beraprost efficiently reduced the production of TNF- α , matrix metalloproteinase 9, and Cyclooxygenase 2 in kidney tissue. In addition, treatment with beraprost efficiently elevated the levels of total superoxide dismutase, glutathione, and nitric oxide, showing significant antioxidative activity.

In a study of mice undergoing liver transplant, Deng et al showed preconditioning with beraprost ameliorated the effects of hepatic ischemia-reperfusion (IR) injury (Deng, Feng et al. 2018). Activated p38 and JNK cascades have been observed in mice undergoing this procedure and are closely associated with aggravated inflammation and apoptotic or autophagic cell death during hepatic IR injury (Li, Wang et al. 2015, Xu, Niu et al. 2017). Deng et al found that beraprost suppressed inflammation, apoptosis, and autophagy, partially inhibiting activation of the p38 and JNK cascades and suppressed the production of inflammatory mediators such as TNF- α and IL1b (Deng, Feng et al. 2018).

Since 2008, beraprost has been studied in Sponsor- and National Institutes of Health-funded studies and has demonstrated increased survival, maintenance of lung function, and reduction in the level of pro-inflammatory cytokines in mice infected with different strains of the Influenza A virus (Daryl Faulds et al, 2014). Using a mouse lethal challenge influenza model, it was determined that CTO1681 is the most active component of beraprost and significantly decreases lung cytokine levels. In addition, CTO1681 has had a positive, statistically significant impact on survival in lethal challenge models. CytoAgents has demonstrated that CTO1681 downregulates the production of multiple proinflammatory cytokines in both ex vivo cytokine release assays in human PBMCs and in vivo murine H1N1/H5N1 influenza infection models (CTO1681 [Investigators Brochure]).

A Phase 1 study (CTO1681 [Investigators Brochure]) in healthy adults was performed to define the electrocardiographic effects, specifically QT prolongation, of CTO1681 in healthy adults. Single, oral therapeutic and supratherapeutic doses (15 and 30 μ g, respectively) of beraprost-314-d (CTO1681) were administered. Peak mean changes in QT interval corrected using Fredericia's formula (QTcF) for heart rate of 7.1 msec (90% confidence interval [CI]: 5.1 to 9.0 msec) and 11.3 msec (90% CI: 9.4 to 13.3 msec) were observed at doses of 15 and 30 μ g, respectively. Neither change in QTcF was of clinical concern. Overall, both doses of CTO1681 were well tolerated in healthy adults; the most common treatment-emergent adverse events (TEAEs) were headache (45.0%), flushing (28.3%), nausea (15.0%), and pain in extremity (13.3%), which were consistent with previous studies of CTO1681.

In addition, in a Phase 1 multiple ascending dose study in healthy adults (Study CTA-1901), the overall safety profile of CTO1681 was found to be consistent with that reported for beraprost sodium. Administration of CTO1681 up to the planned maximum dose of 60 μ g/day was well tolerated. Maximum tolerated dose was not reached in this study. Treatment-emergent AEs were generally mild and transient; no TEAEs were assessed as severe. There were no deaths or other serious adverse events (SAEs) and no TEAEs leading to study discontinuation. The most commonly reported treatment-related TEAEs were nervous system disorders (dizziness, headache), general disorders and administration site conditions (fatigue), and gastrointestinal disorders (abdominal distension, change in bowel habit, constipation, diarrhea). Vascular

disorders, including postural dizziness, hypotension, and orthostatic hypotension, occurred in 16.7% of the study participants, and were all classified as mild in severity. Headache, flushing, dizziness, fatigue, and diarrhea are known to occur in humans taking beraprost sodium (Vizza, Sciomer et al. 2001, Nagaya, Shimizu et al. 2002, Barst, McGoon et al. 2003, Ono, Nagaya et al. 2003a, Ono, Nagaya et al. 2003b).

1.4 Study Rationale

CTO1681 has a broad impact on multiple (20+) cytokines to mitigate hypercytokinemia (CTO1681 [Investigators Brochure]). Early administration of CTO1681, before the initial release of proinflammatory cytokines (eg, IFN- γ , TNF- α , IL-1, IL-2, and IL-6), may mitigate risk of CRS associated with CAR T-cell therapy. Through excess cytokine suppression, CTO1681 could potentially prevent the onset or escalation of CAR T-cell-induced immune-mediated toxicities, thereby reducing associated morbidities and hospitalization. The purpose of this study is to evaluate the safety and tolerability of CTO1681 and determine the preliminary effectiveness of CTO1681, measured mainly by reduction of CAR T-cell-induced toxicities, including CRS and ICANS and other efficacy measures.

This Phase 1b/2a study will enroll adult patients with B-NHL (DLBCL subtype) who are scheduled to receive axicabtagene ciloleucel (ie, YESCARTA[®]) CD19-directed CAR T-cell therapy to manage potential variability between CAR T-cell therapy in response and adverse reaction rates. However, CRS and ICANS are not unique to axicabtagene ciloleucel, and CTO1681 studies in the future may involve additional CAR T-cell treatments.

The Phase 1b portion will include an initial open-label, dose-finding, safety and toxicity assessment on a small sample (n = 9 up to 54) of patients. The Phase 2a portion (n = 100) will be a randomized, proof-of-concept assessment to investigate the potential for benefit of CTO1681, initiated prior to CAR T-cell therapy, versus placebo, in mitigating or reducing CAR T-cell-induced toxicity.

There is a medical need for alternative treatment options to tocilizumab and steroids in B-NHL with CRS and/or ICANS caused by CAR T-cell therapy. CTO1681 has demonstrated a tolerable safety profile in healthy volunteers, consistent with that reported for beraprost sodium. Results from a Phase 1a study (Study CTA-1901) in healthy volunteers support further investigation of CTO1681 to mitigate CAR T-cell therapy toxicities in the DLBCL patient population.

1.5 Rationale for Dose Selection and Schedule

The proposed doses for the Phase 1b multiple ascending dose portion of the study will be 10, 20, and 30 μ g three times daily (TID) (30, 60, and 90 μ g daily). The dose for the proposed Phase 2a portion of the study will be selected based on the findings from the Phase 1b portion of the study. There is substantial evidence from the non-clinical and clinical use history of immediate release formulations of beraprost sodium, and the same active isomer as CTO1681, that these doses will be well tolerated and have the potential for efficacy in the prevention and treatment of CRS.

Beraprost sodium contains 4 stereoisomers, one of which (314d) contains nearly all of the biological activity and is chemically identical to CTO1681. There is extensive worldwide clinical experience with beraprost sodium, and safe and effective doses for its non-CRS indications have been well described. In Japan, the approved doses of immediate-release formulations of

beraprost sodium range from 60 to 180 µg/day (corresponding to 15 to 45 µg/day CTO1681), divided into 3 or 4 doses administered with food, and are often titrated upwards to tolerance. The safety of long-term dosing of higher levels of beraprost sodium, up to 480 µg/day (120 µg/day CTO1681) for up to 36 months, has been reported in studies of patients with pulmonary arterial hypertension ([Vizza, Sciomer et al. 2001](#), [Galie, Humbert et al. 2002](#), [Nagaya, Shimizu et al. 2002](#), [Barst, McGoon et al. 2003](#), [Ono, Nagaya et al. 2003a](#), [Ono, Nagaya et al. 2003b](#), [Durongpisitkul, Laoprasitiporn et al. 2005](#), [Park, Moon et al. 2006](#), [Vizza, Badagliacca et al. 2006](#)).

Findings from the recently completed Phase 1 multiple ascending, repeat-dosing study of CTO1681 ([Study CTA-1901](#)), which evaluated doses of 15 to 60 µg/day in healthy volunteers over a period of 7 days, have confirmed this generally mild safety profile. No SAEs, treatment-emergent AEs (TEAEs) of special interest (defined as clinically significant changes in coagulation parameters or platelets, or persistent/recurrent symptomatic orthostatic hypotension), or TEAEs leading to study discontinuation or dose reduction occurred. Treatment-emergent AEs were exclusively mild (62 AEs) or moderate (4 AEs) in severity, and completely consistent with the safety profile reported prior for beraprost sodium and 314d.

Doses of beraprost sodium that are within the dose equivalencies of CTO1681 for the multiple ascending dose study ([Study CTA-1901](#)) have been shown to cause significant reductions in TNF-α in patients with diabetes ([Fujiwara, Nagasaka et al. 2004](#), [Xu, Pan et al. 2020](#)). It is hypothesized that similar reduction of TNF-α and other proinflammatory cytokines may mitigate the CRS frequently observed with CAR T-cell therapy. Importantly, no clinically significant decreases in cytokine levels (panel of 27 cytokines) were seen in healthy volunteers treated with CTO1681, suggesting that CTO1681 does not completely abolish cytokine responses, but rather mitigates excessive responses, further supporting the safety of CTO1681 and justifying continued clinical evaluation.

2. STUDY OBJECTIVES

2.1 Study Objectives

2.1.1 Phase 1b

- Determine the preliminary safety profile of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.
- Determine the recommended Phase 2 dose (RP2D) and, as appropriate, maximum tolerated dose (MTD) of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.

2.1.2 Phase 2a

- Determine preliminary CTO1681 effectiveness in preventing CRS or reducing CRS severity and duration in comparison to placebo.
- Determine preliminary CTO1681 effectiveness in preventing ICANS or reducing ICANS severity and duration in comparison to placebo.
- Determine expanded safety profile of CTO1681 in patients with DLBCL receiving CAR T-cell therapy.
- Investigate the potential impact of CTO1681 on antitumor activity of CAR T-cell therapy in comparison to historical data and placebo.

3. STUDY PLAN

3.1 Overall Design

This is a Phase 1b/2a study designed to explore the safety and effectiveness of CTO1681 in preventing or reducing CAR T-cell-induced toxicities in patients with DLBCL. The study will be separated into 2 distinct portions designated as Phase 1b and Phase 2a.

After providing written informed consent for Phase 1b or Phase 2a of the study, the patient is considered enrolled. Study participants will be evaluated for inclusion during the Screening period. For all participants, the 19-day screening period will end 2 days before Baseline (Day -3).

Phase 1b

The Phase 1b portion of the study is a multicenter, open-label, dose-escalating, safety and pharmacokinetic (PK) study of multiple ascending doses of CTO1681 in patients with DLBCL who receive commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy. This portion of the study will be conducted using a rolling 6 design to inform dose escalation among cohorts of patients with DLBCL. It will include the following sequential steps for every patient in each cohort: Screening, treatment enrollment, pretreatment with lymphodepleting (LD) chemotherapy, initiation of study drug (CTO1681), CAR T-cell infusion, continuing treatment with CTO1681 (for a total of 15 days), and safety follow-up. Study eligibility will be determined after leukapheresis and prior to LD chemotherapy. Timing of treatment with CTO1681 will occur according to the following schedule in relation to CAR T-cell infusion (CAR T-cell infusion on “Day 0”):

- Days -12 to -3: LD chemotherapy (if warranted per YESCARTA package insert [PI]).
- Day -1: Initiate study drug (CTO1681) approximately 24 hours before CAR T-cell infusion. Study drug will be continued orally (PO), TID for 15 days.

The total daily dose of CTO1681 will be 30, 60, or 90 µg (administered PO in TID increments of 10, 20 or 30 µg/dose, respectively), depending on the cohort into which each patient is enrolled. Study drug must be taken within 30 minutes after a meal or snack.

- Day 0: CAR T-cell infusion (per YESCARTA PI).

Patients will be monitored to document tumor response for up to 6 months using the Lugano Criteria for Malignant Lymphoma ([Section 6.10.9](#)). In addition, patients will be monitored throughout the study for efficacy and safety ([Section 6.7](#) and [Section 6.10](#), respectively).

A rolling 6 design will be implemented to advise on dose escalation. The dose-limiting toxicity (DLT) observation period is defined as 43 days following the first dose of study drug (Day -1), which is 42 days following initiation of CAR T-cell therapy (Day 0), ie, Day +41. Refer to [Section 5.10.1](#) and [Section 5.10.1.1](#) for information regarding DLTs and the management of DLTs, respectively.

Up to 3 dose cohorts will be considered with a maximum of 18 patients enrolled during the rolling 6 dose escalation procedure. Each dose level (30, 60 and 90 µg/day respectively) will be

enrolled with a minimum of 3 and up to 6 patients during the dose escalation part of the study (Table 1). A minimum of 3 patients must complete the DLT observation period (through Day +41) prior to data review for dose escalation. A maximum of 0 DLT observed in 3-5 patients or 1 DLT observed in 6 patients may be recorded to allow escalation to the next higher dose level. Recommendations for the next dose level will be made by the Safety Review Committee (SRC) based on safety, laboratory, and any PK and pharmacodynamic (PD) data that are available at the time of review. If adequate safety data is not available from 1 or more patients in a given cohort, additional patients will be enrolled in that cohort, and only patients for whom there is adequate safety data may be included in the safety evaluation. If 1 DLT has been observed, any new patients will be entered at the same dose level until 6 patients are enrolled and have completed the DLT observation period, or at least 2 DLTs are observed, and therefore MTD has been exceeded.

The MTD is defined as the maximum CTO1681 dose achieved where patients are treated safely. The dose may be escalated to the next level of dose if no more than 0 DLT observed in 3-5 patients or 1 DLT observed in 6 patients. If a dose level is too toxic (2 or more DLTs observed in 3-6 patients), then the MTD stopping rules for that cohort/dose level may be applied.

Any or all of the cohorts deemed safe by the SRC following the dose escalation procedures may be backfilled up to a maximum of 15 additional patients/ cohort at the discretion of the Medical Monitor and Sponsor. The backfill sample will allow further safety, antitumor response, PK, and PD information to better select a RP2D at, or below the MTD for Phase 2a. The maximum sample size for Phase 1b is 54 patients.

Table 1: Dose Escalation and Stopping Rules

Daily Dose Level	Number of Patients in Rolling 6 Dose Escalation	Dose Escalation Rules	MTD Stopping Rules	Maximum Backfill Patients ^a
30 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15
60 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15
90 µg	3-6	0/3 DLT 0/4 DLT 0/5 DLT ≤ 1/6 DLT	≥ 2/3 DLT ≥ 2/4 DLT ≥ 2/5 DLT ≥ 2/6 DLT	12-15

Abbreviations: DLT = dose limiting toxicity; MTD = maximum tolerated dose; SRC = safety review committee

- ^a Backfill patients will be enrolled at the discretion of the Medical Monitor and Sponsor if the dose level is determined safe according to the MTD stopping rules and by recommendation of SRC. A maximum of 18 patients may be treated at each dose level.

A SRC ([Section 3.1.1](#)) will monitor the progress and safety of patients throughout the Phase 1b portion of the study and will determine if dose escalation may occur for each dose level cohort according to the rolling 6 design. In addition, the SRC will advise on the advancement of Phase 1b to the Phase 2a portion of the study and the selection of an RP2D on the basis of safety, including available antitumor response data. The SRC will also advise on the dose schedule and recommended CAR T-cell products for inclusion in Phase 2a.

The study schema is in Figure 2.

Phase 2a

The Phase 2a portion of the study will be conducted as a randomized, double-blind, placebo-controlled study and will enroll approximately 100 patients. Patients will be treated during Phase 2a at the RP2D of CTO1681, established during Phase 1b, or with placebo. Patients will be randomized in a 1:1 ratio between the CTO1681 and placebo arms. Patients will receive commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy. Each patient will follow the same study steps as noted above for the Phase 1b portion (see also Figure 2). Patients enrolled in this Phase 2a portion of the study will be monitored for tumor response for up to 12 months (using the Lugano Criteria for Malignant Lymphoma [[Section 6.10.9](#)]) following CAR T-cell infusion. Sparse PK samples will be collected in all patients in Phase 2a to perform population PK and exploratory exposure-response analyses.

An Independent Data Monitoring Committee (IDMC; [Section 3.1.2](#)) composed minimally of 2 CAR T-cell/hematology oncology experts and an independent biostatistician will be established to monitor safety data during the Phase 2a portion of the study. The IDMC will perform planned safety data reviews after approximately 20, 40, 60, and 80 patients have been followed through Day +41, or have discontinued early from the study. Available data from Phase 1b will be included in each review. The study will be paused or stopped if any of the prespecified pausing or stopping rules are met.

Phase 1b and Phase 2a

For both the Phase 1b and Phase 2a portions of the study, safety, efficacy, PK and PD assessments will be conducted at protocol-specified intervals according to the Schedule of Events for Phase 1b, and Phase 2a ([Appendix 10.1](#) and [Appendix 10.2](#)).

Treatment with CTO1681 will occur according to the schedule detailed in the study schema (Figure 2), all other procedures, assessments, and laboratory tests and their timing for both study phases are documented in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)).

The patient is planned to be treated as inpatient from Baseline/Day -1 to Day +6, at minimum, to allow for proper monitoring. During the period of Days +7 to +13, the patient may be discharged in the absence of CRS/ICANS at the discretion of the Investigator with the approval of the Medical Monitor and Sponsor. If the patient is discharged, then they must return to the clinic for a minimum of two visits during that week (Day +7 to Day +13), which must include the visit on

Day +13) for AE collection and status assessment per the Schedule of Events during the treatment period. In addition, study patients will be hospitalized for observation and management of CAR T-cell-related toxicities based on institutional guidelines (refer to [Section 5.10](#)).

The currently commercially available CD19-directed CAR T-cell therapy for B-NHL (which includes DLBCL), axicabtagene ciloleucel (YESCARTA), will be utilized in both the Phase 1b and Phase 2a portions of this study.

The criteria for pausing or stopping the study are defined in [Section 5.11](#).

3.1.1 Phase 1b – Safety Review Committee

The SRC will be composed of 1 or more study investigators, the study Medical Monitor, the Sponsor's medical representative(s) (or qualified delegate[s]), and ad hoc members as appropriate (eg, statistician, PK and regulatory affairs experts). The SRC will monitor the progress and safety of patients throughout the Phase 1b portion of the study. Regular systematic review of AEs will serve as the basis for pausing or prematurely stopping the study.

Dose-limiting toxicities ([Section 5.10.1](#)) will be the primary criteria for pausing or stopping the study in alignment with the rolling 6 design operational characteristics. The SRC will carefully review any observed AE that may be considered a DLT that first presents more than 5 half-lives after discontinuation of study drug and evaluate if the AE might reasonably be considered related to the study drug; recommendations will appropriately reflect this consideration.

The SRC will review the available safety, laboratory, electrocardiogram (ECG) and PK data through the completion of the DLT evaluation period (ie, Day +41, 43 days after the first dose of CTO1681). The SRC may recommend that the dose may be escalated or, additional patients may be enrolled (up to 6 per dose level), or the SRC may determine the MTD is reached, and no further dose escalation is appropriate. Cumulative safety data will be reviewed at each SRC meeting. After a dose level is determined safe according to the dose escalation procedures and SRC review of safety and PK data (if available), additional backfill patients may be enrolled (up to a maximum of 18 patients per cohort). A cohort will be paused or stopped if any of the pausing or stopping rules (per DLT criteria; [Section 5.11](#)) are met.

The SRC will advise on the advancement to the Phase 2a portion of the study and the selection of an RP2D based on available safety, antitumor response data, PK and PD results. An interim analysis summarizing the completed Phase 1b results will be performed.

The SRC may also advise on any necessary protocol amendments with respect to patient safety throughout the Phase 1b portion of the study.

Further details will be captured in a separate Safety Review Committee Charter.

3.1.2 Phase 2a – Independent Data Monitoring Committee

An IDMC minimally composed of 2 CAR T-cell/hematology oncology experts and an independent biostatistician will be established to monitor safety data during the Phase 2a portion of the study. The IDMC will perform planned safety data reviews after approximately 20, 40, 60, and 80 patients have been followed through Day +41, or have discontinued early from the study. The study will be paused or stopped if any of the prespecified pausing or stopping rules are met ([Section 5.11](#)). The IDMC will carefully review any observed AE that may be considered a DLT

that first presents more than 5 half-lives after discontinuation of study drug and evaluate if it might reasonably be considered related to the study drug; recommendations will appropriately reflect this consideration. An ad hoc meeting of the IDMC will be held in the event of a patient death. In addition, the IDMC will meet more frequently, as necessary, to review SAEs and provide recommendations on when to pause enrollment or terminate the study.

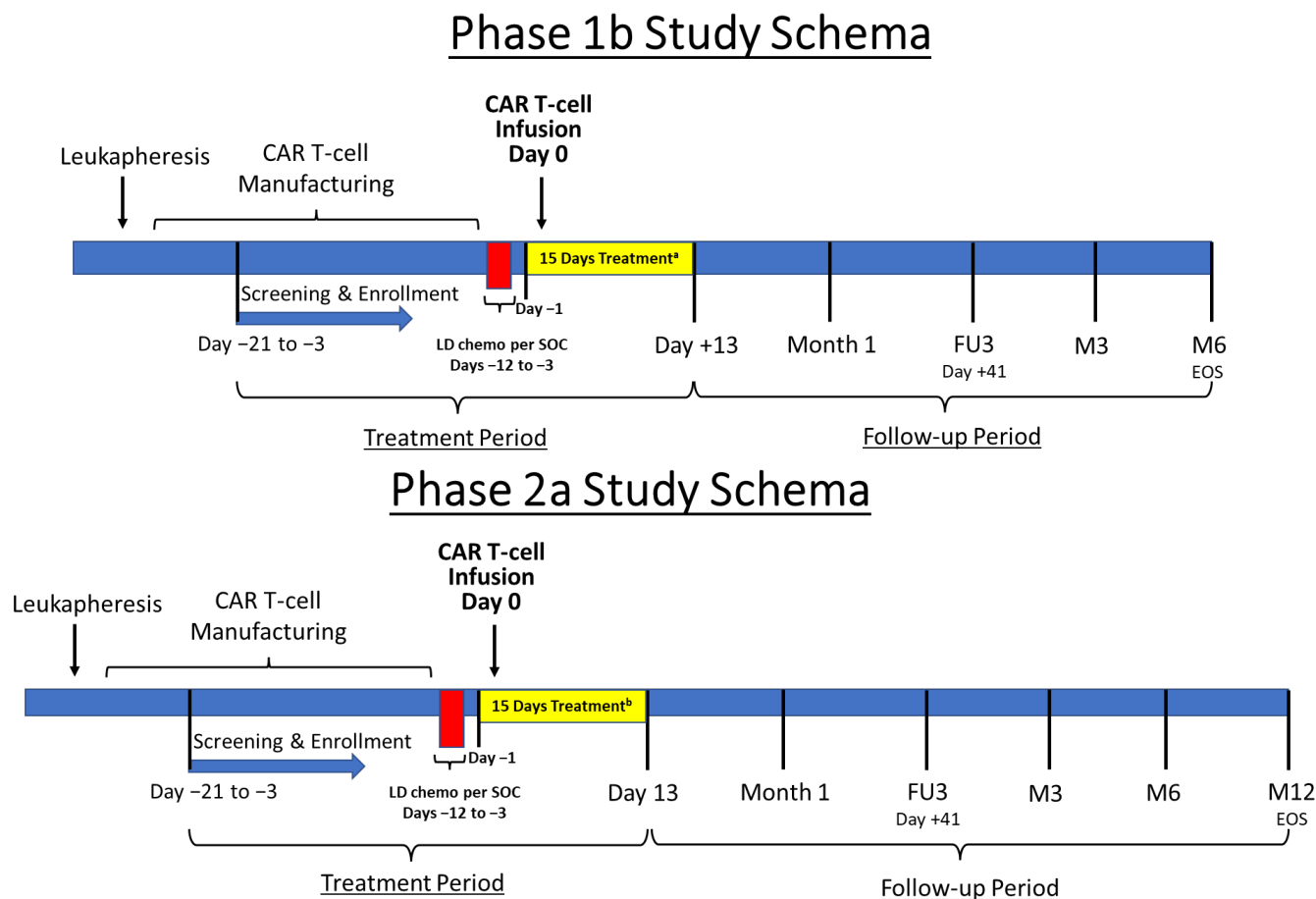
The IDMC will also review available data regarding rates of CR in accordance with a prespecified set of nonbinding antitumor response impact assessments, planned to occur during the 40, 60, and 80 patient safety data reviews (20, 30, and 40 per treatment arm). Patients treated in Phase 1b at the R2PD will be included in these assessments.

The antitumor response assessments will include all patients who have achieved CR, according to the Lugano Criteria ([Section 6.10.9](#)), have a best antitumor response of progressive disease, have prematurely discontinued from the study, or have been followed for at least 3 months since first dose of study drug. Patients who have not achieved CR, have not had progressive disease, and have not been followed for a minimum of 3 months after treatment initiation will not be included in the analysis. This distinction is made to ensure there is sufficient follow-up to allow possible response or disease progression following treatment initiation to avoid an artificial under-reporting of response rate.

At each antitumor response impact assessment, for patients randomized to CTO1681, the CR rate and an exact Clopper-Pearson 95% CI will be computed to compare the upper bound of the CI to a 41% benchmark minimum rate derived from the lower boundary of CR rate 95% CI from the YESCARTA label. The first antitumor response assessment is expected to occur after 20 patients have been enrolled and treated with CTO1681; however, a minimum of 12 eligible patients are required to conduct analysis. Study enrollment may be paused for antitumor response data review as warranted.

Formal charters will be established to specify the rules and scope of responsibilities of the SRC and IDMC.

Figure 2: Phase 1b/2a Study Schema (Key Events)



Abbreviations: CAR = chimeric antigen receptor; chemo = chemotherapy; EOS = end of study; FU = follow-up; LD = lymphodepleting; M = month; OL = open-label; RDBPC = randomized double-blind placebo-controlled; SOC = standard of care.

^a Treatment: Phase 1b = OL CTO1681

^b Phase 2a = 1:1 RDBPC, CTO1681 or placebo

3.2 Study Endpoints

3.2.1 Phase 1b

The primary endpoints of Phase 1b of the study are:

- AEs, including DLTs, and laboratory and ECG abnormalities as characterized by type, frequency, timing, severity (graded per CTCAE, v5.0), seriousness, and relationship to CTO1681.

The exploratory endpoints are:

- Incidence of CRS (any grade).
- Time to resolution of CRS (any grade).
- Incidence of ICANS (any grade).
- Time to resolution of ICANS (any grade).
- Biomarkers relating to the mechanism of action (MOA) of CTO1681 and immune response.
- Frequency and duration of hospitalizations and intensive care unit (ICU) admissions due to CRS or ICANS.
- Use of tocilizumab, steroids, and other anticytokine therapies for treatment of CRS or ICANS (amount and duration).
- PK concentration profile of CTO1681.
- Levels and persistence of CAR T-cells.
- Immune reconstitution (ie, B-cell aplasia).
- Antitumor activity including CR, ORR, duration of response (DOR), and progression-free survival (PFS) according to Investigator review, and OS.

3.2.2 Phase 2a

Endpoints for Phase 2a will be reported as active compared to placebo. In addition, antitumor activity will also be compared to benchmark antitumor response rates, as outlined in [Section 3.1.2](#).

The primary endpoint of Phase 2a of the study is:

- Incidence of Grade ≥ 2 CRS.

The secondary endpoints are:

- Additional assessments of CRS incidence and severity:
 - Incidence of CRS (any grade).
 - Incidence of Grade ≥ 3 CRS.
 - Time to CRS resolution.

- Additional assessments of ICANS incidence and severity:
 - Incidence of ICANS (any grade).
 - Incidence of Grade ≥ 2 ICANS.
 - Incidence of Grade ≥ 3 ICANS.
 - Time to ICANS resolution.
- AEs, laboratory abnormalities, and ECG abnormalities as characterized by type, frequency, timing, severity (graded per CTCAE v5.0), seriousness, and relationship to CTO1681.
- Frequency and duration of hospitalizations and ICU admissions due to CRS or ICANS.
- Use of tocilizumab, steroids, and other anticytokine therapies for treatment of CRS or ICANS.
- Antitumor activity including CR rate, ORR, DOR, and PFS according to Investigator review, and OS.

The exploratory endpoints are:

- Levels and persistence of CAR T-cells.
- PK concentration profile of CTO1681.
- Biomarkers relating to MOA of CTO1681 and immune response.
- Immune reconstitution.

3.3 End of Study and Study Termination

End of study is defined as the time at which all patients have completed or discontinued treatment with CTO1681 (or placebo for the Phase 2a portion of the study) and have been followed for assessment of CAR T efficacy 6 or 12 months (Phase 1b or Phase 2a, respectively), or have died, been lost to follow-up, or withdrawn consent.

The study may be terminated at the discretion of the Sponsor if there is sufficiently reasonable cause. In the event of such action, written notification documenting the reason for study termination will be provided to each Investigator.

Circumstances that may warrant early termination of the study include, but are not limited to:

- Determination of unexpected, significant, or unacceptable risk to patients.
- Insufficient adherence to protocol requirements.
- Plans to modify, suspend, or discontinue the development of study drug.
- Failure to enroll patients at an acceptable rate after exhausting all efforts to improve enrollment.
- Other administrative reasons.

Should the study be terminated prematurely, all study materials must be returned to the Sponsor or Sponsor designee.

4. STUDY POPULATION

4.1 Number of Patients

Phase 1b: Approximately 9-18 patients in dose escalation, and up to a total of 54 patients including backfill expansion will be enrolled.

Phase 2a: Approximately 100 patients will be enrolled, randomized 1:1 to CTO1681 (n = 50) or placebo (n = 50).

4.2 Inclusion Criteria

Patients meeting all of the following criteria will be considered eligible for enrollment in the study:

1. Age 18 years or older.
2. Undergone leukapheresis and is scheduled to receive protocol-specified commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy for DLBCL without corticosteroid prophylaxis for CRS and/or ICANS. All patients must have relapsed or refractory disease to at least 1 prior line of systemic therapy.
3. Met all inclusion criteria for CAR T-cell therapy per institutional guidelines.
4. Adequate organ function defined as:
 - a. Estimated Creatinine Clearance per Cockcroft Gault formula ≥ 60 mL/min.
 - b. Serum alanine aminotransferase/aspartate aminotransferase $\leq 2.5 \times$ ULN.
 - c. Total bilirubin $\leq 1.5 \times$ ULN.
 - d. Left ventricular ejection fraction $\geq 40\%$ on echocardiogram or multigated acquisition and no clinically significant pericardial effusion.
 - e. Platelets $\geq 50,000/\text{mm}^3$.
 - f. Absolute neutrophil count $> 1000/\mu\text{L}$.
 - g. Absolute lymphocyte count $> 100/\mu\text{L}$.
5. Documented measurable lymphoma disease adequate to judge by Lugano Criteria.
6. Eastern Cooperative Oncology Group performance status 0 to 1.
7. Female participants of childbearing potential and all male participants must agree to use Investigator-approved methods of birth control while on study drug and for 30 days thereafter.
8. Patients who are willing to provide written informed consent before the predose procedures, or patients who have a legal representative capable of providing informed consent on their behalf.

4.3 Exclusion Criteria

Patients meeting any of the following criteria will be excluded from the study:

1. Any cytotoxic chemotherapy within 14 days prior to leukapheresis.
2. Clinically significant malabsorption syndromes and swallowing difficulties which are inadequately controlled with medication (eg, odynophagia, dysphagia, gastroesophageal reflux disease) as per Investigator assessment.
3. Grade 2 or greater electrolyte imbalance, per CTCAE v5.0:
 - a. Potassium < 3.0 or > 5.5 mmol/L
 - b. Sodium < 130 or > 150 mmol/L
 - c. Calcium < 8.0 or > 11.5 mg/dL
 - d. Magnesium < 0.5 or > 1.23 mmol/L
4. Clinically significant ECG abnormality at Screening or Baseline (Day -1), including but not limited to, a confirmed QTcF value > 470 msec. Patients with QTcF readings that are borderline or difficult to interpret because of a condition such as bundle branch block, or in those where the end of the T wave is difficult to measure will be excluded. This also includes any Grade 2 or greater conduction block disorder, atrial, or ventricular arrhythmia.
5. History of clinically significant arrhythmia and/or requiring anticoagulation/antiplatelet treatment at therapeutic dose.
6. Any clinically significant (ie, active) cardiovascular disease, including cerebral vascular accident/stroke (< 6 months before enrollment), myocardial infarction (< 6 months before enrollment) or unstable angina, and congestive heart failure ≥ New York Heart Association Classification Class III.
7. Uncontrolled thromboembolic events or recent severe hemorrhage within the last 6 months.
8. Known history of any bleeding disorder.
9. Requirement for ongoing therapeutic doses of anticoagulant therapy, antiplatelet or fibrinolytic agents (low molecular weight heparin prophylaxis is allowed).
10. Baseline systolic blood pressure <100 mmHg.
11. History of autoimmune disease/ graft versus host disease requiring immunosuppressive therapy within the last 2 years. However, physiologic steroids (prednisone equivalent) may be given at a dose of 5 mg or less.
12. Patients who, in the opinion of the Investigator, would be unlikely to comply with study procedures or are otherwise unsuitable for enrollment

4.4 Patient Identification and Registration

Patients who are candidates for treatment enrollment into the study will be evaluated for eligibility by the Investigator to ensure that the inclusion and exclusion criteria ([Sections 4.2](#) and [Section 4.3](#)) have been satisfied.

On identification of an eligible patient, the Investigator or the Investigator's research staff will provide the eligibility information to the Sponsor or designee. At that time, the patient will be registered and assigned a sequential and unique patient number. As confirmation, the Sponsor or designee will provide the Investigator with written verification of each patient's registration. Once a patient number has been assigned, it cannot be reused.

4.5 Withdrawal and Replacement of Patients

4.5.1 Withdrawal

Patients have the right to withdraw from the study at any time for any reason.

Patients must withdraw or be withdrawn from study treatment for any of the following reasons:

- Withdrawal of consent
- Pregnancy
- Death
- Investigator decision
- Termination of study by Sponsor

Patients may withdraw or be withdrawn from study treatment for any of the following reasons:

- AEs
- Protocol deviation
- Noncompliance
- Investigator decision
- Loss to follow-up

When a patient discontinues study drug and/or withdraws from the study, the primary reason(s) for discontinuation or withdrawal must be recorded in the appropriate sections of the electronic case report form (eCRF) and all efforts will be made to complete and report final study observations as thoroughly as possible.

All AEs, irrespective of relationship to CTO1681, should be monitored until they are resolved, have stabilized, have returned to pre-exposure baseline, are determined to be due to another illness, or until a subsequent therapy is initiated. If the patient withdraws from treatment because of an AE, every effort must be made to perform protocol-specified safety follow-up procedures, as outlined in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)).

4.5.2 Replacement

During Phase 1b, patients who are discontinued for any reason other than a DLT, SAE or a severe nonserious AE assessed as related to the study drug by the Investigator may be replaced for the purpose of determining dose escalation or expansion. In the event of poor compliance, additional patients may be enrolled within the dose cohort to allow assessment of safety for the purpose of determining dose escalation or expansion. All patients treated with study medication will be included in analysis as outlined in [Section 7.2](#).

During Phase 2a of the study, patients will not be replaced. Additional patients may be enrolled beyond the planned 100 to account for early withdrawal.

5. STUDY TREATMENT

5.1 Lymphodepleting Chemotherapy

Lymphodepleting chemotherapy will be administered according to the CAR T-cell manufacturer's PI and institutional guidelines. Lymphodepleting chemotherapy is not required for patients with a white blood cell count $\leq 1,000$ cells/ μL within 1 week before planned CAR T-cell infusion but may be performed at the investigator's discretion.

Lymphodepleting chemotherapy will be supplied by the investigative site unless otherwise noted. Guidance on packaging, storage, preparation, administration, and toxicity management associated with the administration of LD chemotherapy should be followed as documented on the current product label.

5.2 Study Drug

For the Phase 1b portion of the study, study drug will consist of CTO1681-IR-OSD-10 μg Immediate Release Tablets.

For the Phase 2a portion of the study, study drug will consist of either CTO1681-IR-OSD-10 μg Immediate Release Tablets or matching placebo tablets.

During Phase 1b of the study, CTO1681 will be administered PO as a 10- μg tablet. In Cohort 1, CTO1681 tablets will be administered PO TID for 15 days at a total daily dose of 30 μg . Cohorts 2 and 3 will receive CTO1681 PO TID 15 days at total daily doses of 60 and 90 μg , respectively.

During Phase 2a of the study, CTO1681 will be orally administered as 10- μg tablets, potentially multiple tablets per dose, depending on the RP2D dose determined by the Phase 1b results. Placebo tablets will be identical in size, shape, color, and packaging to CTO1681 10- μg tablets.

Refer to the investigational product manual for specific details and instructions on storage and administration.

5.2.1 Study Patient Diary

Patients who are discharged during the treatment period will be given a dosing diary for the outpatient portion. Patients should record relevant information regarding the study drug in the diary (eg, confirmation that each daily dose was taken; confirmation study drug was taken within 30 minutes of a meal/snack).

Treatment compliance is described in [Section 5.7](#) and will be assessed based on return of unused drug and the dosing diary.

5.2.2 Dose Modification or Interruption

The dose of CTO1681 may not be modified (neither inpatient or within a cohort), and treatment may not be interrupted except as noted under the section for Management of DLTs ([Section 5.10.1.1](#)).

In patients who develop CRS and/or ICANS, CTO1681 should be continued, except as noted under the section for Management of DLTs ([Section 5.10.1.1](#)).

5.3 CAR T-cell Therapy

Patients will receive commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy, which will be administered according to the manufacturer's PI and institutional guidelines.

5.4 Method of Assigning Patients to Treatment

The Phase 1b portion of the study is open-label.

The Phase 2a portion of the study is randomized, double-blind, and placebo-controlled. Once informed consent is given, patients will be assigned a unique screening identification number. Participants who complete the study Screening assessments and meet all eligibility criteria will be randomly assigned in a 1:1 ratio to receive either CTO1681 or placebo at Baseline (Day -1).

Patients will be assigned a randomization number through the Interactive Web Response System (IWRS) in accordance with the randomization code generated by the Sponsor's authorized unblinded personnel.

5.5 Blinding, Packaging, Labeling, and Storage

5.5.1 Blinding

The Phase 1b portion of the study is open-label. During the placebo-controlled Phase 2a portion of the study, the study staff, including the Investigator, study coordinator, and pharmacist; Sponsor; study team; and all study patients will be blinded as to whether the patients are receiving CTO1681 or placebo. The randomization scheme will not be made available to study staff, the Sponsor, or study patients.

To maintain the study blind and distribute study treatment for the Phase 2a portion of the study, a separate list will be provided to the vendor responsible for study drug packaging to pre-label treatment bottles. This list will also be loaded into the IWRS and the bottle number will be provided to the study site staff during the randomization process and for subsequent dispensations. The bioanalytical laboratory will be provided with the randomization assignment for assay purposes. Bioanalytical results will remain blinded until after database lock.

5.5.2 Procedures for Breaking the Blind

Breaking the blind during the Phase 2a portion is expressly forbidden except in the event of a medical emergency where the identity of the study drug must be known to properly treat the patient. If breaking the blind is required because of a medical emergency, the treatment identity for the unblinded patient will only be revealed by the Sponsor's designee with approval from the Sponsor's Medical Monitor or qualified delegate.

When the blind is broken, the Investigator must record the date and reason for the blind break. The unblinding should be noted in the patient's eCRF.

5.5.3 Packaging and Labeling

CTO1681-IR-OSD-10 µg Immediate Release Tablets and placebo tablets will be provided in plastic bottles containing 45 tablets each. The study number will be printed on each individual

drug label. Labels for Phase 1b will include the identification of the contents as only CTO1681-IR-OSD-10 µg Immediate Release Tablets will be distributed to patients during this open label portion of the study. Drug or placebo bottles will be properly masked as to contents during Phase 2a, the randomized part of the study.

Patient identification number, initials, and the date the bottle was dispensed for each patient will be completed by the study site staff or pharmacist.

Packaging will meet all regulatory requirements. Refer to the investigational product manual for additional details.

5.5.4 Storage

Study drug tablets must be stored at room temperature (20°C to 25°C ([68°F to 77°F]) in their original container, according to the package label. Refer to the label or certificate of analysis for expiry.

All study drug products must be stored in a secure, limited-access location and may be dispensed only by the Investigator or by a member of the staff specifically authorized by the Investigator.

5.6 Duration of Patient Participation

Patient participation during the Phase 1b portion of the study will be approximately 9 months, including Screening.

Patient participation during the Phase 2a portion of the study will be approximately 15 months, including Screening.

5.7 Treatment Compliance

Patients participating in the Phase 1b of the study will be allotted 1, 2 or 3 bottles of CTO1681 10 µg tablets, for 30, 60 or 90 µg/day doses, respectively. In the Phase 2a portion, the CTO1681 or placebo assigned patients will receive 1, 2 or 3 bottles of either CTO1681 10 µg tablets or placebo depending on the randomization assignment and the RP2D carried forward in this portion of the study.

Each bottle will contain 45 tablets of either CTO1681 10 µg tablets or placebo (Phase 2a only).

5.8 Study Drug Accountability

The Investigator or designee will maintain accurate drug accountability records indicating the drug's delivery date to the site, inventory at the study center, administration to each patient, and return to the Sponsor or its designee (or disposal of the drug, if approved by the Sponsor). These records will adequately document that the patients were provided the doses as specified in the protocol and should reconcile all study drug received from the Sponsor. Accountability records will include dates, quantities, batch/serial numbers, expiration dates (if applicable), and patient numbers. The Sponsor or its designee will review drug accountability at the study center on an ongoing basis during monitoring visits. An unblinded monitor may be assigned during the Phase 2a portion to assist in reviewing drug accountability throughout the study.

Study drug must not be used for any purpose other than the present study. Study drug that has been dispensed and returned unused must not be re-dispensed for a different patient.

All unused and used study drug will be retained at the site until they are inventoried by the study monitor. All used, unused, or expired study drug will be returned to the Sponsor or its designee or if authorized, disposed of at the study site per the site's standard operating procedures and must be documented. All material containing CTO1681 will be treated and disposed of as hazardous waste in accordance with governing regulations.

5.9 Prior and Concomitant Medications and Treatment

5.9.1 Prior and Concomitant Medications and Procedures

All medications administered and procedures conducted at study entry within 20 days before the first day of study drug administration (Day -1) and until the end-of-study visit (Month 6 or Month 12 for Phase 1b and Phase 2a, respectively) are to be recorded on the eCRF. In addition, all prior treatments for DLBCL should be recorded.

During the study, LD chemotherapy and CAR T-cell therapy are to be recorded on designated eCRF pages. All other medications taken from the time informed consent is signed should be recorded as concomitant medications on the appropriate eCRF page.

5.9.1.1 Prohibited Concomitant Therapy

The following concomitant therapies are prohibited during participation in the study:

- Bridging chemotherapy is allowed, however, no additional therapy should be given within 7 days of initiation of LD chemotherapy.
- No additional steroids should be given above physiologic dose after leukapheresis.
- Systemic steroids and other immunosuppressive drugs for 3 months after CAR T-cell infusion unless used to manage CAR T-cell related toxicities.
- Systemic steroids or tocilizumab for prevention of CRS/ICANS (prophylactic steroids or tocilizumab) or the treatment of Grade 1 CRS/ICANS. Steroids and/ or tocilizumab should only be given for Grade 2 or higher CRS/ICANS.
- Chemotherapy, immunotherapy, targeted agents, radiation, high-dose corticosteroids, and investigational agents other than LD chemotherapy are not allowed in this protocol except as needed for treatment of disease progression after CAR T-cell therapy.
- Granulocyte macrophage colony-stimulating factor (GM-CSF) can potentially worsen CRS symptoms and should be avoided.
- Short-acting granulocyte colony stimulating factor within 3 days of CAR T-cell infusion or long-acting granulocyte colony stimulating factor within 10 days of CAR T-cell infusion.
- Central nervous system prophylaxis (eg, intrathecal methotrexate) must be stopped more than 1 week prior to CAR T-cell infusion.
- Herbal and natural remedies.

- Inhibitors of BCRP/ABCG2, OAT3, as follows:
 - Ritonavir
 - Nelfinavir
 - Cobicistat
 - Saquinavir
 - Zidovudine
 - Dolutegravir
 - Cabotegravir
 - Daclatasvir
 - Voxilaprevir
 - Dasabuvir
 - Velpatasvir
 - Simeprevir
 - Pibrentasvir
 - Glecaprevir
 - Pritaprevir
 - Ledipasvir
 - Letemovir
 - Favipiravir
- Other potential drugs that can interact with CTO1681 ([CTO1681 \[Investigators Brochure\]](#)).

5.9.1.2 Allowed Concomitant Therapy

Any medication or inactivated vaccine (including over-the-counter or prescription medicines and excluding vitamins) deemed for supportive care and safety of the patient received at the time of enrollment, or received during the study, must be recorded in the eCRF along with reason for use, dates of administration (including start and end dates), and dosage information (including dose and frequency). This may include, but is not limited to, the following:

- CAR T-cell infusion routine prophylaxis medications (eg, diphenhydramine, acetaminophen/paracetamol).
- Per study guidelines systemic steroids: dexamethasone, prednisone or other corticosteroids for the treatment of CRS or ICANS.
- Physiologic steroid replacement (≤ 5 mg/day prednisone or equivalent) topical and inhaled steroids.
- Symptomatic treatments of tumor-associated symptoms (eg, pain and nausea).
- Oxygen therapy and blood products or transfusions.
- Medications to treat chronic diseases (eg, hypertension and hypercholesterolemia).

Any concomitant medication not listed in the protocol may be considered on a case-by-case basis by the Investigator in consultation with the Sponsor or designee, if needed.

5.10 Toxicity Management

5.10.1 Dose-limiting Toxicities

Key DLT criteria are:

- Hypotension: Defined as documented, prolonged (≥ 2 hours) systolic blood pressure < 80 mm Hg unresponsive to basic medical therapy (ie, fluid challenge).
- Bleeding: Any clinically significant bleeding event (graded per Common Terminology Criteria for Adverse Events v5.0 [CTCAE]) Grade ≥ 3 (ie, excluding accidental cuts, periodontal, menstrual, or hemorrhoidal bleeding, or easily controlled epistaxis), including any episode of hemoptysis.
- Hepatic dysfunction: Any occurrence of transaminase elevation of $> 5 \times$ upper limit of normal (ULN; CTCAE Grade ≥ 3) that does not improve to $\leq$ Grade 2 within 7 days.
- QTc prolongation: Development of QTcF ≥ 550 msec or a > 60 msec change from baseline confirmed by repeat 12-lead ECG.
- Ventricular tachycardia (CTCAE Grade ≥ 3): requiring urgent medical intervention.
- Other:
 1. Any other CTCAE Grade ≥ 3 non-hematological toxicity that does not improve to CTCAE $\leq$ Grade 2 within 72 hours.
 2. Any death not due to underlying malignancy (Note: All CTCAE Grade 5 events will trigger a study pause to collect and assess the event for DLT determination. Expedited reporting to the FDA would be included).

Exemptions from DLT criteria are based on anticipated high confounding prevalence in the specific study patient population and will include (per CTCAE):

- Grade 3 or 4 anemia, lymphopenia or thrombocytopenia.
- Grade 3 or 4 fever or febrile neutropenia lasting ≤ 14 days.
- Grade 3 or 4 tumor lysis syndrome lasting ≤ 14 days.
- Grade 3 peripheral sensory neuropathy will not be considered a DLT in subjects with prior history of peripheral sensory neuropathy.
- Grade 3 fatigue and anorexia that does not result in hospitalization, tube feeding or total parenteral nutrition.

5.10.1.1 Management of Key Dose-Limiting Toxicities

The DLTs of hypotension, hepatic dysfunction, bleeding, QTc prolongation, and ventricular tachycardia will be managed as follows:

- DLT of hypotension or hepatic dysfunction: CTO1681 will be interrupted until the DLT resolves. Should the DLT recur, CTO1681 will be discontinued for that patient.

- DLT of bleeding: CTO1681 will be interrupted until the DLT resolves. Should the DLT recur, CTO1681 will be discontinued for that patient.
- DLT of QTc prolongation or ventricular tachycardia:
 - In the case of QTc prolongation, CTO1681 will be interrupted until the DLT resolves. Should the DLT recur, CTO1681 will be discontinued for that patient.
 - In the case of ventricular tachycardia, CTO1681 should be permanently discontinued for that patient.

All other DLTs will be managed according to the institutional SOC.

5.10.2 Infection Prophylaxis

Patients should receive prophylaxis for infection with pneumocystis pneumonia, herpes virus, and fungal infections according to standard institutional practice or National Comprehensive Cancer Network guidelines.

5.10.3 Tumor Lysis Syndrome

All patients with significant malignancy burden and without a contradiction such as allergy should be started on prophylaxis (eg, allopurinol) as per institutional guidelines prior to CAR T-cell infusion. Prophylaxis should be discontinued once the risk of tumor lysis has passed.

5.10.4 B Cell Depletion

It is possible that B cell depletion and hypogammaglobulinemia will occur due to the effects of CAR T-cell therapy on normal B cells. Gammaglobulin will be administered for hypogammaglobulinemia according to institutional guidelines. At a minimum, trough immunoglobulin G (IgG) levels should be kept above 400 mg/dL, especially in the setting of infection ([Hill, Giralt et al. 2019](#)).

5.10.5 Cytokine Release Syndrome

Cytokine Release Syndrome (CRS) should be identified based on clinical presentation. Investigators should evaluate for and treat other causes of fever, hypoxia, and hypotension. If CRS is suspected, manage according to the recommendations in Table 2. The goal of CRS management is to prevent life-threatening conditions while preserving the benefits of antitumor effects. The CRS grading system created by the ASTCT and published by Lee et al. ([Lee, Santomasso et al. 2019](#)) and treatment guidance for CRS for this clinical study to evaluate CTO1681 as a CRS mitigator, are detailed in Table 2. The management guidelines are consistent with the CRS management guidance presented in the YESCARTA package insert for CRS Grades ≥ 2 and provide guidance for use of CTO1681 during this investigational study. For severe or life-threatening CRS, consider intensive-care supportive therapy.

General recommendations and monitoring for CRS and administration of CTO1681 are as follows:

- Patients should be well-hydrated before and while receiving CTO1681.

- CTO1681 should be administered in accordance with the study schedule and dosing instructions.
- Patients should be monitored as inpatients during administration of CTO1681 from Day -1 to Day +13, except if discharged on Day +7 to +13 in the absence of CRS/ICANS or other conditions requiring inpatient care, after thorough evaluation and at the discretion of the Investigator, and with the approval of the Medical Monitor and Sponsor.
- In the event symptoms (fever, hypotension, hypoxia) develop, the patient will be evaluated for/to rule out other causes (eg, infection). Other causes of symptoms should be treated as appropriate.
- Neither steroids nor tocilizumab should be administered prophylactically (prior to CAR T-cell therapy administration or prior to clinical presentation of CRS Grade 2 or higher).

Table 2: Management of Cytokine Release Syndrome

CRS Grade	ASTCT CRS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 0	Prior to any signs or symptoms of CRS	Start on Day -1 (1 day prior to CAR T-cell therapy).	Do NOT administer tocilizumab.	Do NOT administer corticosteroids.	Ensure patient is well-hydrated and temperature control is adequate for patient comfort.
Grade 1	Fever $\geq 38^{\circ}\text{C}$	Continue administration of CTO1681.	Do NOT administer tocilizumab.	Do NOT administer corticosteroids.	Administer supportive therapies as appropriate, for example: Empiric antibiotic coverage. IV fluid hydration. Acetaminophen PRN (not to exceed 3g/24hrs). Hypothermia blanket PRN.

CRS Grade	ASTCT CRS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 2	Fever $\geq 38^{\circ}\text{C}$ AND Hypotension not requiring vasopressors AND/OR Hypoxia requiring low-flow nasal cannula or blow-by oxygen	Continue administration of CTO1681.	Consider administration of tocilizumab 8 mg/kg IV over 1 hour (not to exceed 800 mg). If no clinical improvement in the signs and symptoms of CRS after the first dose, repeat tocilizumab every 8 hours as needed. Limit to a maximum of 3 doses in a 24-hour period; maximum total of 4 doses. If improving, discontinue tocilizumab.	May administer dexamethasone 10 mg intravenously once daily. If improving, continue until severity is Grade 1 or less, then quickly taper corticosteroids as clinically appropriate.	Continue Grade 1 supportive care as appropriate. Also consider: Low-flow nasal cannula or blow-by oxygen for hypoxia. IV fluids for hypotension.

CRS Grade	ASTCT CRS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 3	Fever $\geq 38^{\circ}\text{C}$ AND Hypotension requiring vasopressors with or without vasopressin AND/OR Hypoxia requiring high-flow nasal cannula, facemask, nonrebreather mask, or Venturi mask	Continue administration of CTO1681.	Consider administration of tocilizumab 8 mg/kg IV over 1 hour (not to exceed 800 mg). If no clinical improvement in the signs and symptoms of CRS after the first dose, repeat tocilizumab every 8 hours as needed. Limit to a maximum of 3 doses in a 24-hour period; maximum total of 4 doses. If improving, discontinue tocilizumab.	Administer dexamethasone 10 mg IV three times a day. If improving, manage as appropriate grade above and continue corticosteroids until the severity is Grade 1 or less, then quickly taper as clinically appropriate.	Administer supportive respiratory care for hypoxia as appropriate. Administer vasopressors for hypotension as appropriate. If continued worsening, then consider next line of therapy: e.g. Anakinra, Siltuximab, high dose methylprednisolone, or other agents.

CRS Grade	ASTCT CRS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 4	Fever $\geq 38^{\circ}\text{C}$ AND Hypotension requiring multiple vasopressors (excluding vasopressin) AND/OR Hypoxia requiring positive pressure (CPAP, BiPAP, intubation, mechanical ventilation)	Continue administration of CTO1681.	Consider administration of tocilizumab 8 mg/kg IV over 1 hour (not to exceed 800 mg). If no clinical improvement in the signs and symptoms of CRS after the first dose, repeat tocilizumab every 8 hours as needed. Limit to a maximum of 3 doses in a 24-hour period; maximum total of 4 doses. If improving, discontinue tocilizumab.	Administer methylprednisolone 1000mg IV once per day for 3 days. If not improving for > 24 hours or if patient is deteriorating rapidly, consider methylprednisolone 1000 mg 2-3 times a day. If improving, manage as appropriate grade above and continue corticosteroids until the severity is Grade 1 or less, then taper as clinically appropriate.	Administer vasopressors for hypotension and/or supportive respiratory care for hypoxia as appropriate. Consider additional therapy (e.g., Siltuximab, Anakinra) per institutional guidelines.

Abbreviations: ASTCT = American Society of Transplantation and Cellular Therapy; BiPAP = bilevel/2-level positive airway pressure; C = Celsius; CPAP = continuous positive airway pressure; CRS = cytokine release syndrome; h = hours; IV = intravenous; PRN = as needed.
Source: (Lee, Santomasso et al. 2019).

In Table 2:

- Fever is defined as temperature $\geq 38^{\circ}\text{C}$ not attributable to any other cause. In patients who have CRS then receive antipyretic or anticytokine therapy such as tocilizumab or steroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.
- The CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a patient with temperature of 39.5°C , hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as Grade 3 CRS.
- Low-flow nasal cannula is defined as oxygen delivered at ≤ 6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in pediatric. High-flow nasal cannula is defined as oxygen delivered at > 6 L/minute.

5.10.6 Immune Effector Cell-associated Neurotoxicity Syndrome

Like CRS, ICANS is a potentially life-threatening toxicity syndrome that is often caused by CAR T-cell therapy and frequently occurs within the first week following CAR T-cell therapy. ICANS can progress through increasingly severe manifestations. An ICANS grading system developed by the ASTCT and treatment guidance for ICANS is detailed in Table 3.

General recommendations and monitoring per [Section 5.10.5](#).

If concurrent CRS is suspected, treat CRS according to the recommendations in Table 2.

Table 3: Management of ICANS

ICANS Grade	ASTCT ICANS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 0	Prior to any signs or symptoms of ICANS	Start on Day -1 (1 day prior to CAR T-cell therapy).	Do NOT administer tocilizumab.	Do NOT administer corticosteroids	Ensure patient is well-hydrated and temperature control is adequate for patient comfort.
Grade 1	ICE score: 7-9 Consciousness: Awakens spontaneously Seizure: None Motor findings: None Elevated ICP/cerebral edema: None	Continue administration of CTO1681.	Do NOT administer tocilizumab.	Do NOT administer corticosteroids.	Consider levetiracetam for seizure prophylaxis. Supportive care and neurological work-up including MRI and/or CT scan, EEG and neurology consult per institutional protocols.

ICANS Grade	ASTCT ICANS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 2	ICE score: 3-6 Consciousness: Awakens to voice Seizure: None Motor findings: None Elevated ICP/cerebral edema: None	Continue administration of CTO1681.	Consider administration for treatment of concurrent CRS, if appropriate. See CRS management guidelines.	Administer dexamethasone 10 mg IV four times a day. If improving, continue corticosteroids until the severity is Grade 1 or less, then quickly taper as clinically appropriate.	Consider levetiracetam for seizure prophylaxis.

ICANS Grade	ASTCT ICANS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 3	ICE score: 0-2 Consciousness: Awakens only to tactile stimulus Seizure: Any clinical seizure the resolves rapidly or nonconvulsive seizures on EEG that resolve with intervention Motor findings: None Elevated ICP/cerebral edema: Focal/local edema on neuroimaging	Continue administration of CTO1681.	Consider administration for treatment of concurrent CRS, if appropriate. See CRS management guidelines.	Administer methylprednisolone 1000 mg IV once daily. If cerebral edema is suspected (on head CT/MRI): Consider methylprednisolone 1000 mg/ day in divided doses for 3 days (edema in brainstem or thalamus); or 1 day (if other brain areas only) or administer per institutional guidelines. If improving, continue corticosteroids until the severity is Grade 1 or less, then quickly taper as clinically appropriate.	Consider levetiracetam for seizure prophylaxis.

ICANS Grade	ASTCT ICANS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
Grade 4	ICE score: 0 (patient is unable to perform ICE) Consciousness: Patient is unarousable or requires vigorous or repetitive tactile stimuli to arouse. Stupor or coma Seizure: Life-threatening prolonged seizure (>5 min); or repetitive clinical or electrical seizures without return to baseline in between Motor findings: Deep focal motor weakness such as hemiparesis or paraparesis	Continue administration of CTO1681.	Consider administration for treatment of concurrent CRS, if appropriate. See CRS management guidelines.	Administer methylprednisolone 1000 mg IV twice daily. If improving, continue corticosteroids until the severity is Grade 1 or less, then quickly taper as clinically appropriate. If not improving, consider 1000 mg of methylprednisone IV 3 times a day or alternate therapy.	Consider levetiracetam for seizure prophylaxis. If Grade 4 ICANS is refractory for > 24 hours or if patient is deteriorating rapidly, consider next line of therapy: e.g. Anakinra, Siltuximab, cyclophosphamide, or other agents per institutional protocol.

ICANS Grade	ASTCT ICANS Consensus Grading	CTO1681 Administration	Tocilizumab Administration	Corticosteroid Administration	Other Supportive Therapies
	Elevated ICP/cerebral edema: Diffuse cerebral edema on neuroimaging; decerebrate or decorticate posturing; or cranial nerve VI palsy; or papilledema; or Cushing's triad				

Abbreviations: ASTCT = American Society of Transplantation and Cellular Therapy; CT = computed tomography; EEG = electroencephalogram; h = hours; ICANS = immune effector cell-associated neurotoxicity syndrome; ICE = immune effector cell-associated encephalopathy; ICP = intracranial pressure; IV = intravenous; MRI = magnetic resonance imaging; VI = six.
Source: (Lee, Santomasso et al. 2019).

In Table 3:

- ICANS grade is determined by the most severe event (ICE score, level of consciousness, seizure, motor findings, raised ICP/cerebral edema) not attributable to any other cause; for example, a patient with an ICE score of 3 who has a generalized seizure is classified as Grade 3 ICANS.
- A patient with an ICE score of 0 may be classified as grade 3 ICANS if awake with global aphasia, but a patient with an ICE score of 0 may be classified as Grade 4 ICANS if unarousable.
- Depressed level of consciousness should be attributable to no other cause (eg, no sedating medication).
- Tremors and myoclonus associated with immune effector cell therapies may be graded according to CTCAE v5.0, but they do not influence ICANS grading.
- Intracranial hemorrhage with or without associated edema is not considered a neurotoxicity feature and is excluded from ICANS grading. It may be graded according to CTCAE v5.0.

5.10.6.1 Fever and Neutropenia

Evaluation for the source of an infection should be performed per institutional guidelines. Fevers should be treated with acetaminophen and comfort measures. Nonsteroidal anti-inflammatory drugs and corticosteroids should be avoided. Patients who are neutropenic and febrile should receive broad-spectrum antibiotics. Maintenance intravenous fluids (normal saline) should be started on most patients with high fevers, especially if oral intake is poor or if the patient is tachycardic. An even daily fluid balance should be sought in patients who are not hypotensive and not experiencing active tumor lysis syndrome.

Filgrastim should be used according to published guidelines (eg, Infectious Disease Society of America).

5.10.6.2 Blood Product Support for Anemia and Thrombocytopenia

Treatment of anemia and thrombocytopenia should be provided per institutional guidelines. All blood products should be irradiated. Using complete blood counts as a guide, the patient should receive platelets and packed red blood cells as needed. Attempts should be made to maintain a hemoglobin level > 8.0 gm/dL and a platelet count $> 20,000/\text{mm}^3$. Leukocyte filters should be used for all blood and platelet transfusions to decrease sensitization to transfused white blood cells and decrease the risk of Cytomegalovirus infection.

5.10.6.3 Deep Vein Thrombosis Prophylaxis

Prevention of deep vein thrombosis should be administered per institutional guidelines. Deep vein thrombosis prophylaxis should be administered to all patients with reduced mobility during hospitalization per institutional guidelines. Low molecular weight heparin is encouraged provided there are no contraindications (eg, recent surgery, bleeding diathesis, platelet count $< 50,000/\mu\text{L}$) based on benefit/risk. Noninvasive mechanical intermittent pneumatic compression

devices for deep vein thrombosis prophylaxis should be used in those who cannot receive anticoagulants due to increased bleeding risk or other concerns ([Lyman, Carrier et al. 2021](#)).

5.11 Criteria for Pausing or Stopping the Study

If any of the events referenced below as Pausing criteria or Stopping criteria occur within 43 days after the first exposure to the study drug (ie, Day +41), the SRC (in Phase 1b) or the IDMC (in Phase 2a) will be immediately informed, and an ad hoc meeting scheduled to review all available safety data. A pause to study recruitment will commence at that time. The committee will then provide recommendations as to how to proceed.

5.11.1 Pausing Criteria

Occurrence of any DLT will prompt an ad hoc meeting of the SRC (Phase 1b) or IDMC (Phase 2a):

- Hypotension: Defined as documented, prolonged (≥ 2 hours) systolic blood pressure < 80 mm Hg unresponsive to basic medical therapy (ie, fluid challenge).
- Bleeding: Any clinically significant bleeding event (graded per CTCAE v5.0) Grade ≥ 3 (ie, excluding accidental cuts, periodontal, menstrual, or hemorrhoidal bleeding, or easily controlled epistaxis), including any episode of hemoptysis.
- Hepatic dysfunction: Any occurrence of transaminase elevation of $> 5 \times$ upper limit of normal (ULN; CTCAE Grade ≥ 3) that does not improve to $\leq$ Grade 2 within 7 days.
- QTc prolongation: Development of QTcF ≥ 550 msec or a >60 msec change from baseline, confirmed by repeat 12-lead electrocardiogram (ECG).
- Ventricular tachycardia: (CTCAE Grade ≥ 3) requiring urgent medical intervention.
- Other:
 1. Any other CTCAE Grade ≥ 3 non-hematological toxicity that does not improve to CTCAE $\leq$ Grade 2 within 72 hours.
 2. Any death not due to underlying malignancy (Note: All CTCAE Grade 5 events will trigger a study pause to collect and assess the event for DLT determination. Expedited reporting to the FDA would be included).

5.11.2 Stopping Criteria

The SRC (Phase 1b) or IDMC (Phase 2a) will recommend stopping the study if:

- CTCAE Grade 4 toxicity (unmanageable, unexpected) occurring in $> 25\%$ of dosed patients that is possibly, probably, or definitely related to CTO1681 (excluding fever, hematological toxicities, and CRS/ICANS).
- Death resulting from a CTO1681-associated DLT not related to underlying disease, CAR T-cell therapy, or an extrinsic event (unrelated to the investigational product or study procedures). Specifically:
 - Fatal bleeding event.
 - Refractory hypotension resulting in death.
 - Refractory ventricular tachycardia resulting in death.

Note: Safety stopping criteria for Phase 2a will also take into account the data from subjects from Phase 1b who were treated at the RP2D as well as with lower dosages.

The Sponsor may decide to pause, stop, or make adaptations to the study based on recommendations of the SRC/IDMC and the totality of evidence. The SRC may recommend that the study be paused or stopped if it is determined that patient safety may be compromised by continuing the study. The IDMC may recommend that the study be stopped based on safety or reduced antitumor response (refer to [Section 3.1.2](#)).

6. STUDY ASSESSMENTS AND PROCEDURES

The Schedule of Events for Phase 1b and Phase 2a of the study are provided in [Appendix 10.1](#) and [Appendix 10.2](#), respectively. A detailed listing of assessments by visit day is provided in [Appendix 10.3](#).

After obtaining written informed consent, patients will undergo screening evaluations to determine eligibility within 20 days before Baseline/ Day –1 (ie, Day –21). Screening may occur across multiple days but should be completed by Day –3.

Study Day –1/ Baseline is defined as the day the patient receives their first dose of the study drug, approximately 24 hours before CAR T-cell infusion on Day 0.

The patient is planned to be treated as inpatient from Baseline/Day –1 to Day +6, at minimum, to allow for proper monitoring. During the period of Days +7 to +13, the patient may be discharged in the absence of CRS/ICANS at the discretion of the Investigator with the approval of the Medical Monitor and Sponsor. If the patient is discharged, then they must return to the clinic for a minimum of two visits that week, which must include Day +13 for AE collection and status assessment per the Schedule of Events during the treatment period.

Whenever more than 1 assessment is scheduled for the same nominal time, the assessments should be performed in the order of least invasive to most invasive assessment (eg, vital signs, ECG, blood draw[s]) unless otherwise specified in the protocol.

Note: Records of all assessments should be promptly entered into the patients eCRF to facilitate SRC/ IDMC review ([Section 3.1.1](#) and [Section 3.1.2](#)).

6.1 Informed Consent and Confirmation of Eligibility

A complete description of the study is to be presented to each potential patient and a signed and dated informed consent is to be obtained before any study-specific procedures are performed.

The patient's eligibility will be confirmed at Screening, and before the first dose of study drug. Study eligibility will be based upon the patient satisfying the study inclusion/ exclusion criteria ([Section 4.2](#) and [Section 4.3](#)).

6.2 Enrollment and Randomization

Patients will be enrolled and randomized (Phase 2a only) onto the study during the Screening period ([Section 5.4](#)) following confirmation of eligibility.

6.3 Demographics and Medical History

Subject demographic data, including date of birth, sex, age, race and ethnicity, will be obtained at Screening.

A medical and surgical history (see [Section 5.9](#)) will be obtained at Screening for all subjects, this includes all relevant prior medical history and current medical conditions. All medications administered and procedures conducted at study entry within 28 days before the first day of study drug administration (Day –1), and until the end-of-study visit (Month 6 or Month 12 for Phase 1b and Phase 2a, respectively) are to be recorded on the eCRF. In addition, all prior treatments for DLBCL should be recorded.

Concomitant medication(s) will be monitored and recorded throughout the study.

6.4 Lymphodepleting Chemotherapy

Lymphodepleting Chemotherapy should be performed between Study Days –12 to –3. However, if the patient has a white blood cell count $\leq 1,000$ cells/ μ L within 1 week before CAR T-cell infusion, LD chemotherapy is not required, but may be performed at the investigator's discretion.

6.5 CAR T-cell Infusion

Patients will receive CAR T-cell infusion (commercially available axicabtagene ciloleucel CD19-directed CAR T-cell therapy) on Day 0, which will be administered according to the manufacturer's PI.

6.6 CTO1681 Administration

CTO1681 will be administered per [Section 5.2](#).

CTO1681 dosing must be initiated at the Baseline visit (Day –1), approximately 24 hours (± 1 hour) before CAR T-cell infusion and will be dosed TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals on Day –1 to Day +13.

Treatment compliance ([Section 5.7](#)) will be measured on all dosing days (Day –1 to Day +13). For Day +7 to Day +13, patients who are not hospitalized will be given a dosing diary ([Section 5.2.1](#)) to record any relevant information regarding the study drug, and the diary will be reviewed by site staff for compliance during inpatient visits, and required date recorded in the patients eCRF ([Section 8.6.2](#)).

6.7 Efficacy Measurements

6.7.1 Cytokine Release Syndrome and Immune Effector Cell-associated Neurotoxicity Syndrome Grading

Grading of CRS and ICANS will be conducted daily from Day 0 (and as clinically indicated, per ASTCT criteria (see [Section 5.10.5](#) and [Section 5.10.6](#)).

For Day +7 to Day +13, if patients are not hospitalized, CRS/ ICANS assessment will be performed only during clinic visits irrespective of timing of CTO1681 dose.

CRS Consensus Grading

In grading CRS, a CRS severity scale associated with antibody therapeutics was published by National Cancer Institute investigators. Appreciating the scale needed to be adapted for other therapeutics, to define mild, moderate, severe, and life-threatening events; account for overlapping symptoms; and guide treatment recommendations, a revised CRS grading system was created during a meeting supported by ASTCT and published by Lee et al (Table 2).

ICANS Grading

Immune effector cell-associated neurotoxicity syndrome is a clinical neuropsychiatric syndrome that can occur following immunotherapy, particularly associated with T-cell engaging therapies

and can often occur concurrently with or immediately after CRS. The ASTCT created and published an ICANS grading system (Table 3).

6.7.2 Hospitalizations and Intensive Care Unit Admissions

The frequency and duration of hospitalizations and ICU admissions due to CRS or ICANS will be recorded.

6.7.3 Administration of Anticytokine Therapies

Use of tocilizumab, steroids, and other anticytokine therapies (eg, siltuximab, anakinra, ruxolitinib, dasatinib, and lenzilumab) for treatment of CRS or ICANS will be documented.

6.8 Pharmacokinetic Measurements

Serial blood samples for PK assessments will be drawn according to the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)) for patients during the Phase 1b and Phase 2a portions of the study. Those samples will be used to determine the PK concentration profile of CTO1681.

In the Phase 1b portion, blood samples for preparation of plasma will be collected for analysis on:

- Day –1 (first [morning] dose): at time = 0 (before eating and within 1-hour prior to the morning dose), and 0.5 ($\pm$ 10 minutes), 1 ($\pm$ 10 minutes), 1.5 ($\pm$ 10 minutes), 2 ($\pm$ 10 minutes), 3 ($\pm$ 10 minutes), and 4 ($\pm$ 10 minutes) hours after the first (morning) dose of CTO1681.
- Day –1 (second and third doses): at 0.5 hours ($\pm$ 10 minutes) predose and 1-hour postdose for the second and third doses of CTO1681.
- Day 0, Day +2, Day +4, Day +6 and Day +13: 1-hour ($\pm$ 10 minutes) after the first morning dose of CTO1681.

In the Phase 2a portion, sparse blood samples for the preparation of plasma will be collected for all patients to perform population PK and exploratory exposure-response analyses. Samples should be taken approximately 1 hour ($\pm$ 10 minutes) after the first morning dose.

Where PK samples are scheduled at the same time as ECG assessments and cytokine draws, assessments should take place in the following order: (1) ECG measurements, (2) PK draw, and (3) cytokine draw.

Pharmacokinetic samples should be processed within 1 hour of collection and frozen at -80°C until shipment. Further details regarding collection, handling and processing of PK samples will be contained in a separate study manual.

6.9 Biomarkers and Immune Reconstitution Measurements

For patients in the Phase 1b and Phase 2a portions of the study, biomarkers relating to the MOA of CTO1681 and immune response, as well as immune reconstitution will be collected through blood sampling (see [Section 7.5.6](#)).

Biomarkers relating to the MOA of CTO1681 and immune response include: cytokines (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF), C-reactive protein (CRP), and ferritin. In addition, blood

samples for IgG, a lymphocyte subsets enumeration (TBNK) panel and immunophenotyping will be performed to determine immune reconstitution, and will be obtained at the timepoints specified in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)).

Cytokine Sampling

Patients should avoid strenuous physical activity for 24 hours prior to Cytokine collection.

For both Phase 1b and Phase 2a, blood sampling for cytokines will occur pre- and approximately 1 hour post morning CTO1681 dose on Day +1 to Day+6. For Screening, LD chemotherapy, Day -1, Day 0, Day +7 to Day +13, Follow-up 1, Month 1, Month 3, Month 6, and Month 12 (Phase 2a only), sampling should occur at approximately the same time of day that the post-morning CTO1681 sampling will occur during the inpatient treatment period. On days where PK sampling is also scheduled for this timepoint (Days 0, +2, +4, +6 and +13), the cytokine sample should be drawn immediately following the PK sample.

Samples should be processed within 1 hour of collection and frozen at -80°C until shipment. Further details regarding collection, handling and processing of PK samples will be contained in a separate study manual.

6.10 Safety Measurements

The timing of safety assessments is described in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). Additional safety assessments may be performed when clinically indicated, at the Investigator's discretion, and based on local requirements.

All clinical and laboratory safety assessments will be evaluated by the Investigator for clinical significance. New clinically significant findings or medical history conditions that have worsened from baseline in severity will be reported as AEs.

6.10.1 Complete and Symptom Driven Physical Examination

Complete physical examinations including a complete neurological examination will be performed at Screening and baseline and at all other time points as specified in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). A physical examination may be performed at any time, if clinically indicated, based on signs and symptoms at the discretion of the Investigator.

A complete physical examination will include, at a minimum, assessment of the following body systems: skin, head, ears, eyes, nose, mouth/throat/neck, thyroid, lymph nodes, respiratory system, cardiovascular system, gastrointestinal system, neurological condition, blood and lymphatic systems, and the musculoskeletal system.

Additional physical examinations, including neurologic examinations, may be performed as needed and when patients are seen in clinic.

6.10.2 Height, Weight and BMI

The patient's height should be measured without shoes. Weight will be measured as defined per the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)), with body mass index calculation only to be taken at Screening, Day -1, Follow-up 1 (Day +14), Follow-up 3 (Day +41), Month 1 (Day +30), Month 3 (Day +90), Month 6 (Day +180) and Month 12 (Day +360, Phase 2a only).

6.10.3 Vital Signs

Vital signs measurements will include blood pressure (BP) (systolic and diastolic), pulse rate, oral temperature, respiratory rate, and oxygen saturation and will be measured during all study visits at the time points specified in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). In addition, vital signs should be collected at any time, if clinically indicated, based on signs and symptoms at the discretion of the Investigator.

Vital signs measurements will be collected after the patient has been in a supine resting position for at least 10 minutes.

Inpatient vital signs for Day -1 to Day +13 must be collected at the timepoints specified below:

- Day -1: pre-morning CTO1681 administration (first dose); 30 minutes ($\pm$ 10 minutes) and 1 ($\pm$ 10 minutes) and 2 hours ($\pm$ 10 minutes) post-dose for all 3 doses.
- Day 0 to Day +13: pre-CAR T-cell infusion (Day 0 only); pre-morning CTO1681 dose and 2 hours ($\pm$ 15 minutes) post-morning CTO1681 dose and Q6-8 h thereafter.

Outpatient vital signs will be collected for Day +7 to Day +13 only during clinic visits irrespective of timing of CTO1681 dose. Unless otherwise indicated or per Investigator judgement, vital signs will be taken once at all outpatient study visits.

6.10.3.1 Temperature Monitoring

Temperature monitoring will take place during the vital signs assessments, it is not required to take additional temperature measurements unless indicated or per Investigator's instructions.

6.10.4 Performance Status

Determination of Eastern Cooperative Oncology Group (ECOG) performance status will be performed at the time points specified in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). The patient's performance status will be assessed using the ECOG performance status tool (see [Appendix 10.5](#)).

6.10.5 Twelve-Lead Electrocardiogram

Twelve-lead ECGs will be obtained for all patients at the time points outlined in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)).

Patients must avoid strenuous physical activity prior to the ECG assessment, and rest in a supine resting position for at least 10 minutes prior to the collection of the ECG.

The screening ECG does not need to be repeated if performed within the screening window (Day -21 to Day -3) by the study institution. On Day -1, ECGs will be recorded in triplicate at predose -30 minutes ($\pm$ 5 minutes) and -15 minutes ($\pm$ 5 minutes), and at 1-hour postdose ($\pm$ 10 minutes) for the morning dose. On Days 0 to Day +6, and Day +8, Day +10, Day +12, and Day +13, ECGs will be recorded in triplicate predose and 1-hour postdose ($\pm$ 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion. Electrocardiograms will be evaluated by the Investigator and centrally.

Electrocardiograms will be reviewed and interpreted by the Investigator. Electrocardiograms from both Phase 1b and Phase 2a will also be analyzed centrally by a core ECG laboratory to provide evidence of the lack of the clinically relevant effects of CTO1681 on ECG parameters.

6.10.6 Two-Dimensional Echocardiogram

If a 2-dimensional (2D) echocardiogram has not been performed within 60 days before Screening, it must be performed. A 2D echocardiogram also may be performed during the study at the Investigator's discretion.

6.10.7 Hematology, Chemistry Panel and Urinalysis

Samples for laboratory assessments (hematology, serum chemistries, coagulation, urinalysis, pregnancy tests etc.) are to be collected per the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). A complete listing of required clinical laboratory tests is presented in [Section 6.10.7.1](#).

6.10.7.1 Clinical Laboratory Measurements

Clinical laboratory blood test for safety will be performed at a central laboratory.

Clinical laboratory evaluations will be conducted at the time points outlined in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)). In addition, all clinically significant laboratory abnormalities, as assessed by the Investigator, will be followed by repeat testing and further investigated according to the judgement of the Investigator.

The following safety laboratory tests are to be evaluated by the Investigator:

Hematology:	CBC with differential
Coagulation:	PT, PTT and INR
Serum chemistry:	Sodium, potassium, blood urea nitrogen or urea, bicarbonate (venous), creatinine, calcium, chloride, magnesium, phosphorus, albumin, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, gamma glutamyl transferase, total and direct bilirubin, lactate dehydrogenase, uric acid, and glucose.
Urinalysis (dipstick):	pH, specific gravity, bilirubin, blood, glucose, ketones, leukocyte esterase, nitrite, protein, and urobilinogen.
Pregnancy (urine)^a:	β-hCG.
Other:	TBNK panel, CRP, ferritin, IgG, Anti-CD19 CAR T-cells, CAR copy number, immunophenotyping.

Abbreviations: β hCG = beta human chorionic gonadotropin; CAR = chimeric antigen receptor; CBC = complete blood count; CRP = C-reactive protein; INR = international normalized ratio; PT = prothrombin time; PTT = partial thromboplastin time; TBNK = quantitative lymphocyte subsets

^a Rapid urine pregnancy tests will be collected from females of childbearing potential at Screening only.

6.10.8 Disease Assessment

Disease assessment will be performed by computerized tomography (CT) scan or positron emission tomography-CT scan. Screening scans are accepted if performed within the 30 days leading up to Screening visit. Magnetic resonance imaging as additional imaging will be performed only if deemed necessary per Investigator's assessment or institutional guidelines. If bridging therapy is administered, repeat imaging is required before LD chemotherapy.

6.10.9 Lugano Criteria for Malignant Lymphoma

The Lugano Criteria are the standard response criteria currently in use for lymphoma. The criteria are based on positron emission tomography (PET) or bidimensional tumor measurements on CT for non-fluorodeoxyglucose (FDG) avid lymphomas, or when PET imaging is not available ([Cheson, Fisher et al. 2014](#)). These criteria represent a revision of the Cheson criteria, published in 2014 ([Barrington, Mikhaeel et al. 2014](#)). The most important changes in classification criteria regarding imaging ([Barrington, Mikhaeel et al. 2014](#), [Cheson, Fisher et al. 2014](#)) were:

- Replacement of the dichotomous PET evaluation with the Deauville 5-point scale (5-PS).
- Introduction of the interim PET scan during treatment.

The 5-PS is a purely visual qualitative assessment that allows a more differentiated classification than was possible with the previous method of evaluating FDG uptake. The criteria for the 5-PS are summarized in Table 4.

The interim PET examination makes it possible to adjust treatment earlier in some cases. This allows lack of treatment response to be addressed more quickly so that corresponding treatment escalation can be initiated or, in the case of early treatment response, treatment can be deescalated to minimize toxicity and secondary diseases ([Spaepen, Stroobants et al. 2002](#), [Hutchings, Mikhaeel et al. 2005](#), [Hutchings, Loft et al. 2006](#), [Cerci, Pracchia et al. 2010](#)).

These response criteria will be used to assess response in patients to CAR T-cell therapy (Table 4).

Table 4: Lugano Criteria for Malignant Lymphoma

Response	CT: Radiological Response	PET/CT: Metabolic Response
CR	All criteria met: • No new lesions • Reduction ≤ 1.5 cm GTD of target lesions • No extranodal manifestation Spleen/liver: • Spleen and liver normal sized New lesions: None	• Score 1/2/3 according to the 5-PS without residual disease: <u>Score Criterion^a</u> 1. No ^{18}F-FDG uptake 2. Uptake $\leq$ mediastinum 3. Uptake $>$ mediastinum and $\leq$ liver 4. Uptake moderately above liver level 5. Uptake significantly above liver level and/or new lesion(s) - X New uptake region(s), most likely not part of lymphoma

Response	CT: Radiological Response	PET/CT: Metabolic Response
		New lesions: None
PR	- Reduction of SPD of the max. 6 measurable target lesions (nodal/extranodal) $\geq 50\%$ - Lesions that are too small/not measurable are assigned a size of 0.5×0.5 cm - Lesions that are not visible are assigned a size of 0×0 cm Spleen/liver: - Length reduction $> 50\%$ of the enlarged spleen New lesions: none	Score 4/5 according to 5-PS with reduced ^{18}F-FDG uptake with respect to baseline and residual lymph nodes of any size New lesions: none
SD	- Reduction of SPD of the max. 6 measurable target lesions (nodal/extranodal) $< 50\%$ - Progressive disease criteria not met New lesions: none	Score 4/5 according to 5-PS New lesions: None
Progressive disease	At least 1 criterion met: Abnormal lymph nodes must meet the following criteria: - GTD > 1.5 cm and - PPD increase $\geq 50\%$ and - Increase of GTD or SAD by 0.5 cm for ≤ 2.0 cm 1.0 cm for > 2.0 cm Spleen/liver: - In the case of existing splenomegaly: Increase in the size of the spleen $> 50\%$ - In the case of newly occurring splenomegaly: Increase in the length of the spleen ≥ 2.0 cm with respect to baseline New lesions: - Recurrence of lesions that had already returned to the normal range Or - New nodal lesion > 1.5 cm regardless of the axis Or - New extranodal lesions > 1.0 cm regardless of the axis or ≤ 1.0 cm in the case of clear assignment to lymphoma	Score 4/5 according to 5-PS with increase in ^{18}F-FDG uptake with respect to baseline New lesions: New ^{18}F-FDG-avid typical lymphoma lesions

Abbreviations: 5-PS = 5-point scale; CR = complete response; CT = computed tomography; FDG = fluorodeoxyglucose; GTD = great transverse diameter; PET = positron emission tomography; PPD = product of the perpendicular diameters; PR = partial response; SAD = short axis diameter; SD = stable disease; SPD = sum of the product of the diameters.

^a Deauville 5-PS.

Source: (Skusa, Weber et al. 2020).

6.10.10 Overall Response Rate

Overall response rate is defined as the proportion of patients who achieved a confirmed best response of CR or partial response (PR) as their best response, as assessed by the Lugano Criteria (Table 4).

6.10.11 Duration of Response

Duration of response is determined for patients with an objective response. Duration of response is measured from the time response criteria are first met until the first date of recurrent or progressive disease, or death due to any cause.

6.10.12 Progression-free Survival

Progression-free survival is measured from the start of treatment (Phase 1b) or date of randomization (Phase 2a) until the first date of recurrent or progressive disease, or death due to any cause.

6.10.13 Overall Survival

Overall survival is defined as the time from start of treatment to the date of death due to any cause.

6.10.14 Levels and Persistence of CAR T-cells

Levels of CAR⁺ T-cells in serum post CAR T-cell infusion will be measured from Days 0 to 180 in the Phase 1b portion of the study, and Days 0 to 360 in the Phase 2a portion of the study according to the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#))

6.10.15 Adverse Events

6.10.15.1 Definition of Adverse Events

Adverse Events are defined by FDA and other health authorities as any untoward medical occurrence associated with the use of a drug in humans, whether or not considered drug-related. 21 CFR 312 defines “adverse event” and related terms, along with safety reporting requirements for studies conducted under an investigational new drug application (IND). An AE (also referred to as an adverse experience) can:

- Be any unfavorable and unintended sign (eg, an abnormal laboratory finding), symptom, or disease temporally associated with the use of a drug, whether or not considered related to the drug.
- Arise from any use of the drug (eg, use in combination with another drug) and from any route of administration, formulation, or dose, including an overdose.
- Arise from development of a new disease or the worsening of an existing disease or recurrence of an intermittent medical condition not present at baseline.
- Be any deterioration in laboratory value and/or other clinical test that is associated with symptoms or leads to a change in study treatment or concomitant treatment or discontinuation from study treatment.

- Arise related to protocol procedures (including those occurring prior to initiating treatment with a study drug).

Disease progression or death due to progression of the underlying malignancy will not be considered an AE (or a SAE) in this study but will be collected as an outcome or reason for discontinuation, as appropriate. “Lack of efficacy” or “failure of expected pharmacological action” per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AEs or SAEs if they fulfil the definition of an AE or SAE. For medical or surgical procedures (eg, debulking, endoscopy, appendectomy), the condition that leads to the procedure is the AE. Adverse events (or SAEs) considered as complications of disease progression should be reported.

6.10.15.2 Treatment-emergent Adverse Event

A treatment-emergent AE in this study is any AE or worsening of an existing disease that occurs after the first dose of CTO1681 through the end of treatment plus 28 days, or any event that was subsequently considered drug-related by the Investigator through the end of the study. This period is also the DLT observation period, defined as 43 days following the first dose of study drug (Day -1) which is 42 days following initiation of CAR T-cell therapy (Day 0), ie, Day +41.

6.10.15.3 Suspected Adverse Reaction

A suspected adverse reaction is any AE for which there is a reasonable possibility that the drug caused the AE. For the purposes of expedited safety reporting, “reasonable possibility” means there is evidence to suggest a causal relationship between the drug and the AE. A suspected adverse reaction implies a lesser degree of certainty about causality than an adverse reaction. Adverse reaction means any adverse event caused by a drug. Adverse reactions are a subset of all suspected adverse reactions where there is reason to conclude that the drug caused the event.

6.10.15.4 Unexpected Adverse Event

An unexpected AE is an AE for which the nature/description, specificity or severity of the event is not consistent with the applicable product information (eg, the Investigator’s Brochure [IB] and/or the IND).

6.10.15.5 Serious Adverse Event

An SAE is an AE occurring after informed consent is given until the end-of study visit (unless assessed as treatment-related by the Investigator; treatment-related SAEs should be reported irrespective of time since last dose), whether the patient is receiving investigational product or placebo, that results in 1 or more of the following:

- Death
- Is life-threatening
- Inpatient hospitalization or prolongation of existing hospitalization
- Persistent or significant disability or incapacity

- Congenital abnormality or birth defect

An important medical event that may not result in death, be life-threatening, or require hospitalization, may be considered an SAE when, based on appropriate medical judgement, the event may jeopardize the patient and may require medical or surgical intervention to prevent 1 of the outcomes listed in the definition of an SAE. Death is an outcome of an SAE and not an SAE itself. When death is an outcome, the underlying medical diagnosis or suspected diagnosis that contributed to the fatal outcome should be reported as the SAE term (eg, “pulmonary embolism”). Generally, only 1 such event should be reported. If the cause of death is unknown, then “Death, unknown cause” will be reported as the SAE term. If the cause of death later becomes available (eg, after autopsy), “Death, unknown cause” should be replaced by the established cause of death.

An AE is life-threatening if the patient was at immediate risk of death from the event as it occurred (ie, it does not include a reaction that, if it had occurred in a more serious form, might have caused death). For example, drug-induced hepatitis that resolved without evidence of hepatic failure would not be considered life-threatening even though drug-induced hepatitis can be fatal.

Adverse events requiring hospital admission should be considered SAEs. Hospitalization for elective surgery or routine clinical procedures that are not the result of an AE (eg, elective surgery for a pre-existing condition that has not worsened) need not be considered AEs or SAEs. If anything untoward is reported during the procedure, that occurrence must be reported as an AE, either “serious” or “nonserious” according to the usual criteria.

Events that require an escalation of care when the patient is already hospitalized should be recorded as a SAE. Examples of such events include movement from routine care in the hospital to the ICU or if that event resulted in a prolongation of the existing planned hospitalization.

An SAE is incapacitating or disabling if the experience results in a substantial and/or permanent disruption of the patient’s ability to carry out normal life functions.

All SAEs that occur after informed consent is given, before treatment, during treatment, or before the end-of-study visit (Month 6 for Phase 1b, Month 12 for Phase 2a), whether or not they are related to the study, must be recorded on the appropriate eCRF page or on the SAE Report Form. Serious AEs assessed by the Investigator as related to study drug should be reported, even after the end-of-study visit.

6.10.16 Procedures for Reporting Adverse Events and Serious Adverse Events

The Investigator is responsible for ensuring each patient is carefully monitored for the development of any AEs. This information should be obtained in the form of non-leading questions (eg, “How are you feeling?”) and from signs and symptoms detected during each examination, observations of study personnel, and spontaneous reports from patients.

After informed consent has been obtained but prior to initiation of study drug, only SAEs caused by a protocol-mandated intervention should be reported. After initiation of study drug, all AEs and SAEs should be reported until the end of study visit (Month 6 for Phase 1b, Month 12 for Phase 2a) or until study discontinuation/termination. All AEs (serious and nonserious)

spontaneously reported by the patient and/or in response to an open question from study personnel or revealed by observation, physical examination, or other diagnostic procedures will be recorded on the AE eCRF. Any clinically relevant deterioration in laboratory assessments or other clinical findings is considered an AE and must be recorded on the AE eCRF. When possible, signs and symptoms indicating a common underlying pathology should be noted as one comprehensive event rather than as individual events.

All SAEs are to be reported immediately (no later than within 24 hours of awareness) by the Investigator to the Sponsor's designee for Pharmacovigilance and/ or Medical Affairs. Contact information for the representative will be provided in a separate Safety Management Plan. All SAEs must be reported whether or not they are considered causally related to CTO1681. Serious AEs will be reported using the eCRF or on a paper/PDF SAE report form, and the information collected will include, at minimum: the patient number, event term(s), severity, and causality (ie, relationship to study drug) assessments by the Investigator and a narrative description of the event. Follow-up information regarding the SAE may be requested by the Sponsor or its designated Pharmacovigilance Manager and/or Medical Monitor.

If there are serious and unexpected adverse drug reactions associated with the use of CTO1681, the Sponsor or its designee will notify the appropriate regulatory agency(ies) and all participating Investigators on an expedited basis. The local Institutional Review Board (IRB)/Independent Ethics Committee (IEC) will be promptly notified based on local regulations where required by the IRB/IEC of all serious, unexpected adverse drug reactions involving risk to human subjects.

All AEs, whether serious or not, will be described in the source documents and on the AE page of the eCRF. All new events, as well as those that worsen in severity or frequency relative to baseline, which occur after signing the informed consent through 28 days following the last dose of study drug (ie, Day +41), must be recorded. Adverse events that are ongoing at the time of treatment discontinuation should be followed through the end-of-study visit. Serious AEs judged by the Investigator to be related to CTO1681 must be reported any time the Investigator becomes aware of such an event, even if this occurrence is after the last dose of study drug.

Information to be reported in the description of each AE includes:

- A medical diagnosis of the event (if a medical diagnosis cannot be determined, a description of each sign or symptom characterizing the event should be recorded)
- The date of onset of the event
- The date of resolution of the event
- Whether the event is serious or not
- Severity of the event (see below for definitions)
- Relationship of the event to study treatment (see below for definitions)
- Action taken
- Outcome

Severity of all AEs, including clinically significant treatment-emergent laboratory abnormalities, will be graded according to the National Cancer Institute (NCI) CTCAE, version 5.0 ([Appendix 10.4](#)). Adverse events not listed in the NCI CTCAE will be graded as follows:

- **Grade 1: Mild.** The event is noticeable to the patient but does not interfere with routine activity.
- **Grade 2: Moderate.** The event interferes with routine activity but responds to symptomatic therapy or rest.
- **Grade 3: Severe.** The event significantly limits the patient's ability to perform routine activities despite symptomatic therapy.
- **Grade 4: Life-threatening.** An event in which the patient was at risk of death at the time of the event.
- **Grade 5: Fatal.** An event that results in the death of the patient.

Relationship to study drug administration will be determined by the Investigator based on clinical judgement and according to the following criteria:

- **Not related:** An AE will be considered “not related” to the use of the study drug if there is no plausible temporal or causal relationship between the study drug and AE (eg, exposure to the study treatment did not occur, or the occurrence of the AE is not reasonably related in time, or the AE is considered unlikely to be related to the study treatment).
- **Unlikely related:** An AE will be considered “unlikely related” to the use of the study drug if there is an improbable but not impossible temporal or causal relationship between the study drug and the AE. Disease or other drugs provide plausible explanations.
- **Possibly related:** An AE will be considered “possibly related” to the use of the study drug if there is a reasonable possibility of a temporal association to the study drug and AE that follows a suspected pattern of response, but other etiologies are likely to be the cause or an alternative cause may be present. However, involvement of the trial treatment cannot be excluded, based on the current information.
- **Probably related:** An AE will be considered “probably related” to the use of the study drug if there is a reasonable probability of a temporal relationship to the study drug and AE that follows a known or suspected pattern of response; other etiologies are possible but less likely to be the cause, or an alternative cause may be present based on the current information.
- **Definitely related:** An AE will be considered “definitely related” to the use of the study drug if there is a reasonable possibility of a temporal and causal relationship between the study drug and AE that follows a known pattern of response and no alternative cause is present. “Reasonable possibility” means there is evidence to suggest a causal relationship between the study drug and the AE.

For the safety analyses, all AEs that are classified as related will be considered treatment-related AEs.

Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to the study drug treatment administration, will be considered and investigated.

The Investigator may change their opinion of causality in light of follow-up information.

6.10.17 Follow-up of Adverse Events

After the initial AE/SAE report, the Investigator is required to proactively follow each patient at subsequent visits/contacts. All SAEs will be followed until resolution (ie, return to baseline state of health), stabilization (ie, Investigator does not expect any further improvement or worsening of patient's condition), the patient is lost to follow-up, or the patient has died.

6.10.18 Regulatory Reporting Requirements for Serious Adverse Events

Prompt notification of SAEs by the Investigator to the Sponsor is essential so that legal obligations and ethical responsibilities towards the safety of patients and the safety of a study intervention under clinical investigation are met.

The Sponsor has a legal responsibility to notify regulatory agencies about the safety of a study intervention under investigation. The Sponsor will comply with regulatory requirements relating to the regulatory authority, IRB/IEC, and Investigators.

Investigator safety reports must be prepared for suspected unexpected serious adverse reactions according to local regulatory requirements and Sponsor policy and forwarded to the Investigators, as necessary.

An investigator who receives an investigator safety report describing an SAE or other specific safety information from the Sponsor will review and file it along with the IB, and will notify the IRB/IEC, as appropriate, according to local requirements.

6.10.18.1 Pregnancy

Pregnancy will not be recorded as an AE or SAE, unless a complication relating to the pregnancy occurs (eg, spontaneous abortion, which may qualify as an SAE). However, any pregnancy in a participating female patient or partner of a male patient that occurs during this study or within 4 weeks after the last dose of CTO1681 must be reported in the eCRF or on a Pregnancy Notification Form within 24 hours of being notified of the pregnancy (see separate Safety Management Plan). The Investigator must follow up and document the course and outcome of all pregnancies even if the patient is discontinued from the study or if the study has finished. The female patient or partner of a male patient should receive any necessary counseling regarding the risks of continuing the pregnancy and the possible effects on the fetus. Monitoring should continue until conclusion of the pregnancy.

All outcomes of pregnancy must be reported by the Investigator to the Sponsor or its designated Pharmacovigilance/Medical Affairs representative in the eCRF or on a Pregnancy Notification Form within 30 days after he/she has gained knowledge of the delivery or elective abortion.

Any SAE that occurs during pregnancy must be recorded on the SAE report form (eg, maternal serious complications, spontaneous or therapeutic abortion, ectopic pregnancy, stillbirth,

neonatal death, congenital anomaly, or birth defect) and reported within 24 hours in accordance with the procedure for reporting SAEs ([Section 6.10.16](#)).

All patients, male and female, must agree to use Investigator-approved contraception from time of enrollment until 28 days after the last dose of CTO1681 (ie, Day +41).

7. STATISTICAL PROCEDURES

7.1 Sample Size Estimation

Phase 1b

For the Phase 1b portion of the study, the maximum sample size of 54 (maximum 18 per dose level) was selected to support the dose escalation and RP2D selection of CTO1681 dose level in accordance with the rolling 6 dose escalation procedure and backfill expansion to ensure sufficient safety data for Phase 2a dose selection.

Phase 2a

The Phase 2a portion of the study was designed with sample size selected to provide sufficient power for the primary endpoint.

With a sample size of 100 patients (50 per group), there is 80% power with a 1-sided type I error rate of 0.05 to detect a difference in proportion of 40% of control patients experiencing Grade ≥ 2 CRS versus 16% of CTO1681 patients.

Additional patients may be enrolled to account for early withdrawal in Phase 2a.

7.2 Populations for Analysis

The following analysis populations will be used for presentation of the data:

- Modified Intent-to-treat (mITT) Population: The mITT Population will include all patients who receive both CAR T-cell therapy and at least 1 dose of study drug (CTO1681 or placebo [Phase 2a only]). Patients will be grouped by dose group for Phase 1b analyses and as randomized for Phase 2a analyses.
- Safety Population: The Safety Population will include all patients who receive at least 1 dose of study drug (CTO1681 or placebo [Phase 2a only]). Patients will be grouped as treated.
- Dose Limiting Toxicity Population: The DLT Population will include all patients in Phase 1b who receive both CAR T-cell therapy and at least 1 dose of CTO1681. Patients who withdraw before experiencing a DLT and before completing the DLT observation period (through Day +41, 43 days after the first dose of CTO1681 or placebo [Phase 2a only]) for reasons unrelated to study procedures, conduct, or treatment, will not be included in this population.

Additional analysis populations may be defined in the statistical analysis plan (SAP). A standard intent-to-treat population, including all randomized patients, may be analyzed as an evaluation of mITT robustness.

7.3 Procedures for Handling Missing, Unused, and Spurious Data

All available data will be included in data listings and tabulations. Data that are potentially spurious or erroneous will be examined according to standard data management operating procedures.

In general, missing data will be treated as missing and no data imputation will be applied, unless otherwise specified.

Further details of missing data handling/censoring rules will be documented in the SAP.

7.4 Interim Analyses

An end-of-Phase 1b analysis will be performed once all subjects in the last cohort have either discontinued from the study or completed the DLT evaluation period (ie, Day +41, 43 days after the first dose of CTO1681).

An end-of-Phase 2a analysis will be performed on completion of Phase 2a once all subjects have either discontinued from the study or completed the DLT evaluation period (ie, Day +41, 43 days after the first dose of CTO1681).

7.5 Statistical Methods

7.5.1 General Methods

Demographic, safety, PK, PD, and efficacy parameters will be summarized by phase of study, presented overall and by dose group for Phase 1b, and overall and by treatment group for Phase 2a, and will be supported by graphical presentations, as appropriate. Descriptive summaries will include the number of patients (n), mean, standard deviation, minimum, and maximum for continuous parameters, and frequency and percentage for categorical parameters. Longitudinal data will be presented by appropriate time intervals depending on the endpoint being summarized. Time-to-event endpoints will be analyzed using Kaplan-Meier methods and summarized with quartiles (25th, 50th [medial], and 75th percentiles). Confidence intervals will be presented, as appropriate.

All data will be provided in by-patient listings.

- **Phase 1b:** In keeping with the exploratory nature of a Phase 1b study, no inferential hypothesis testing will be performed. All endpoint summaries will be presented with descriptive statistics. Tabulations will be summarized by dose level and overall. Summaries will be performed on the mITT, Safety, and DLT Populations.
- **Phase 2a:** Primary and secondary endpoint analyses will be performed on the mITT populations. Analysis of safety parameters will be performed on the Safety Population.

All statistical analyses will be described in a detailed SAP that will be executed prior to the unblinding of the Phase 2a data. Analysis of the Phase 1b portion of the study occurring before the SAP finalization will be described in a separate Phase 1b/interim analysis plan. In the event of discrepancies between the protocol and the SAP, the SAP will be used to determine correct analyses and methodologies.

7.5.2 Disposition of Patients

A tabulation of the disposition of patients will be presented, including the number screened, number treated in Phase 1b, number randomized in Phase 2a, number treated, number in Phase 2a, number in each analysis population by phase, number of patients who completed treatment by phase, and number of patients who completed the study by phase. The reasons for

treatment and study discontinuation will be reported. Entry criteria and protocol deviations will be listed.

7.5.3 Baseline Evaluations

Demographic and baseline disease characteristics data will be summarized using descriptive statistics. Data to be tabulated will include sex, age, race, ethnicity, height, weight, and body mass index, as well as disease-specific information.

7.5.4 Efficacy Analysis

The primary Phase 2a endpoint, incidence of Grade ≥ 2 CRS, will be summarized as the frequency and proportion of patients with the event and compared between treatment groups using the Fisher's exact test assessed at a 1-sided type I error rate of 0.05. Exact 90% CI will be presented.

Incidence of CRS (any grade), Grade ≥ 3 CRS, ICANS (any grade), Grade ≥ 2 ICANS, and Grade ≥ 3 ICANS will be summarized in a similar manner to Grade ≥ 2 CRS. Comparisons between treatment groups will be made for exploratory purposes using Fisher's exact test. Ordinal variables, including maximum severity of CRS, will be analyzed using a Wilcoxon-Mann-Whitney test, as appropriate.

Time to CRS resolution and time to ICANS resolution will be analyzed using Kaplan-Meier methods and compared between treatment groups using a logrank test.

Frequency and duration of hospitalization and ICU admissions due to CRS or ICANS, and use of tocilizumab, steroids, and anti-cytokine therapy for treatment of CRS or ICANS will be summarized using descriptive statistics.

7.5.5 Pharmacokinetic Analysis

Descriptive statistics (ie, number of patients, mean, standard deviation, geometric mean, coefficient of variation, median, minimum, and maximum) will be used to summarize PK concentration parameters (ie, maximum serum concentration [$C_{\max}$], time to $C_{\max}$ [$t_{\max}$], area under the curve [AUC], and half-life [$t_{1/2}$]) through noncompartmental analysis for patients in the Phase 1b and Phase 2a portions of the study. The relationships between dose and both $C_{\max}$ and AUC will be explored graphically for dose-proportionality.

7.5.6 Biomarker and Immune Reconstitution Analysis

The objectives, assessments and analysis criteria of biomarkers and reconstitution analysis are defined below:

Objectives	Assessment	Analysis
Levels and persistence of CAR T-cells	CAR copy number, anti-CD19 CAR T, CD19	Level of CAR+ T-cells in serum post CAR T-cell infusion, maximum CAR+ T-cell level attained, time to maximum level (peak), time at which there were no detectable CAR+ T-cells, AUC of CAR T-cell expansion from Day 0 to Day +30

Immune reconstitution	IgG antibodies	Serum IgG levels over time (baseline to last assessment) eg, < 400 mg/dL, > 400 mg/dL
Biomarkers relating to MOA of CTO1681 and immune response	Cytokines, ferritin, CRP, TBNK panel, immunophenotyping	Cytokines, ferritin, CRP, TBNK levels over time (baseline to last assessment), change in immunophenotype/peripheral blood molecular characterization, composition of T-cells, levels of normal B cells over time

Abbreviations: AUC = area under the curve; CAR = chimeric antigen receptor; CRP = c-reactive protein; IgG = immunoglobulin G; MOA = mode of action; TBNK = lymphocyte subsets enumeration

7.5.7 Safety Analysis

7.5.7.1 Safety Parameters

Analysis of safety parameters will include summary of AEs, clinical laboratory measurements, vital signs, ECG results, and concomitant medications.

Adverse events will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) for purposes of summarization. All AEs occurring during the study will be included in by-patient data listings and tabulated by MedDRA system organ class and preferred term. Adverse events will be summarized overall, by relationship and by severity. Events leading to death, SAEs, and events resulting in treatment discontinuation will be listed and, if warranted by the data, tabulated.

Complete response rate and ORR (best responses of CR or PR) will be summarized as the frequency and proportion. Clopper-Pearson exact 95% CI will be reported. Duration of response, PFS, and OS will be summarized using Kaplan-Meier methods. Median and 95% CI for DOR, PFS, and OS will be presented. Censoring rules will be detailed in the SAP.

Exposure to study drug and study drug compliance will be tabulated.

The actual value and change from baseline to each on-study evaluation will be summarized for vital sign measurements and laboratory test results. Clinically significant laboratory abnormalities will be listed.

Twelve-lead ECG results will be summarized descriptively, including the number and percentage with normal, abnormal, and clinically significant abnormal results at Baseline and each on-study evaluation. Clinically significant or abnormal changes in QTc intervals will be reported as AEs. Additionally, actual values and changes from baseline in PR, QRS, HR, QT, and QTcF intervals will be summarized for each dose group in Phase 1b and placebo-corrected for subjects in Phase 2a. If data allow, concentration-QTc analysis may also be undertaken to exclude under safety conditions that CTO1681 causes a QTc effect > 10 ms. The number and frequency of QTcF prolongation will be presented along with prolonged changes from baseline in QTcF intervals.

The use of concomitant medications, coded using World Health Organization Drug Dictionary, will be summarized, and included in by-patient listing.

Further detail will be provided in the SAP.

7.5.7.2 Justification for the Rolling 6 Design in Determining Dose Escalation

In accordance with FDA input from PIND interactions, it was determined that a Rolling 6 dose escalation procedure may be implemented with appropriate safety monitoring and decision rules for cohort sizes of up to 6 and a target toxicity of 25% while providing simplicity in its implementation.

Similar to the traditional 3 + 3 design, the Rolling 6 Design has been utilized in many dose-finding studies, including Phase I pediatric oncology trials, as an approach that both decreases suspension of study enrollment and the overall duration of the trial. The Rolling 6 Design is an algorithm-based extension of the 3 + 3 design. Patients can be continuously enrolled at the same dose level, starting with two up to a total of six. The dose level allocated to a new patient is based on the number of patients currently enrolled, the number of patients experiencing DLT and the number of patients whose evaluation is pending at the time of new patient entry. De-escalation occurs when two or more DLTs occur at a dose level, whereas escalation can be performed when 3/3, 4/4, 5/5 or $\geq 5/6$ patients are evaluated without DLT. Once six patients have been enrolled at the current dose level, enrollment will stop until at least five of those six patients have completed the evaluation. The rolling 6 design can reduce the total enrollment time by allowing continuous enrollment of *at least 3* and *up to 6* patients. A minimum of 3 patients must complete the DLT observation period (through Day +41) prior to data review for dose escalation.

Simulations were performed to assess the operating characteristics of the Rolling 6 Design as planned for the CTA-2101 Study. The operating characteristics of the Rolling 6 Design are included in Table 5. Each row represents different scenarios of the true toxicity rates for each of the three dose levels being evaluated. The probability that each dose level will be selected as the MTD is provided for each of the different dose-toxicity rate scenarios.

For example, the first row represents the scenario where Dose Level 1, Dose Level 2, and Dose Level 3 have toxicity rates of 5%, 15%, and 25%, respectively. In that scenario, there is 77.3% chance that Dose Level 3 will be chosen as the MTD when the targeted DLT rate is 25%.

Table 5: Operating Characteristics of the Rolling 6 Design

True Toxicity Rate			Chances the Dose Level is MTD Dose When the Targeted DLT Rate is 25%		
Dose Level 1 Toxicity	Dose Level 2 Toxicity	Dose Level 3 Toxicity	Dose Level 1	Dose Level 2	Dose Level 3
5%	15%	25%	2.8	19.9	77.3
10%	20%	30%	8.2	24.4	67.4
15%	25%	35%	18.9	34.3	46.8
20%	30%	40%	29	34.1	36.9

True Toxicity Rate			Chances the Dose Level is MTD Dose When the Targeted DLT Rate is 25%		
Dose Level 1 Toxicity	Dose Level 2 Toxicity	Dose Level 3 Toxicity	Dose Level 1	Dose Level 2	Dose Level 3
25%	35%	45%	38.4	36.6	25

Abbreviations: MTD = maximum tolerated dose; N = sample size

^a Indicates the maximum toxicity allowed in the study.

Moreover, to safeguard the safety of patients going from the 2nd to the 3rd dose level where potentially higher toxicity may be anticipated, the SRC and Sponsor safety data reviews will be carried out as well as continually monitoring adverse events throughout the study. Furthermore, the Medical Monitor for the study will be reviewing all adverse events in real time with the Investigator(s) to assess safety on an ongoing basis from the first patient onward and will inform the SRC of events in real time. An ad-hoc SRC meeting may be requested by the Medical Monitor and/or the Investigator, and convened at any time during the study, to support close safety monitoring of the study patients.

As mentioned above, the rolling 6 design is utilized in many prospective clinical trials when entering a population, including oncology populations. Simulation work shows that the rolling 6 design is not associated with an increased risk in toxicity over other models, including the traditional 3+3 design ([Skolnik, Barrett et al. 2008](#)). This proposed design, and the associated safety monitoring plan using the Medical Monitor and SRC oversight, is anticipated to allow a reasonable study timeline with maintaining strict requirements for toxicity oversight and management. If excessive toxicity is observed at any dose in the study, the dosing may need to be revised, and data continue to be scrutinized during backfill of each cohort for a total of 18 patients per cohort, pending DLT requirements, in the same manner as prior for monitoring of safety and management of safety dosing (see Criteria for Pausing or Stopping the Study, [Section 5.11](#)).

7.5.8 Procedures for Reporting Deviations to Original Statistical Analysis Plan

All deviations from the original version of the SAP will be provided in the final clinical study report.

8. ADMINISTRATIVE REQUIREMENTS

8.1 Good Clinical Practice

The study will be conducted in accordance with the International Council on Harmonisation (ICH) for Good Clinical Practice (GCP) guidelines and the appropriate regulatory requirement(s). The Investigator will be thoroughly familiar with the appropriate use of the study drug as described in the protocol and IB. Essential clinical documents will be maintained to demonstrate the validity of the study and the integrity of the data collected. Master files should be established at the beginning of the study, maintained for the duration of the study, and retained according to the appropriate regulations.

8.2 Ethics

8.2.1 Institutional Review Board/Independent Ethics Committee

The study will be conducted in accordance with ethical principles founded in the Declaration of Helsinki and the Belmont Report.

Institutional Review Board (IRB)/Independent Ethics Committee (IEC) approval must be obtained and submitted in writing to the Sponsor before patient enrollment into the study may begin. The IRB/IEC will review all appropriate study documentation in order to safeguard the rights, safety, and wellbeing of the patients. The protocol, IB, informed consent, advertisements (if applicable), written information given to the patients, safety updates, annual progress reports, and any revisions to these documents will be provided to the IRB/IEC. The IRB/IEC is to be notified of any amendment to the protocol in accordance with local requirements. Progress reports and notifications of serious unexpected adverse drug reactions will be provided to the IRB/IEC according to local regulations and guidelines.

8.3 Patient Information and Informed Consent

The Investigator at each center will ensure the patient is given full and adequate oral and written information about the nature, purpose, possible risk, and benefit of the study. Patients must also be notified that they are free to discontinue from the study at any time. The patient should be given the opportunity to ask questions and allowed time to consider the information provided.

After the study has been fully explained, written informed consent will be obtained from either the patient or his/her guardian or legal representative before study participation.

The patient's signed and dated informed consent must be obtained before conducting any study-related procedures. The Investigator must maintain the original, signed consent form. A copy of the signed form must be given to the patient.

The method of obtaining and documenting the informed consent and the contents of the consent will comply with ICH-GCP and all applicable regulatory requirement(s).

8.4 Privacy and Confidentiality

In order to maintain patient privacy, all source documents/eCRFs, study drug accountability records, study reports and communications will identify the patient by initials and the assigned

patient number. The Investigator will grant monitor(s) and auditor(s) from the Sponsor or its designee and regulatory authority(ies) direct access to the patient's original medical records for verification of data gathered on the source documents/eCRFs and to audit the data collection process. The patient's confidentiality will be maintained and will not be made publicly available to the extent permitted by the applicable laws and regulations.

8.5 Protocol Compliance

The Investigator will conduct the study in compliance with the protocol. Modifications to the protocol should not be made without agreement of both the Investigator and the Sponsor except as defined below. Changes to the protocol will require written IRB/IEC approval/favorable opinion prior to implementation, except when the modification is needed to eliminate an immediate hazard(s) to patients. The IRB/IEC may provide, if applicable, where regulatory authority(ies) permit, expedited review and approval/favorable opinion for minor change(s) in ongoing studies that have the approval/favorable opinion of the IRB/IEC. The Sponsor or designee will submit all protocol modifications to the regulatory authority(ies) in accordance with the governing regulations.

When immediate deviation from the protocol is required to eliminate an immediate hazard(s) to patients, the Investigator will contact the Sponsor or Medical Monitor, if circumstances permit, to discuss the planned course of action. Any departures from the protocol must be fully documented in the source documents/eCRF.

8.6 Data Management

All data for the patients recruited for the study will be promptly entered into the eCRFs via an Electronic Data Capture system provided by the Sponsor or designee. Only authorized site staff may enter data onto the eCRFs. If an entry error is made, the corrections to the eCRFs will be made according to eCRF guidelines by an authorized member of the site staff.

Electronic case report forms will be checked for correctness against source document data by the Sponsor's monitor. If any entries into the eCRF are incorrect or incomplete, the monitor will ask the Investigator or the study site staff to make appropriate corrections, and the corrected eCRF will again be reviewed for completeness and consistency. Any discrepancies will be noted in the eCRF system by means of electronic data queries. Authorized site staff will be asked to respond to all electronic queries according to the eCRF guidelines.

8.6.1 Source Data/Electronic Case Report Form Completion

Source documents/eCRFs will be completed for each study patient. It is the Investigator's responsibility to ensure the accuracy, completeness, and timeliness of the data reported in the patient's source document/eCRF. The source document/eCRF should indicate the patient's participation in the study and should document the dates and details of study procedures, AEs and patient status.

The Investigator, or designated representative, should complete the source document/eCRF as soon as possible after information is collected, preferably on the same day that a study patient is seen for an examination, treatment, or any other study procedure. Any outstanding entries must

be completed immediately after the final examination. An explanation should be given for all missing data.

The Investigator must sign and date the Investigator's Statement at the end of the source document/eCRF to endorse the recorded data.

8.6.2 Patient Diary

The patient dosing diary ([Section 5.2.1](#)) will be brought by the patient to each study visit so review of compliance with study drug can be completed and for entry of required data by study staff into the eCRF. Upon study completion, all patient dosing diaries will be kept on site.

8.7 Direct Access to Source Data

The study will be monitored by the Sponsor or designee. Monitoring will be done by personal and/or remote visits from a representative of the Sponsor (site monitor) and will include on-site and/or electronic review of the source documents/eCRFs for completeness and clarity, cross-checking with source documents, and clarification of administrative matters will be performed. The review of medical records will be performed in a manner to ensure that patient confidentiality is maintained.

The site monitor will ensure that the investigation is conducted according to protocol design and regulatory requirements by frequent communications (letter, telephone, e-mail, and fax).

All unused study drug and other study materials are to be returned to the Sponsor or designee and/or appropriately destroyed/disposed of after the study has been completed.

Regulatory authorities, the IEC/IRB, and/or the Sponsor or designee's clinical quality assurance group may request access to all source documents, eCRFs, and other study documentation for an on-site audit or inspection. Direct access to these documents must be guaranteed by the Investigator, who must provide support at all times for these activities.

8.8 Record Retention

The Investigator will maintain all study records according to ICH-GCP and applicable regulatory requirement(s). Records will be retained for at least 2 years after the last marketing application approval or 2 years after formal discontinuation of the clinical development of the investigational product or according to applicable regulatory requirement(s). If the Investigator withdraws from the responsibility of keeping the study records, custody must be transferred to a person willing to accept the responsibility. The Sponsor must be notified in writing if a custodial change occurs.

8.9 Liability and Insurance

The Sponsor has subscribed to an insurance policy covering, in its terms and provisions, its legal liability for injuries caused to participating persons and arising out of this research performed strictly in accordance with the scientific protocol as well as with applicable law and professional standards.

8.10 Publication of Study Findings and Use of Information

All information regarding CTO1681 supplied by the Sponsor or designee to the Investigator is privileged and confidential information. The Investigator agrees to use this information to accomplish the study and will not use it for other purposes without consent from the Sponsor. It is understood that there is an obligation to provide the Sponsor with complete data obtained during the study. The information obtained from the clinical study will be used towards the development of CTO1681 and may be disclosed to regulatory authority(ies), other Investigators, corporate partners, other Sponsor partners (eg, investigators and auditors), or consultants as required.

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10. APPENDICES

10.1 Schedule of Events Phase 1b

Table 6: Schedule of Events: Phase 1b

	Treatment Period ^a																Follow-up Period						
Activity	Screenin g (Post Leuka- pheresis)	LD Chemo - therap y	Base- line D -1	D0-D+13														FU1 b	FU2	M1	FU3	M3	M6 EOS
																							ET ^b
Study Day	D -21 to D-3	~D -12 to D-3	D -1 ^c	D0	D+1	D+2	D+3	D+4	D+5	D+6	D+7	D+8	D+9	D+1 0	D+1 1	D+1 2	D+1 3	D+14	D+20	D+30	D+41	D+9 0	D+180
Visit Window (days)	-7D	SOC	-24 h ^d	±1 for D0-D+13														±1	±1	±2	±2	±3	± 7
Informed consent	X																						
Inclusion/ exclusion criteria	X		X																				
Enrollment ^e	X																						
Demographics	X																						
Height (without shoes)	X																						
Weight ^f	X		X															X		X	X	X	X
Medical/ medication history	X		X																				
Urine pregnancy testing (if applicable)	X																						
Physical examination ^g	X		X	X						X								X	X	X	X	X	X
ECOG ^h	X		X																				X
ECG (12-lead) ⁱ	X		X	X	X	X	X	X	X	X		X		X		X	X						
Vital signs ^{k, l, m}	X	X ^j	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2D echocardiogram ⁿ	X																						
CBC with differentials ^o	X	X	X	X		X		X		X		X		X		X	X			X	X	X	X
Chemistry panel ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X

	Treatment Period ^a																Follow-up Period						
Activity	Screenin g (Post Leuka- pheresis)	LD Chemo - therap y	Base- line D -1	D0-D+13														FU1 b	FU2	M1	FU3	M3	M6 EOS
																							ET ^b
Study Day	D -21 to D-3	~D -12 to D-3	D -1 ^c	D0	D+1	D+2	D+3	D+4	D+5	D+6	D+7	D+8	D+9	D+10	D+11	D+12	D+13	D+14	D+20	D+30	D+41	D+90	D+180
Visit Window (days)	-7D	SOC	-24 h ^d	±1 for D0-D+13														±1	±1	±2	±2	±3	± 7
Urinalysis	X		X																				
PT/PTT/INR ^o	X	X	X	X		X		X		X		X		X		X	X			X	X	X	X
C-reactive protein ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X
TBNK panel	X			X																X		X	X
IgG antibodies	X		X	X		X		X		X		X		X		X	X			X	X	X	X
IFN-γ, TNF-α, IL-1, IL-2, IL-6, and GM-CSF ^p	X	X ^q	X	X	X	X	X	X	X	X		X		X			X	X		X		X	X
Ferritin ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X
Anti-CD19 CAR T-cells				X						X							X			X		X	X
CAR copy number ^r				X		X		X		X							X			X		X	X
LD chemotherapy ^s		X																					
CTO1681 ^{t,u}			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X						
CAR T-cell infusion				X																			
Disease assessment ^v	X		X																	X	X	X	X
CRS assessments ^w				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ICANS assessment ^x				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
PK blood sample collection ^y			X	X		X		X		X							X						
Blood immunophenotyping (flow cytometry)	X			X						X							X			X		X	X
Dosing compliance			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X						
Concomitant medications				Collected throughout study																			

	Treatment Period ^a																Follow-up Period						
Activity	Screenin g (Post Leuka- pheresis)	LD Chemo - therap y	Base- line D -1	D0-D+13													FU1 b	FU2	M1	FU3	M3	M6 EOS	
																						ET ^b	
Study Day	D -21 to D-3	~D -12 to D-3	D -1 ^c	D0	D+1	D+2	D+3	D+4	D+5	D+6	D+7	D+8	D+9	D+10	D+11	D+12	D+13	D+14	D+20	D+30	D+41	D+90	D+180
Visit Window (days)	-7D	SOC	-24 h ^d	±1 for D0-D+13													±1	±1	±2	±2	±3	±7	
AE monitoring				Ongoing throughout study																			

Abbreviations: 2D = 2-dimensional; AE = adverse event; ASTCT = American Society for Transplantation and Cellular Therapy; CAR = chimeric antigen receptor; CBC = complete blood count; CD = cluster of differentiation; CRP = C-reactive protein; CRS = cytokine release syndrome; CT = computed tomography; D = day; ECG = electrocardiogram; ECOG = Eastern Cooperative Oncology Group; EOS = end-of-study; ET = early termination; FU = follow-up; GM-CSF = granulocyte-macrophage colony-stimulating factor;

ICANS = immune effector cell-associated neurotoxicity syndrome; ICE = immune effector cell-associated encephalopathy; IFN-γ = interferon gamma; h = hours; IgG = immunoglobulin G; IL = interleukin; INR = international normalized ratio; LD = lymphodepleting; M = month; min = minutes;

PK = pharmacokinetic(s); PT = prothrombin time; PTT = partial thromboplastin time; Qx h = every x hours; SOC = standard of care;

SoE = schedule of events; TBNK = lymphocyte subsets enumeration; TID = three times daily; TNF-α = tumor necrosis factor alpha.

- ^a The patient is planned to be treated as inpatient from Baseline/ D-1 to D+6, at minimum, to allow for proper monitoring. During the period of D+7 to D+13, the patient may be discharged in the absence of CRS/ICANS at the discretion of the Investigator with the approval of the Medical Monitor and Sponsor. If the patient is discharged, then they must return to the clinic for a minimum of 2 visits per week, which must include D+13, for AE collection and status assessment per the SoE during the treatment period.
- ^b If treatment is terminated early, the patient should proceed to FU1. In addition, the ET visit should be performed for patients who discontinue from the study prior to completing all study-required procedures.
- ^c The following assessments should be completed predose on D-1: vital signs, ECG, and PK sample collection.
- ^d The following assessments may be completed up to 24 h before the first dose of CTO1681: weight, medical/medication history; safety laboratory testing (urinalysis, CBC, chemistry, PT/PTT/INR, CRP); physical examination; ECOG performance status; sampling for IFN-γ, TNF-α, IL-1, IL-2, IL-6, and GM-CSF; IgG antibodies; ferritin; and disease assessment. Assessments should be repeated only if abnormal, clinically significant or at the Investigator's discretion.
- ^e After obtaining informed consent, inclusion and exclusion criteria will be assessed and eligibility will be confirmed. Enrollment will then occur within 24 h before initiation of LD chemotherapy.
- ^f Body mass index calculation at Screening, D-1, FU1, FU3, M1, M3, and M6.
- ^g Physical examination: A complete physical examination (including complete neurological examination) will be performed at Screening, D-1, D0, D+6, FU1, FU2, and FU3, and at M1, M3, and M6. Additional physical examinations, including neurologic examinations, may be performed as needed and when patients are seen in clinic.
- ^h ECOG performance status to be completed at Screening, D-1, and M6.

- ⁱ The Screening ECG does not need to be repeated if performed within the Screening window (D-21 to D-3) by the study institution. On D-1, ECGs will be performed in triplicate predose at -30 and -15 min (± 5 min), and at 1 h postdose (± 10 min) for the morning doses. For D0 to D+6, and D+8, D+10, D+12 and D+13 ECGs will be recorded in triplicate: predose and 1 h postdose (± 10 min) for the morning doses. Additional ECGs may be performed at the Investigator's discretion. Patients should avoid strenuous physical activity, and rest in a supine position for at least 10 min before each ECG recording. ECGs will be evaluated by the Investigator and centrally.
- ^j Vital signs (ie, systolic and diastolic blood pressure, pulse rate, oral temperature, respiratory rate, and oxygen saturation) must be performed post-LD chemotherapy per SOC.
- ^k Vital signs collection:
 - On D-1: Following hospitalization, vital signs must be collected pre-morning CTO1681 administration (first dose); 30 min (± 10 min) and 1 (± 10 min) and 2 h (± 10 min) post-dose for all 3 doses;
 - On D0: Vital signs to be collected pre-CAR T-cell infusion;
 - On D0 through D+13: pre-morning CTO1681 dose and 2 h (± 15 min) post-morning CTO1681 dose and every Q6-8 h thereafter or more frequently at the Investigator's discretion.
- ^l For D+7 to D+13, if patients are not hospitalized, vital sign monitoring will be performed only during clinic visits irrespective of timing of CTO1681 dose unless otherwise indicated or per Investigator judgement. Vital signs will be measured at least once at all other outpatient visits.
- ^m Blood pressure will be measured after the patient has rested quietly for 10 min in the supine position.
- ⁿ If a 2D echocardiogram has been performed within 60 days before Screening, it need not be repeated. A 2D echocardiogram may be performed during the study at the Investigator's discretion.
- ^o Blood samples must be collected pre-morning CTO1681 dose on D-1, D0, D+2, D+4, D+6, D+8, D+10, D+12 and D+13, and at FU3, M1, M3, and M6. For D+7 to D+13, if patients are not hospitalized, blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose; cytokine and PK draws should be taken as per instructions in footnotes 'P' and 'Y,' respectively.
- ^p Patients should avoid strenuous physical activity for at least 24 h prior to cytokine collection. Blood sampling for cytokines will occur pre- and approximately 1 h post- morning CTO1681 dose on D+1, D+2, D+3, D+4, D+5 and D+6. For Screening, LD, D-1, D0, D+8, D+10 and D+13, FU1, M1, M3, and M6, sampling should occur at approximately the same time of day that the post-morning CTO1681 sampling will occur during the treatment period. Samples should be collected immediately following PK sample collection (where applicable), processed within 1 h of collection, and frozen at -80°C until shipment. See Laboratory Manual for further details.
- ^q Cytokines will be collected 1 day after completion of LD chemotherapy.
- ^r If patients are not hospitalized, CAR copy number will be performed only during clinic visits D0, D+2, D+4, D+6, D+13, M1, M3 and M6, irrespective of timing of CTO1681 dose.
- ^s If patient has a white blood cell count $\leq 1,000$ cells/ μL within 1 week before CAR T-cell infusion, LD is not required and may be performed at the investigator's discretion.
- ^t CTO1681 dosing must be initiated at the Baseline visit (D-1), approximately 24 h (± 1 h) before CAR T-cell infusion.
- ^u CTO1681 will be dosed TID within 30 min after starting to eat a meal or snack; 5 h (± 30 min) preferred between meals on D-1 to D+13.
- ^v Disease assessment by CT scan or positron emission tomography-CT scan. Screening scans are accepted if performed within the 30 days leading up to the Screening visit. Magnetic resonance imaging as additional imaging will be performed only if deemed necessary per Investigator's assessment or institutional guidelines. . If bridging therapy is administered, repeat imaging is required before LD chemotherapy.
- ^w CRS assessment (determination if CRS is present and, if so, assessment of grade) to be conducted per the SoE and as clinically indicated, per ASTCT criteria. For D+7 to D+13, if patients are not hospitalized, CRS assessment will be performed only during clinic visits irrespective of timing of CTO1681 dose.

- ^x ICANS assessment (determination if ICANS is present and, if so, assessment of grade), to include ICE score to be conducted per the SoE and as clinically indicated, per ASTCT criteria. For D+7 to D+13, if patients are not hospitalized, ICANS assessment will be performed only during clinic visits irrespective of timing of CTO1681 dose.
- ^y PK blood sample collection:
 - On D–1 PK samples to be collected at time = 0 (before eating and within 1 h prior to the morning dose of CTO1681) and 0.5 ($\pm$ 10 min), 1 ($\pm$ 10 min), 1.5 ($\pm$ 10 min), 2 ($\pm$ 10 min), 3 ($\pm$ 10 min), and 4 ($\pm$ 10 min) h after the first (morning) dose of CTO1681.
 - On D–1 PK sampling is to occur at 0.5 h ($\pm$ 10 min) predose and 1 h postdose for the second and third doses of CTO1681.
 - On D0, D+2, D+4, D+6, and D+13, PK sampling will occur 1 h ($\pm$ 10 min) after the first morning dose of CTO1681.Where samples occur at the same nominal time as ECG assessments (1 h postdose; D–1, D0, D+2, D+4, D+6, and D+13), PK samples should take place immediately following the ECG assessment. PK samples should be processed within 1 h of collection and frozen at -80°C until shipment.

10.2 Schedule of Events Phase 2a

Table 7: Schedule of Events: Phase 2a

	Treatment Period ^a																	Follow-up Period						
Activity	Screenin g (Post Leuka- pheresis)	LD Chemo - therap y	Base- line D -1	D0-D+13														FU1 ^b	FU2	M1	FU3	M3	M 6	M12 EOS
				ET ^b																				
Study Day	D -21 to D -3	~D -12 to D -3	D -1 ^c	D0	D+1	D+2	D+3	D+4	D+5	D+6	D+7	D+8	D+9	D +10	D +11	D +12	D +13	D+14	D+20	D+30	D+41	D+9 0	D+18 0	D+360
Visit Window (days)	-7D	SOC	-24 h ^d	±1 for D0-+D13														±1	±1	±2	±2	±3	± 7	± 7
Informed consent	X																							
Inclusion/ exclusion criteria	X		X																					
Enrollment and randomization ^e	X																							
Demographics	X																							
Height (without shoes)	X																							
Weight ^f	X		X															X		X	X	X	X	X
Medical/ medication history	X		X																					
Urine pregnancy testing (if applicable)	X																							
Physical examination ^g	X		X	X						X								X	X	X	X	X	X	X
ECOG ^h	X		X																				X	X
ECG (12-lead) ⁱ	X		X	X	X	X	X	X	X	X		X		X		X	X							
Vital signs ^{k, l, m}	X	X ^j	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2D echocardiogram ⁿ	X																							
CBC with differentials ^o	X	X	X	X		X		X		X		X		X		X	X			X	X	X	X	X
Chemistry panel ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X	X
Urinalysis	X		X																					

	Treatment Period ^a																Follow-up Period							
Activity	Screenin g (Post Leuka- pheresis)	LD Chemo - therap y	Base- line D -1	D0-D+13														FU1 ^b	FU2	M1	FU3	M3	M 6	M12 EOS
				ET ^b																				
Study Day	D -21 to D -3	~D -12 to D -3	D -1 ^c	D0	D+1	D+2	D+3	D+4	D+5	D+6	D+7	D+8	D+9	D +10	D +11	D +12	D +13	D+14	D+20	D+30	D+41	D+9 0	D+18 0	D+360
Visit Window (days)	-7D	SOC	-24 h ^d	±1 for D0-+D13														±1	±1	±2	±2	±3	± 7	± 7
PT/PTT/INR ^o	X	X	X	X		X		X		X		X		X		X	X			X	X	X	X	X
C-reactive protein ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X	X
TBNK panel	X			X																X		X	X	X
IgG antibodies	X		X	X		X		X		X		X		X		X	X			X	X	X	X	X
IFN-γ, TNF-α, IL-1, IL-2, IL-6, and GM-CSF ^p	X	X ^q	X	X	X	X	X	X	X	X		X		X			X	X		X		X	X	X
Ferritin ^o	X		X	X		X		X		X		X		X		X	X			X	X	X	X	X
Anti-CD19 CAR T-cells				X						X							X			X		X	X	X
CAR copy number ^r				X		X		X		X							X			X		X	X	X
LD chemotherapy ^s		X																						
CTO1681/placebo ^{t,u}			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X							
CAR T-cell infusion				X																				
Disease assessment ^v	X		X																	X	X	X	X	X
CRS assessments ^w				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ICANS assessment ^x				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
PK blood sample collection ^y			X	X		X		X		X							X							
Blood immunophenotyping (flow cytometry)	X			X						X							X			X		X	X	X
Dosing compliance			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X							
Concomitant medications				Collected throughout study																				
AE monitoring				Ongoing throughout study																				

Abbreviations: 2D = 2-dimensional; AE = adverse event; ASTCT = American Society for Transplantation and Cellular Therapy; CAR = chimeric antigen receptor; CBC = complete blood count; CD = cluster of differentiation; CRP = C-reactive protein; CRS = cytokine release syndrome; CT = computed tomography; D = day; ECG = electrocardiogram; ECOG = Eastern Cooperative Oncology Group; EOS = end-of-study; ET = early termination; FU = follow-up; GM-CSF = granulocyte-macrophage colony-stimulating factor; h = hours; ICANS = immune effector cell-associated neurotoxicity syndrome; ICE = immune effector cell-associated encephalopathy; IFN- γ = interferon gamma; IgG = immunoglobulin G; IL = interleukin; INR = international normalized ratio; LD = lymphodepleting; M = month; min = minutes; PK = pharmacokinetic(s); PT = prothrombin time; PTT = partial thromboplastin time; Qx h = every x hours; SOC = standard of care; SoE = schedule of events; SRC = safety review committee; TBNK = lymphocyte subsets enumeration; TID = three times daily; TNF- α = tumor necrosis factor alpha.

- ^a The patient is planned to be treated as inpatient from Baseline/D -1 to D+6, at minimum, to allow for proper monitoring. During the period of D+7 to D+13, the patient may be discharged in the absence of CRS/ICANS at the discretion of the Investigator with the approval of the Medical Monitor and Sponsor. If the patient is discharged, then they must return to the clinic for a minimum of 2 visits per week, which must include D+13 for AE collection and status assessment per the SoE during the treatment period.
- ^b If treatment is terminated early, the patient should proceed to FU1. In addition, the ET visit should be performed for patients who discontinue from the study prior to completing all study-required procedures.
- ^c The following assessments should be completed predose on D-1: vital signs, ECG, and PK sample collection.
- ^d The following assessments may be completed up to 24 h before the first dose of CTO1681/placebo: weight, medical/medication history; safety laboratory testing (urinalysis, CBC, chemistry, PT/PTT/INR, CRP); physical examination; ECOG performance status; sampling for IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF; IgG antibodies; ferritin; and disease assessment. Assessments should be repeated only if abnormal, clinically significant or at the Investigator's discretion.
- ^e After obtaining informed consent, inclusion and exclusion criteria will be assessed and eligibility will be confirmed. Enrollment and randomization will then occur within 24 h before initiation of LD chemotherapy.
- ^f Body mass index calculation at Screening, D-1, FU1, FU3, M1, M3, M6, and M12.
- ^g Physical examination: A complete physical examination (including complete neurological examination) will be performed at Screening, D-1, D0, D+6, FU1, FU2, and FU3, and at M1, M3, M6, and M12. Additional physical examinations, including neurologic examinations, may be performed as needed and when patients are seen in clinic.
- ^h ECOG performance status to be completed at Screening, D-1, M6, and M12.
- ⁱ The Screening ECG does not need to be repeated if performed within the Screening window (D-21 to D-3) by the study institution. On D-1, ECGs will be performed and recorded in triplicate predose at -30 and -15 min ($\pm$ 5 min), and at 1 h postdose ($\pm$ 10 min) for the morning doses. For D0 to D+6, and D+8, D+10, D+12 and D+13 ECGs will be recorded in triplicate: predose and 1 h postdose ($\pm$ 10 min) for the morning doses. Additional ECGs may be performed at the Investigator's discretion. Patients should avoid strenuous physical activity, and rest in a supine position for at least 10 min before each ECG recording. ECGs will be evaluated by the Investigator and centrally.
- ^j Vital signs (ie, systolic and diastolic blood pressure, pulse rate, oral temperature, respiratory rate, and oxygen saturation) must be performed post-LD chemotherapy per SOC.
- ^k Vital signs collection:
 - On D-1: Following hospitalization, vital signs must be collected pre-morning CTO1681 administration (first dose); 30 min ($\pm$ 10 min) and 1 ($\pm$ 10 min) and 2 h ($\pm$ 10 min) post-dose for all 3 doses;
 - On D0: Vital signs to be collected pre-CAR T-cell infusion;
 - On D0 through D+13: pre-morning CTO1681 dose and 2 h ($\pm$ 15 min) post-morning CTO1681 dose and every Q6-8 h thereafter or more frequently at the Investigator's discretion.

- ^l For D+7 to D+13, if patients are not hospitalized, vital sign monitoring will be performed only during clinic visits irrespective of timing of CTO1681 dose unless otherwise indicated or per Investigator judgement. Vital signs will be measured at least once at all other outpatient visits.
- ^m Blood pressure will be measured after the patient has rested quietly for 10 min in the supine position.
- ⁿ If a 2D echocardiogram has been performed within 60 days before Screening, it need not be repeated. A 2D echocardiogram may be performed during the study at the Investigator's discretion.
- ^o Blood samples must be collected pre-morning CTO1681/placebo dose on D-1, D0, D+2, D+4, D+6, D+8, D+10, D+12 and D+13, and at FU3, M1, M3, M6, and M12. For D+7 to D+13, if patients are not hospitalized, blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose; cytokine and PK draws should be taken per instructions in footnotes 'P' and 'Y,' respectively.
- ^p Patients should avoid strenuous physical activity for at least 24 h prior to cytokine collection. Blood sampling for cytokines will occur pre- and approximately 1 h post morning CTO1681 dose on D+1, D+2, D+3, D+4, D+5 and D+6. For Screening, LD, D-1, D0, D+8, D+10, and D+13, FU1, M1, M3, M6 and M12, sampling should occur at approximately the same time of day that the post-morning CTO1681 sampling will occur during the treatment period. Samples should be collected immediately following PK sample collection (where applicable), processed within 1 h of collection, and frozen at -80°C until shipment. See Laboratory Manual for further details.
- ^q Cytokines will be collected 1 day after completion of LD chemotherapy.
- ^r If patients are not hospitalized, CAR copy number will be performed only during clinic visits D0, D+2, D+4, D+6, D+13, M1, M3, M6 and M12, irrespective of timing of CTO1681 dose.
- ^s If patient has a white blood cell count $\leq 1,000$ cells/ μ L within 1 week before CAR T-cell infusion, LD is not required and may be performed at the Investigator's discretion.
- ^t CTO1681/placebo dosing must be initiated at the Baseline visit (D-1), approximately 24 h (± 1 h) before CAR T-cell infusion.
- ^u CTO1681/placebo will be dosed TID within 30 min after starting to eat a meal or snack; 5 h (± 30 min) preferred between meals on D-1 to D+13.
- ^v Disease assessment by CT scan or positron emission tomography-CT scan. Screening scans are accepted if performed within the 30 days leading up to the Screening visit. Magnetic resonance imaging as additional imaging will be performed only if deemed necessary per Investigator's assessment or institutional guidelines. If bridging therapy is administered, repeat imaging is required before LD chemotherapy.
- ^w CRS assessment (determination if CRS is present and, if so, assessment of grade) to be conducted per the SoE and as clinically indicated, per ASTCT criteria. For D+7 to D+13, if patients are not hospitalized, CRS assessment will be performed only during clinic visits irrespective of timing of CTO1681 dose.
- ^x ICANS assessment (determination if ICANS is present and, if so, assessment of grade), to include ICE score to be conducted per the SoE and as clinically indicated per ASTCT criteria. For D+7 to D+13, if patients are not hospitalized, ICANS assessment will be performed only during clinic visits irrespective of timing of CTO1681 dose.
- ^y Sparse PK samples in all patients in Phase 2a to perform population pharmacokinetics and exploratory exposure-response analyses. PK blood sample collection should take place approximately 1 h (± 10 min) postdose immediately following ECG assessment (where applicable). PK samples should be processed within 1 h of collection and frozen at -80°C until shipment.

10.3 Assessments by Visit Day

The following guidance is provided only to assist the site. The assessments found in the Schedule of Events ([Appendix 10.1](#) and [Appendix 10.2](#)) should be followed and performed in the order that is the SOC for your institution unless otherwise stated in the protocol.

Concomitant medications and adverse events are to be monitored throughout the study.

10.3.1 Screening Visit (Day –21 to Day –3)

At the screening visit (post leukapheresis but before LD chemotherapy), information will be collected, and subjects will have clinical evaluations as follows:

- Obtain patient informed consent.
- Collect demographics, medical/ medication history.
- Measure height (without shoes) and weight.
- Collect vital signs (supine): BP (systolic and diastolic), pulse rate, oral temperature, respiratory rate, and oxygen saturation (performed post LD chemotherapy per SOC).
- Perform a complete physical examination (including neurological examination).
- Obtain an ECOG performance status.
- Collect a supine 12-Lead ECG: Repeat 3 times consecutively.
- Collect blood samples for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, ferritin, and IgG.
- Collect blood samples for TBNK panel.
- Collect blood samples for blood immunophenotyping.
- Collect urine for urinalysis and pregnancy testing (if applicable).
- Perform 2D echocardiogram: note that this is not required if 2D echocardiogram was obtained within 60 days before screening.
- Collect blood sample for cytokine assay: IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF: Attempts should be made to draw the sample in the morning. The sample should be processed within 1-hour of collection, and frozen at –80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Determine disease assessment by CT scan or positron emission tomography-CT scan. Screening scans are accepted if performed within the 30 days of screening. Magnetic resonance imaging as additional imaging only if deemed necessary per Investigator's assessment or institutional guidelines.
- Determine qualification for the study: ie, assess inclusion/ exclusion criteria.
- If patient is qualified for study participation then they should be enrolled and randomized (Phase 2a only) within 24 hours prior to LD chemotherapy.

- Schedule LD chemotherapy and next study visit.

10.3.2 Lymphodepleting Chemotherapy (Study Days –12 to –3)

On Study Days –12 to –3, the following procedures and evaluations will be performed:

- Collect vital signs (supine): BP (systolic and diastolic), pulse rate, oral temperature, respiratory rate, and oxygen saturation.
- Collect blood samples for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, PT, PTT and INR.
- Conduct lymphodepleting chemotherapy.
- Cytokine blood sample should be collected 1 day after completion of LD chemotherapy. Collect blood sample for cytokine assay: IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF: Attempts should be made to draw the sample in the morning, samples should be processed within 1-hour of collection, and frozen at –80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Schedule date for hospitalization check in and next study visit (Day –1).

10.3.3 Baseline (Day –1)

The following assessments may be completed up to 24 hours before the first dose of CTO1681 (assessments repeated only if abnormal, clinically significant or at the Investigator's discretion):

- Collect medical/ medication history.
- Collect blood for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, ferritin, and IgG.
- Collect blood for cytokine measurements. This may be done the day before or on the same day as the first dose of CTO1681. Blood sample to be drawn 1-hour post morning dose (if same day) or at approximately that same time of day (if done the day before). Blood draw should be done following PK sample collection. Blood samples should be processed within 1-hour of collection, and frozen at –80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Collect urine for urinalysis
- Measure weight (including BMI calculation).
- Conduct complete physical examination (including neurological examination).
- Obtain an ECOG performance status.
- Determine disease assessment by CT scan or positron emission tomography-CT scan.

After check in for hospitalization and at the Day –1 visit, the following procedures and evaluations will be performed.

- Reassess inclusion/ exclusion criteria and study qualification.

- Collect blood for PK: PK sample should be collected at time = 0 (before eating and within 1 hour prior to the morning dose of CTO1681). The PK blood sample should be processed within 1 hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.
- Perform supine 12-Lead ECGs in triplicate predose at -30 minutes (± 5 minutes) and -15 minutes (± 5 minutes).
- Assess vital signs (supine):
 - Measure BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature and oxygen saturation.
 - Collect at pre-morning CTO1681 administration (first dose) and at 30 minutes (± 10 minutes) and 1-hour (± 10 minutes) and 2 hours (± 10 minutes) post dose for all 3 doses.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals.
- PK blood sample collection:

For Phase 1b:

- 30 minutes (± 10 minutes), 1-hour (± 10 minutes; immediately following ECG assessment), 1.5 hours (± 10 minutes), 2 hours (± 10 minutes), 3 hours (± 10 minutes), and 4 hours (± 10 minutes) after the first (morning) dose.
- Second and third dose: blood sampling is to occur at 30 minutes (± 10 minutes) predose, and 1-hour postdose for the second and third doses.

For Phase 2a: blood samples to be collected 1-hour (± 10 minutes) postdose, immediately following the ECG assessment.

The PK blood samples should be processed within 1-hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.

- If not done the day before, collect blood samples for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): Blood sample to be drawn 1-hour post morning dose, following PK sample collection. Blood samples should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Collect supine 12-Lead ECG at 1-hour postdose (± 10 minutes) for the morning dose. Additional ECGs may be performed at the Investigator's discretion.
- Record dosing compliance.

10.3.4 Day 0

The following 1-hour postdose assessments must be performed in the following sequence: (1) 12-Lead ECG, (2) PK draw, and (3) cytokine draw.

At the Day 0 visit (± 1 day), the following evaluations will be performed:

- Collect vital signs (supine) predose of CTO1681: BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation.
- Conduct complete physical examination (including neurological examination).
- Collect blood for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, ferritin, IgG. Blood samples must be collected pre-morning CTO1681 dose.
- Collect blood sample for TBNK panel.
- Collect blood sample for anti-CD19 CAR T-cells.
- Collect blood sample for blood immunophenotyping.
- Collect blood sample for CAR copy number (performed irrespective of timing of CTO1681 dose).
- Collect supine 12-Lead ECG: Repeat 3 times consecutively at predose.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours ($\pm$ 30 minutes) preferred between meals.
- Collect supine 12-lead ECG: Repeat 3 times consecutively at 1-hour postdose ($\pm$ 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Collect PK blood sample: samples collected 1-hour ($\pm$ 10 minutes) post morning dose, immediately following ECG assessment. PK samples should be processed within 1-hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.
- Collect blood samples for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): Blood sample drawn approximately 1-hour postdose, following PK sample, and should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Administer CAR T-cell infusion according to the manufacturer's PI.
- Collect vital signs (supine) pre CAR T-cell infusion: BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Collect vital signs (supine) 30 minutes ($\pm$ 10 minutes), 1 hour ($\pm$ 10 minutes) and 2 hours ($\pm$ 15 minutes) post-morning dose of CTO1681. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- Obtain CRS and ICANS assessments.
- Record dosing compliance.

10.3.5 Day +1, Day +3 and Day +5

At the Day +1, +3 and +5 visits ($\pm$ 1 day), the following evaluations will be performed:

- Collect vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours ($\pm$ 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours ($\pm$ 30 minutes) preferred between meals.
- Collect supine 12-Lead ECG: Repeat 3 times consecutively at predose and 1-hour postdose ($\pm$ 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Collect blood samples for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): blood sample should be drawn prior to the morning dose and 1-hour postdose and processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Obtain CRS and ICANS assessments.
- Record dosing compliance.

10.3.6 Day +2 and Day +4

The following 1-hour postdose assessments must be performed in the following sequence: (1) 12-Lead ECG, (2) PK draw, and (3) cytokine draw.

At the Day +2 and Day +4 visits ($\pm$ 1 day), the following evaluations will be performed:

- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours ($\pm$ 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- Collect blood for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG and ferritin. Blood samples must be collected pre-morning CTO1681 dose.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours ($\pm$ 30 minutes) preferred between meals.
- Collect supine 12-Lead ECG: Repeat 3 times consecutively at predose and 1-hour postdose ($\pm$ 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Collect PK blood sample: blood samples collected 1-hour ($\pm$ 10 minutes) post morning dose, immediately following ECG assessment. PK samples should be processed within 1-hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.
- Collect blood samples for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): blood sample to be drawn prior to the morning dose and 1-hour postdose, following PK

sample, and should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.

- Collect blood for CAR copy number (performed irrespective of timing of CTO1681 dose)
- Obtain CRS and ICANS assessments.
- Record dosing compliance.

10.3.7 Day +6

The following 1-hour postdose assessments must be performed in the following sequence: (1) 12-Lead ECG, (2) PK draw, and (3) cytokine draw.

At the Day +6 visit (± 1 day), the following evaluations will be performed:

- Conduct complete physical examination (including neurological examination).
- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours (± 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- Collect blood for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG, and ferritin. Blood samples must be collected pre-morning CTO1681 dose.
- Collect supine 12-Lead ECG: Repeat 3 times consecutively at predose and 1-hour postdose (± 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals.
- Collect PK blood sample: blood samples collected 1-hour (± 10 minutes) post morning dose, immediately following ECG assessment. PK samples should be processed within 1-hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.
- Collect blood samples for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): blood sample to be drawn prior to the morning dose and 1-hour postdose, following PK sample, and should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Collect blood for immunophenotyping.
- Collect blood for Anti-CD19 CAR T-cells assay.
- Collect blood for CAR copy number (performed irrespective of timing of CTO1681 dose).
- Obtain CRS and ICANS assessments. If no CRS or ICANS noted then discharge is allowed if approved by Investigator, Medical Monitor and Sponsor. If patient is discharged schedule next study visit.

- Record dosing compliance.

10.3.8 Day +7

If patient is not hospitalized (ie, Day +7 through +13), vital signs and blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose.

At the Day +7 visit (± 1 day), the following evaluations will be performed:

- If the patient is hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours (± 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- If the patient is not hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken one time during the clinic visit, but may be repeated at the Investigator's discretion.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals.
- Obtain CRS and ICANS assessments if the patient is hospitalized or is seen for a clinic visit.
- Record dosing compliance: if patient is not hospitalized, study patient diaries ([Section 5.2.1](#)) should be collected and reviewed for completeness and compliance.
- If patient is not hospitalized schedule the next study visit.

10.3.9 Day +8 and Day +10

If patient is not hospitalized (ie, Day +7 through +13), vital signs and blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose. The 1-hour postdose 12-Lead ECG should be carried out prior to cytokine blood draw.

At the Day 8 visit (± 1 day), the following evaluations will be performed:

- If the patient is hospitalized measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours (± 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- If the patient is not hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken one time during the clinic visit, but may be repeated at the Investigator's discretion.
- Collect blood sample for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG and ferritin. Blood samples must be collected pre-morning CTO1681 dose.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals.

- Collect supine 12-Lead ECG: Repeat 3 times consecutively at 30 min predose and 1-hour postdose (± 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Collect blood sample for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): Sample drawn 1-hour postdose, following 12-Lead ECG, and should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Obtain CRS and ICANS assessments if the patient is hospitalized or is seen for a clinic visit.
- Record dosing compliance: if patient is not hospitalized, study patient diaries ([Section 5.2.1](#)) should be collected and reviewed for completeness and compliance.
- If patient is not hospitalized schedule the next study visit.

10.3.10 Day +9 and Day +11

If patients are not hospitalized (ie, Day +7 through +13), vital signs and blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose.

At the Day +9 and Day +11 visits (± 1 day), the following evaluations will be performed:

- If the patient is hospitalized measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours (± 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- If the patient is not hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken one time during the clinic visit, but may be repeated at the Investigator's discretion.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours (± 30 minutes) preferred between meals.
- Obtain CRS and ICANS assessments if the patient is hospitalized or is seen for a clinic visit.
- Record dosing compliance: if patient is outpatient, study patient diaries ([Section 5.2.1](#)) should be collected and reviewed for completeness and compliance.
- If patient is not hospitalized schedule the next study visit.

10.3.11 Day +12

If patients are not hospitalized (ie, Day +7 through +13), vital signs and blood samples will be collected only during clinic visits irrespective of timing of CTO1681 dose.

At the Day +12 visits (± 1 day), the following evaluations will be performed:

- If the patient is hospitalized measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be

taken pre-morning CTO1681 dose and 2 hours ($\pm$ 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.

- If the patient is not hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken one time during the clinic visit, but may be repeated at the Investigator's discretion.
- Collect blood sample for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG and ferritin. Blood samples must be collected pre-morning CTO1681 dose.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours ($\pm$ 30 minutes) preferred between meals.
- Collect supine 12-Lead ECG: Repeat 3 times consecutively at 30 minutes predose and 1-hour postdose ($\pm$ 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Obtain CRS and ICANS assessments if the patient is hospitalized or is seen for a clinic visit.
- Record dosing compliance: if patient is not hospitalized, study patient diaries ([Section 5.2.1](#)) should be collected and reviewed for completeness and compliance.
- If patient is not hospitalized schedule the next study visit.

10.3.12 Day +13

If patients are not hospitalized, they must return for a visit on Day +13.

The following 1-hour postdose assessments must be performed in the following sequence:

(1) 12-Lead ECG, (2) PK draw, and (3) cytokine draw.

At the Day +13 visit ($\pm$ 1 day), the following evaluations will be performed:

- If the patient is hospitalized measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken pre-morning CTO1681 dose and 2 hours ($\pm$ 15 minutes) post-morning dose. Collect vital signs every 6-8 hours or more frequently at the Investigator's discretion.
- If the patient is not hospitalized, measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation. Readings should be taken one time during the clinic visit, but may be repeated at the Investigator's discretion.
- Collect blood sample for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG, and ferritin. Blood samples must be collected pre-morning CTO1681 dose.
- CTO1681 tablets will be dosed at the protocol specified dose TID within 30 minutes after starting to eat a meal or snack; 5 hours ($\pm$ 30 minutes) preferred between meals.

- Collect supine 12-Lead ECG: Repeat 3 times consecutively at 30 minutes predose and 1-hour postdose (± 10 minutes) for the morning doses. Additional ECGs may be performed at the Investigator's discretion.
- Collect PK blood sample: samples collected 1-hour (± 10 minutes) post morning dose, immediately following ECG assessment. PK samples should be processed within 1-hour of collection and frozen at -80°C until shipment. See [Section 6.8](#) and Laboratory Manual for further details.
- Collect blood sample for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): Sample drawn 1-hour postdose, following PK sample, and should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Collect blood for blood immunophenotyping.
- Collect blood sample for Anti-CD19 CAR T cells assay.
- Collect blood sample for CAR copy number (performed irrespective of timing of CTO1681 dose).
- Perform CRS and ICANS assessments.
- Record dosing compliance: if patient is outpatient, study patient diaries ([Section 5.2.1](#)) should be collected and reviewed for completeness and compliance.
- Schedule the next visit.

10.3.13 Follow-up 1 (Day +14) Visit

At the Follow-up 1 (Day +14 [± 1 day]) visit, the following evaluations will be performed:

- Measure weight (including BMI).
- Conduct complete physical examination (including neurological examination).
- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation.
- Collect blood sample for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): samples drawn at the same time of day that the post-morning CTO1681 samples are expected to be collected during the treatment period. Samples should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Perform CRS and ICANS assessments.
- Schedule next study visit.

10.3.14 Follow-up 2 (Day +20) Visit

At the Follow-up 2 (Day +20) visit (± 1 day), the following evaluations will be performed:

- Conduct complete physical examination (including neurological examination).

- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation.
- Perform CRS and ICAN assessments.
- Schedule next study visit.

10.3.15 Follow-up 3 (Day +41) Visit

At the Follow-up 3 (Day +41) visit (± 2 days), the following evaluations will be performed:

- Measure weight (including BMI).
- Conduct complete physical examination (including neurological examination).
- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation.
- Collect blood sample for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG and ferritin.
- Determine disease assessment by CT scan or positron emission tomography-CT scan.
- Perform CRS and ICANS assessments.
- Schedule next study visit.

10.3.16 Month 1; Month 3; and Month 6, End of Study/ Early Termination Visits (Phase 1b = Month 6; Phase 2a = Month 12)

At the Month 1 Visit (Day 30 ± 2); Month 3 Visit (Day 90 ± 3); and Month 6 visit (Day 180 ± 7), End of Study/ Early Termination Visit (Phase 1b = Day 180 ± 7 ; Phase 2a = Month 12/ Day 360 ± 7), the following evaluations will be performed:

- Measure weight (including BMI).
- Conduct complete physical examination (including neurological examination).
- Measure vital signs (supine): BP (systolic and diastolic), pulse rate, respiratory rate, oral temperature, and oxygen saturation.
- Collect blood sample for clinical laboratory measurements ([Section 6.10.7.1](#)): CBC with differentials, chemistry panel, PT, PTT and INR, CRP, IgG and ferritin.
- Collect blood sample for TBNK panel.
- Collect blood sample for blood immunophenotyping.
- Collect blood sample for cytokine assay (IFN- γ , TNF- α , IL-1, IL-2, IL-6, and GM-CSF): samples drawn at the same time of day that the post-morning CTO1681 samples are expected to be collected during the treatment period. Samples should be processed within 1-hour of collection, and frozen at -80°C until shipment. See [Section 6.9](#) and Laboratory Manual for further details.
- Collect blood sample for Anti-CD19 CAR T cells assay.

- Collect blood sample for CAR copy number assay.
- Perform ECOG assessment:
 - Phase 1b: performed at Month 6/ End of Study/ Early Termination (Day 180 ± 7) visits only.
 - Phase 2a: performed at Month 6 (Day 180 ± 7) and Month 12/ End of Study/ Early Termination (Day 360 ± 7) visits only.
- Determine disease assessment by CT scan or positron emission tomography-CT scan.
- Perform CRS and ICANS assessments.
- Schedule any remaining study visits.

10.4 National Cancer Institute Common Terminology Criteria for Adverse Events, Version 5.0

The National Cancer Institute CTCAE, version 5.0, can be accessed using the following link:

https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_5x7.pdf

10.5 Eastern Cooperative Oncology Group Performance Status Scoring

Grade	Symptomatology
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, eg, light housework, office work
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair
5	Dead

Source: ([Oken, Creech et al. 1982](#)).